

FATIMA JINNAH WOMEN UNIVERSITY



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Reg no: 2023-BCS-073

Course: Computer Network (Lab)

Section: BCS-3(B)

Instructor: Sir Majid

PROJECT REPORT

Installation of VMWARE

Configuration of VMWARE

Microsoft Server R2 on VMWARE

Microsoft Window on VMWARE for domain controller

Create an Active Directory

Steps for Installing and Configuring VMware Workstation 15:

1. Install VMware Workstation 15:

Download VMware Workstation 15:

Go to the VMware Workstation download page and download the appropriate version (ensure it matches your system architecture: 32-bit or 64-bit).

Run the Installer:

Open the downloaded file to start the installation process.

Follow the installation wizard, accept the license agreement, and click through the necessary steps.

Choose whether to enable shortcuts and desktop icons as per your preferences, and complete the installation.

Launch VMware Workstation:

After installation, launch **VMware Workstation** from your desktop or Start Menu.

2. Create a New Virtual Machine in VMware Workstation 15:

Create a New Virtual Machine:

From the VMware Workstation menu, select **File > New Virtual Machine**.

Choose the **Typical (recommended)** configuration and click **Next**.

Select Installation Media:

Choose the installation media:

If you are installing **Windows Server 2012 R2**, select the **Installer disc image file (ISO)** and browse to the ISO file of **Windows Server 2012 R2**.

For **Windows**, select the respective Windows ISO.

Select Operating System:

For **Windows Server 2012 R2** choose **Microsoft Windows > Windows Server 2012 R2**.

Click **Next**.

Name the Virtual Machine and Choose Location:

Provide a name for your VM

Choose the location where you want to store the VM files.

Click **Next**.

Configure Virtual Machine Hardware:

Choose the **Number of Processors** and **Memory**

Create a Virtual Hard Disk

Click **Next**, then **Finish** to complete the VM creation.

3. Install Windows Server 2012 R2 (Domain Controller Setup):

Start the Virtual Machine:

After creating the VM, click **Power on this virtual machine** to start the installation.

Install Windows Server 2012 R2:

Follow the on-screen steps to install **Windows Server 2012 R2**:

Choose language, time, and keyboard preferences.

Select **Windows Server Standard (Desktop Experience)** and click **Next**.

Accept the license agreement and click **Next**.

Choose **Custom: Install Windows only** and select the virtual disk to install the OS.

Complete Installation:

After the installation completes, the server will reboot, and you will be prompted to set an **Administrator password**.

Set a strong password for the Administrator account and log in after the server reboots.

4. Install Active Directory Domain Services:

Open Server Manager:

Once you're logged into the Windows Server, open **Server Manager** from the Start Menu or Taskbar.

Add Roles and Features:

Click **Manage > Add Roles and Features** to start the wizard.

Install AD DS:

In the wizard, select **Role-based or feature-based installation** and click **Next**.

Select the **Server** (usually the local server) and click **Next**.

On the **Select server roles** screen, check the box for **Active Directory Domain Services**.

Click **Next** until you reach the **Confirm installation selections** page, then click **Install**.

Promote the Server to Domain Controller:

Once AD DS is installed, a notification will appear in **Server Manager** prompting you to promote the server to a Domain Controller.

Click on the **Promote this server to a domain controller** link.

Configure Domain:

In the **Active Directory Domain Services Configuration Wizard**:

Select **Add a new forest**.

Enter the **Root domain name**

Set the **Forest functional level** and **Domain functional level** (typically **Windows Server 2012 R2**).

Set the **Directory Services Restore Mode (DSRM)** password (used for Directory Services recovery).

Complete the Promotion:

Follow the wizard, then click **Next** and **Install**.

The server will need to reboot.

5. Verify the Domain Controller Installation:

Login:

After the server reboots, log in as **Administrator** using the password set during the promotion process.

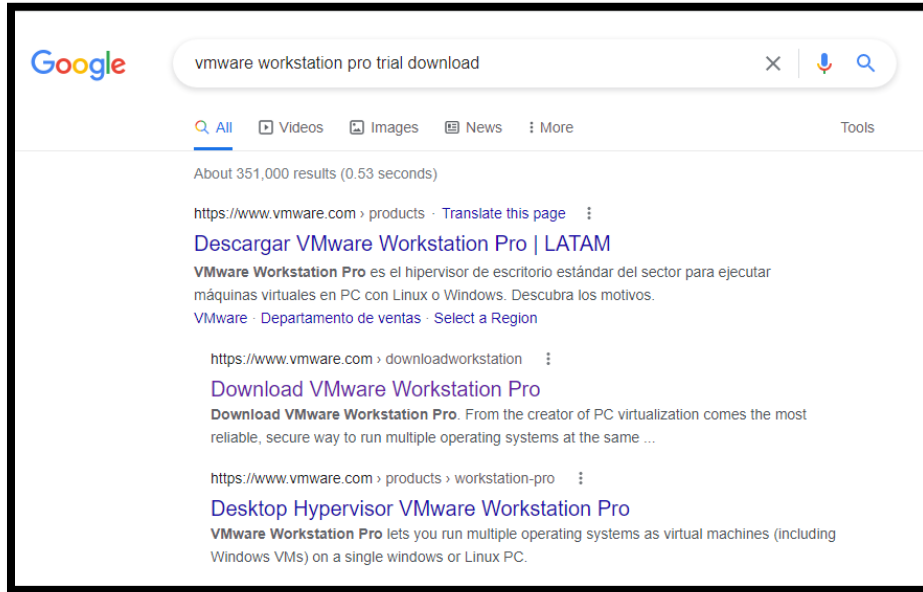
Check Active Directory:

Open **Active Directory Users and Computers** from **Server Manager > Tools**.

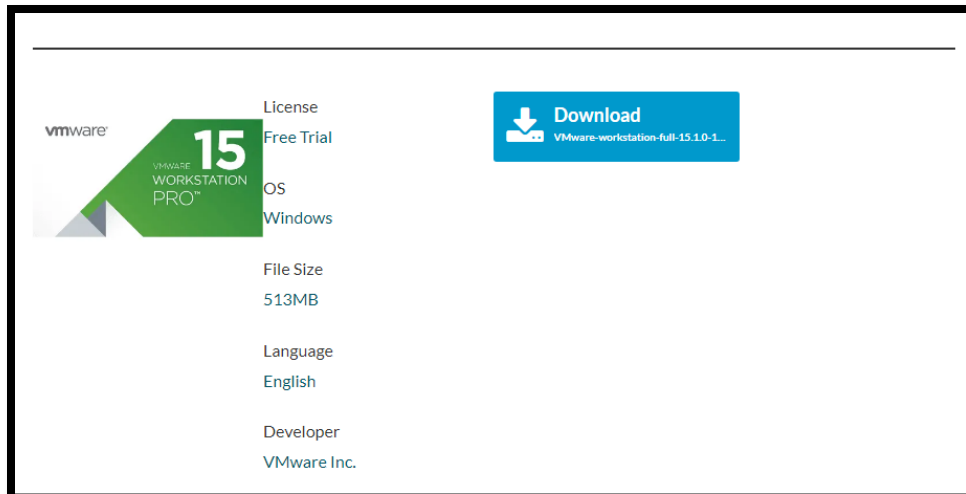
Verify that your domain is listed, and you can manage users, groups, and organizational units.

Screenshots

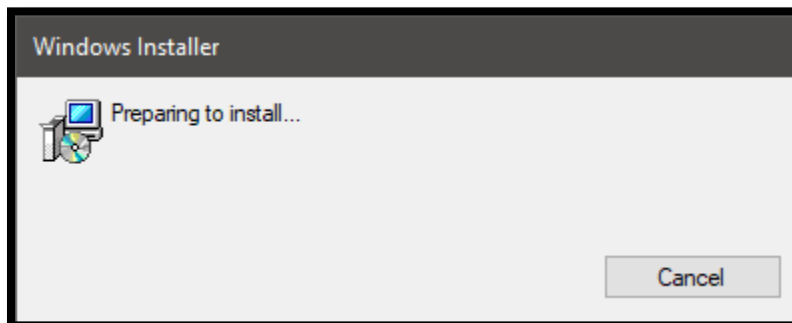
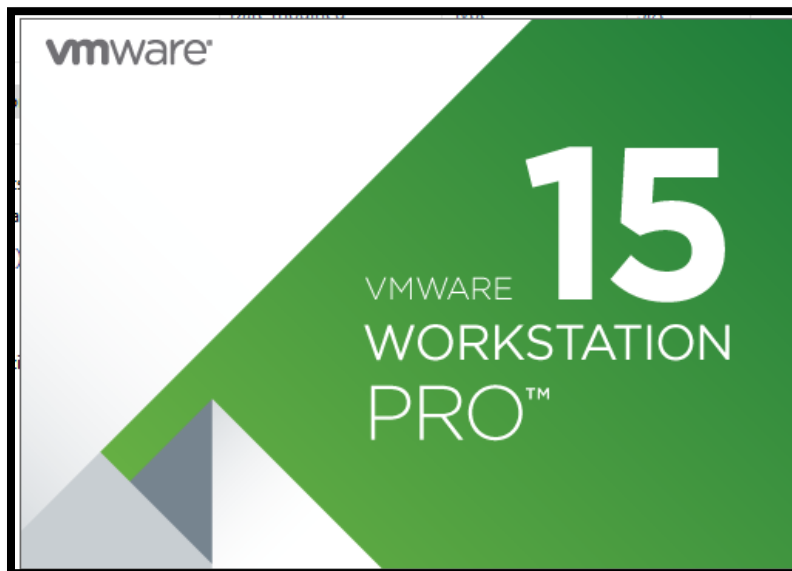
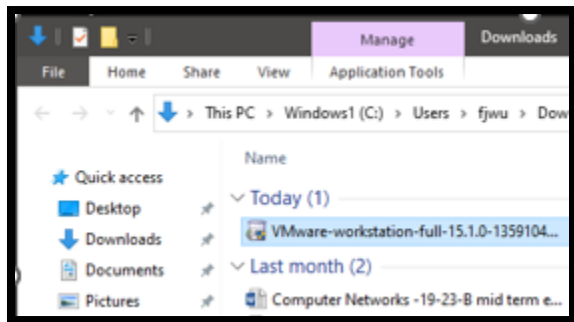
1.Installation of VMWARE



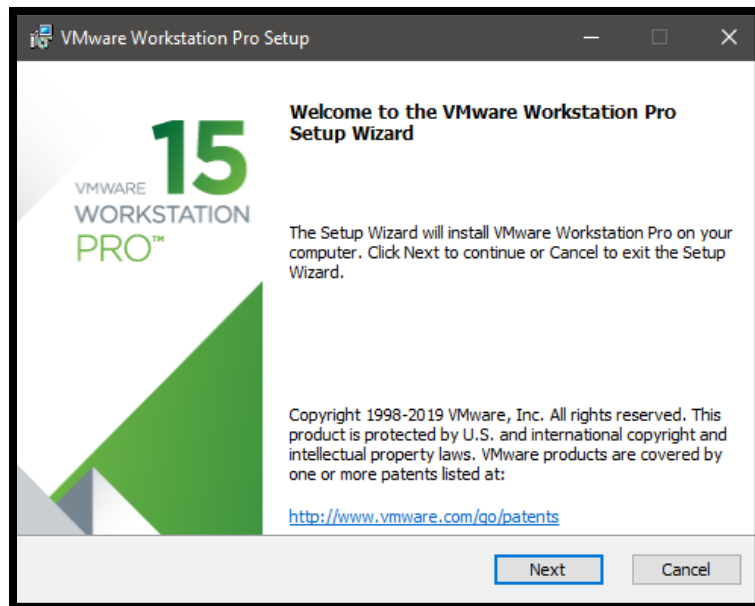
Click on Download VMware Workstation pro



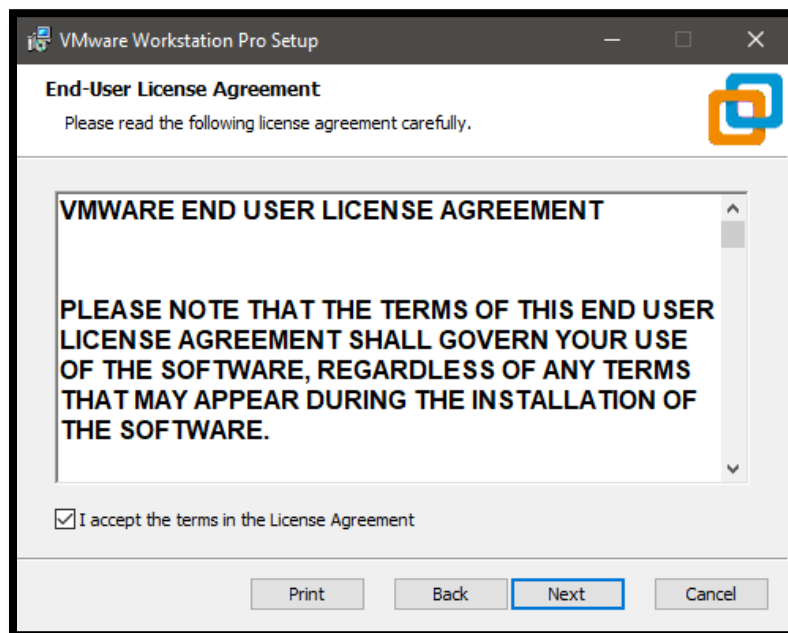
After Downloading is complete, go to Downloads and click on the setup.



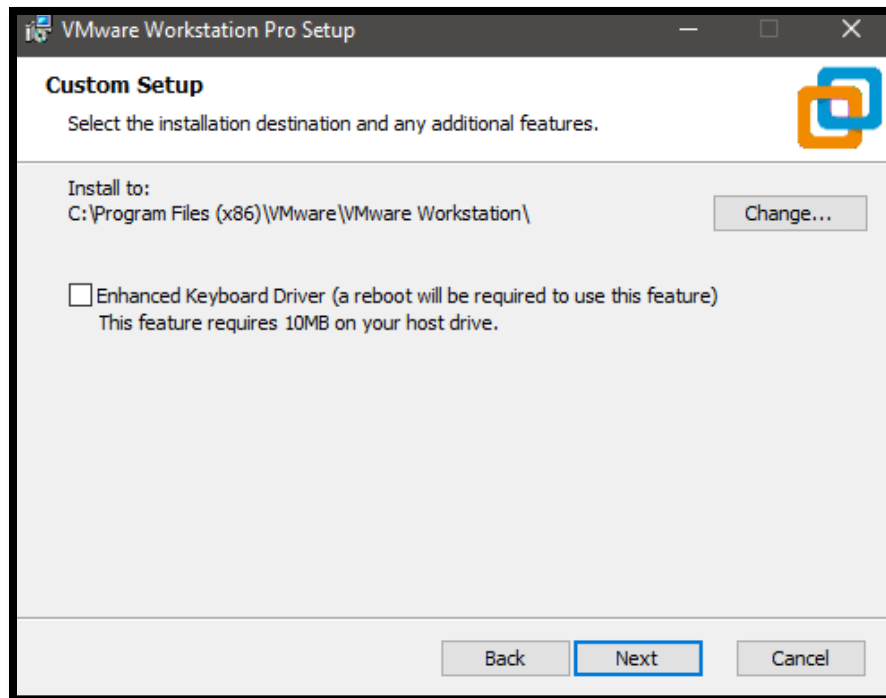
Click on next



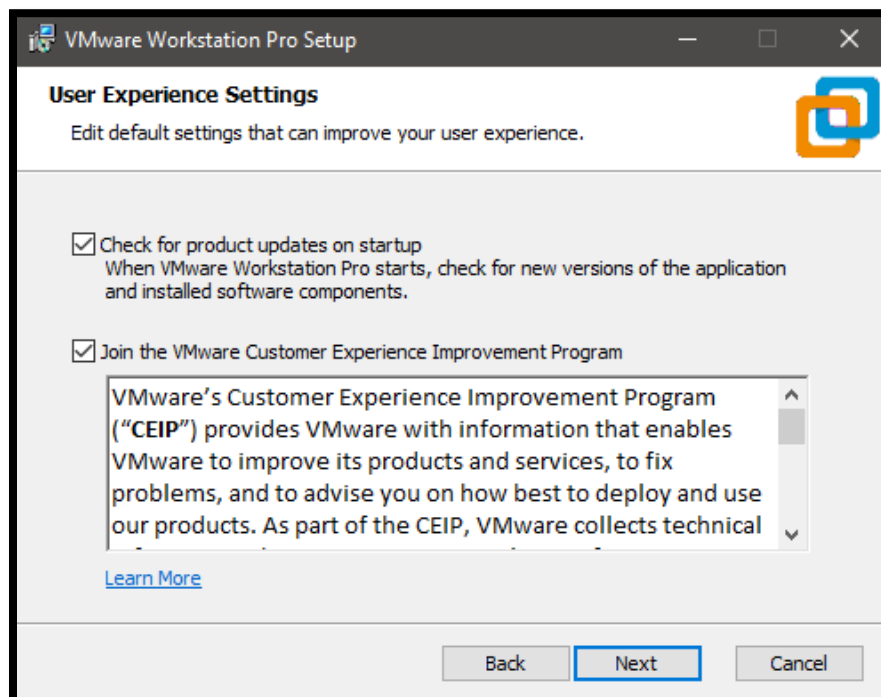
Check the I accept the terms in License Agreement and then click on next.



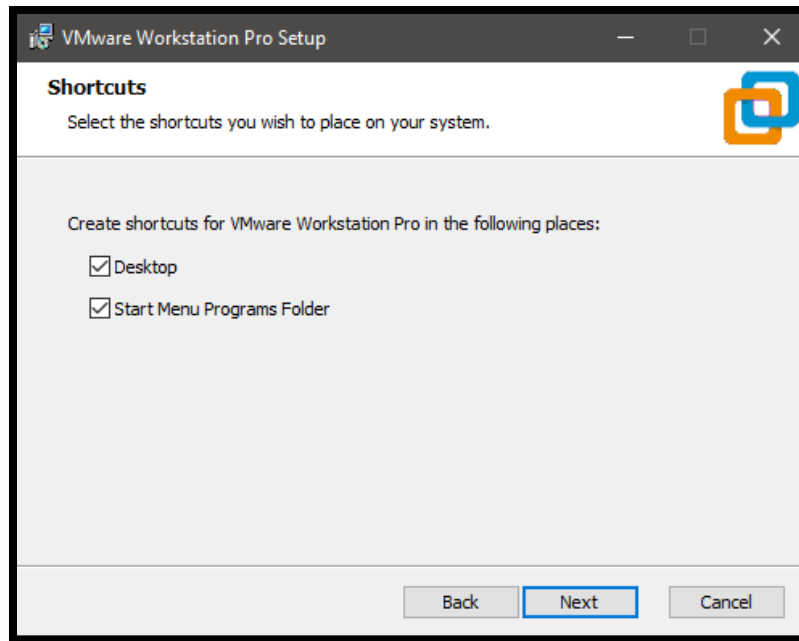
Click on next.



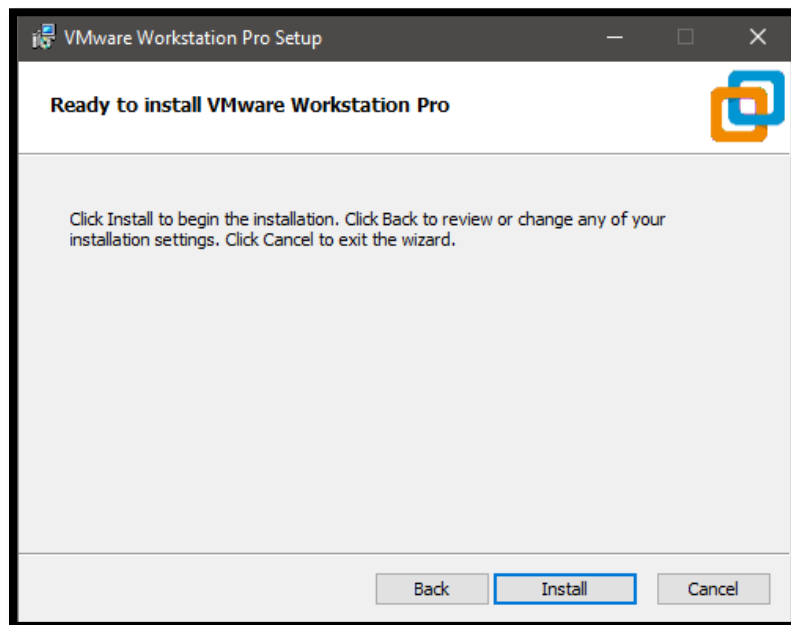
Check both options and click on next.



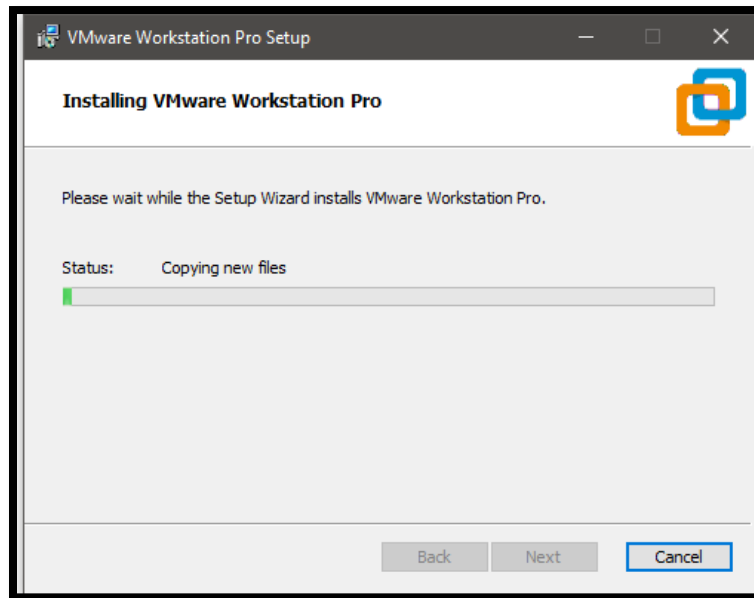
Create shortcut of VMware.



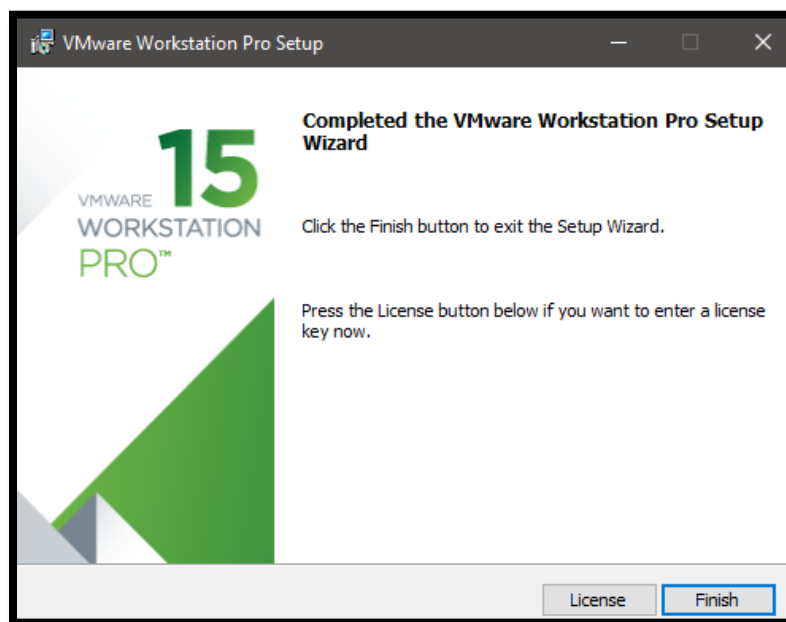
Click on install



Wait for the installation



Click on finish

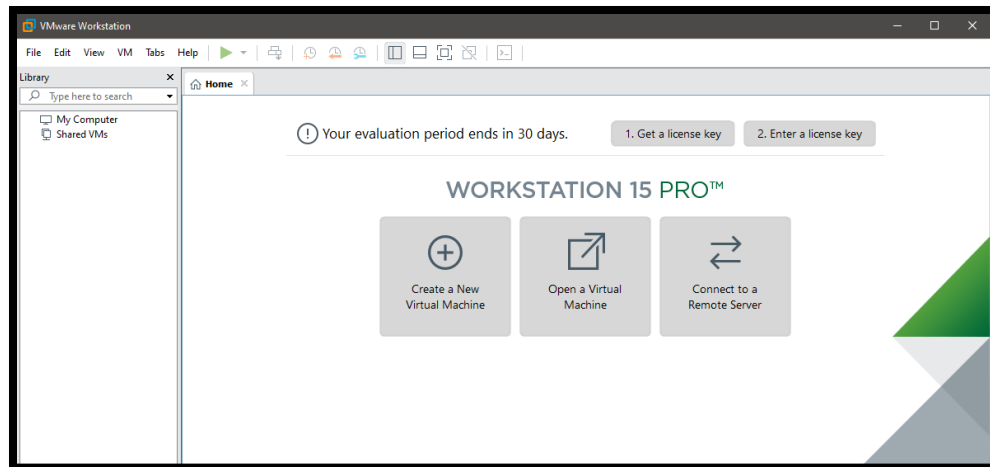


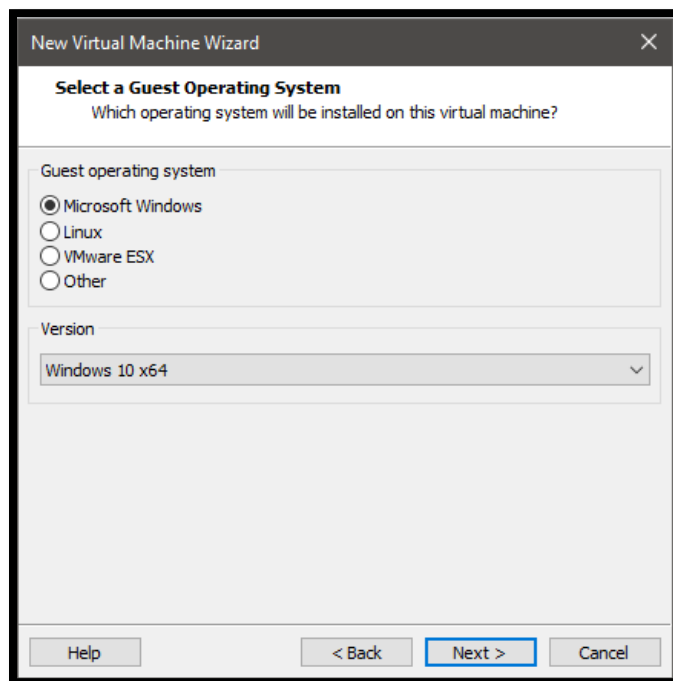
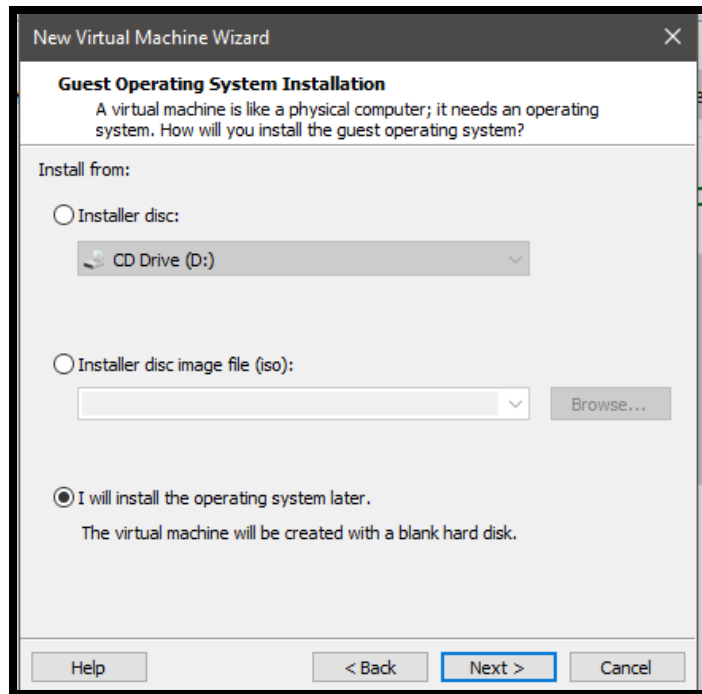
Click on continue



Then click on finish







New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

Location:

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

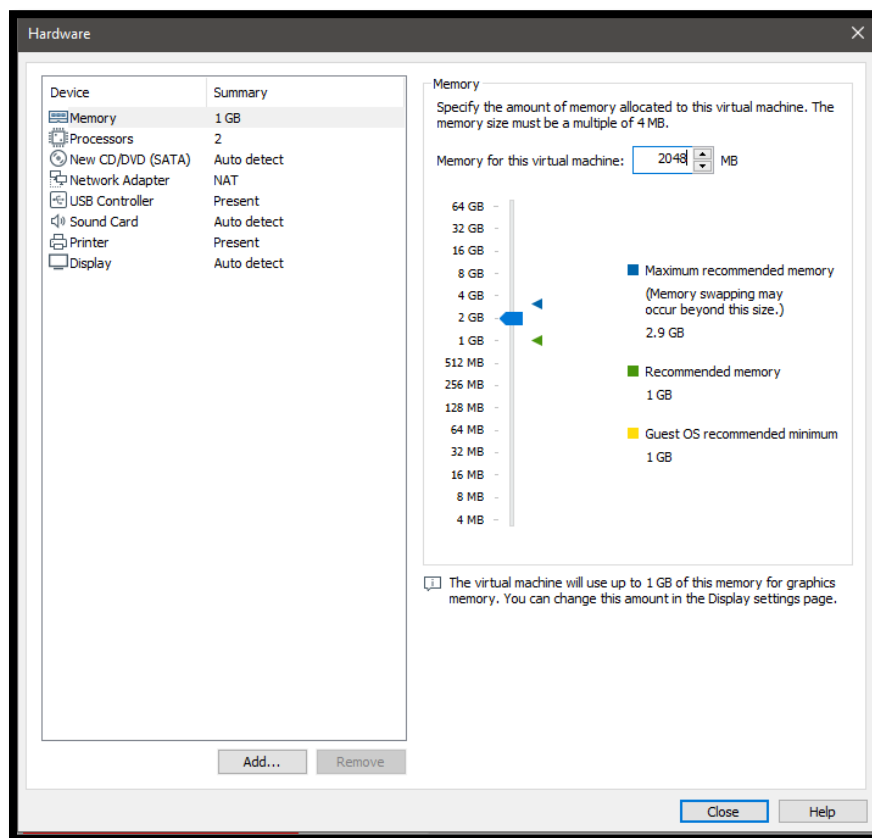
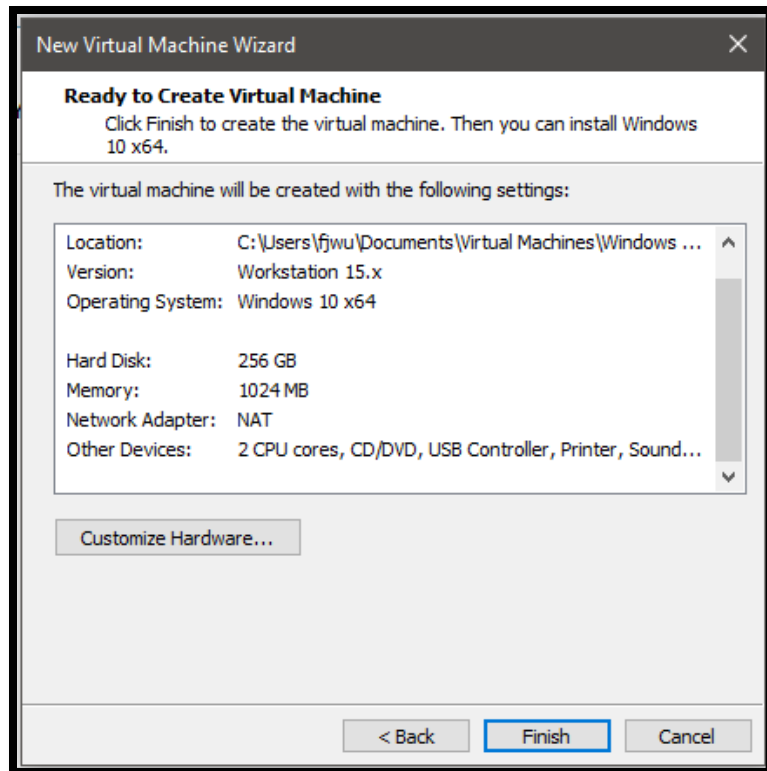
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

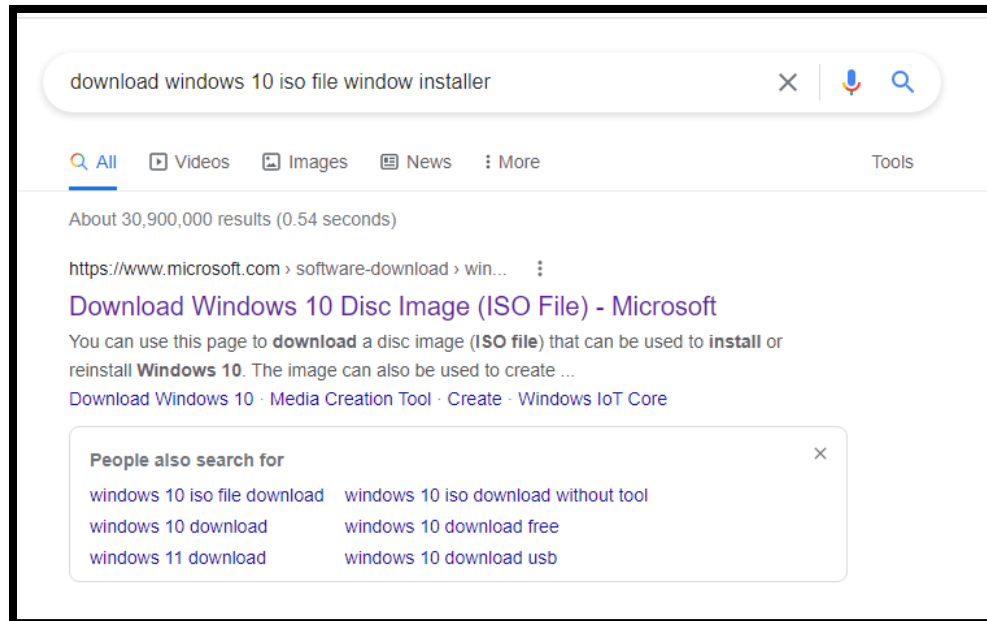
Recommended size for Windows 10 x64: 60 GB

☒ Store virtual disk as a single file
☐ Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

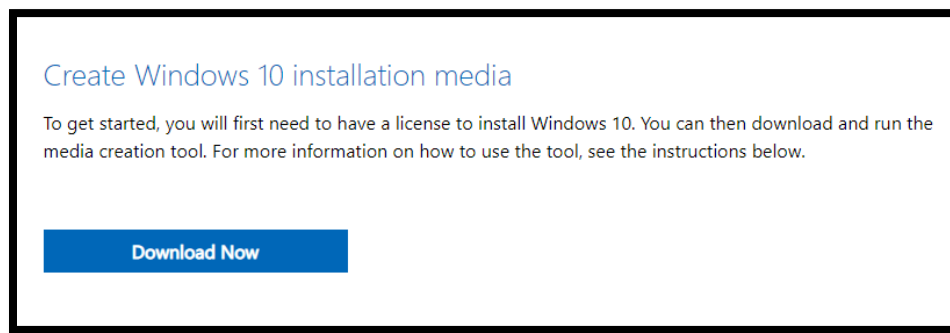
Help < Back Next > Cancel

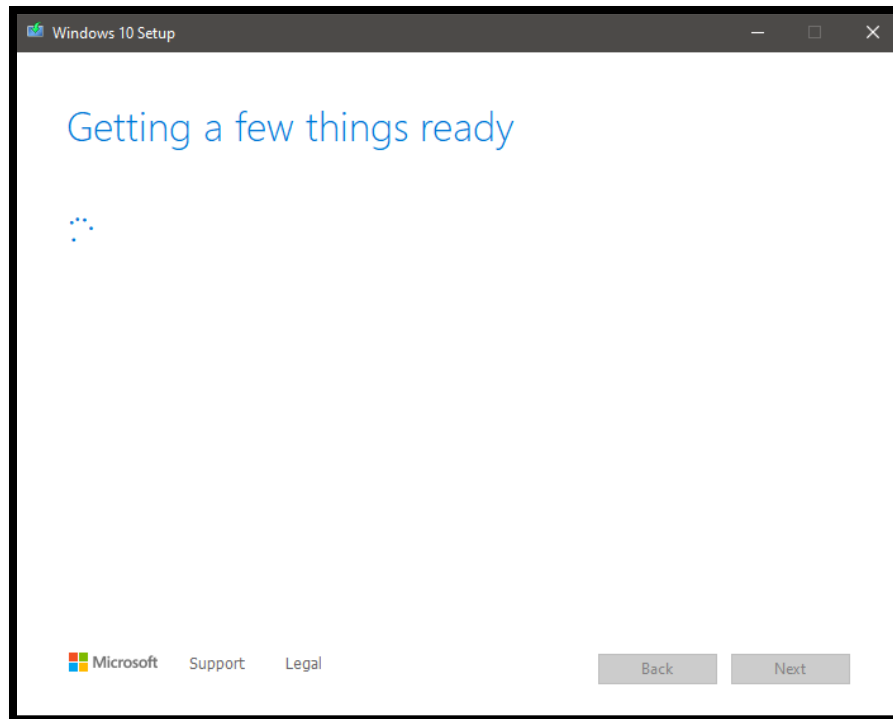


Download windows 10 iso file window installer

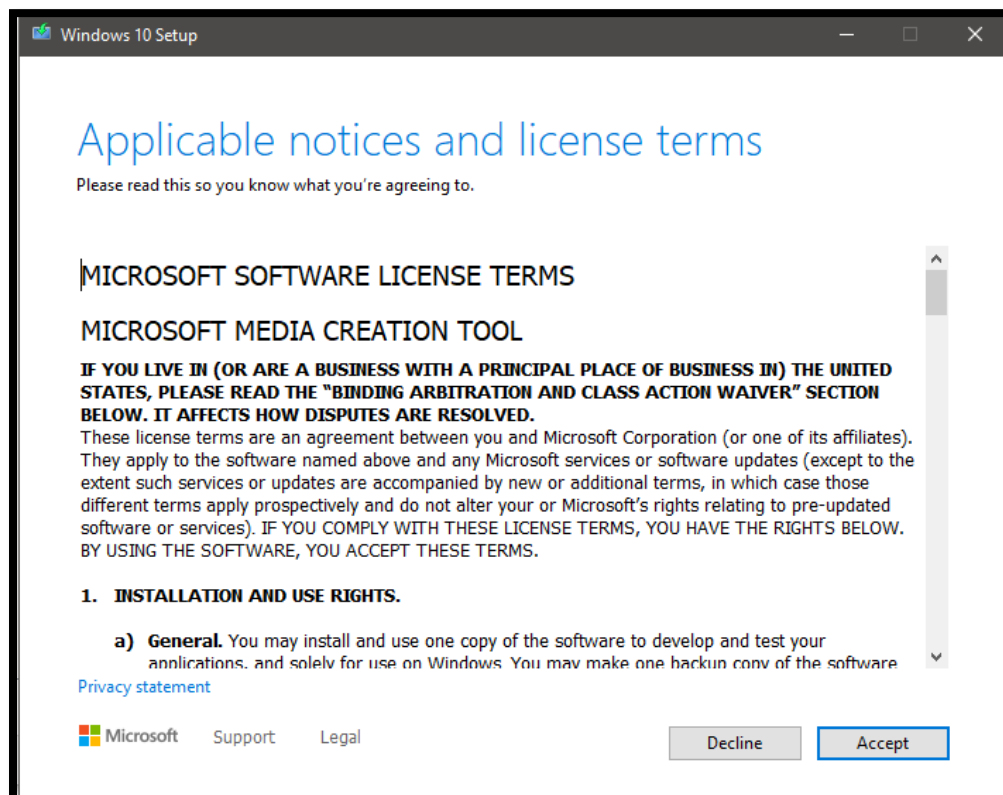


Click on Download

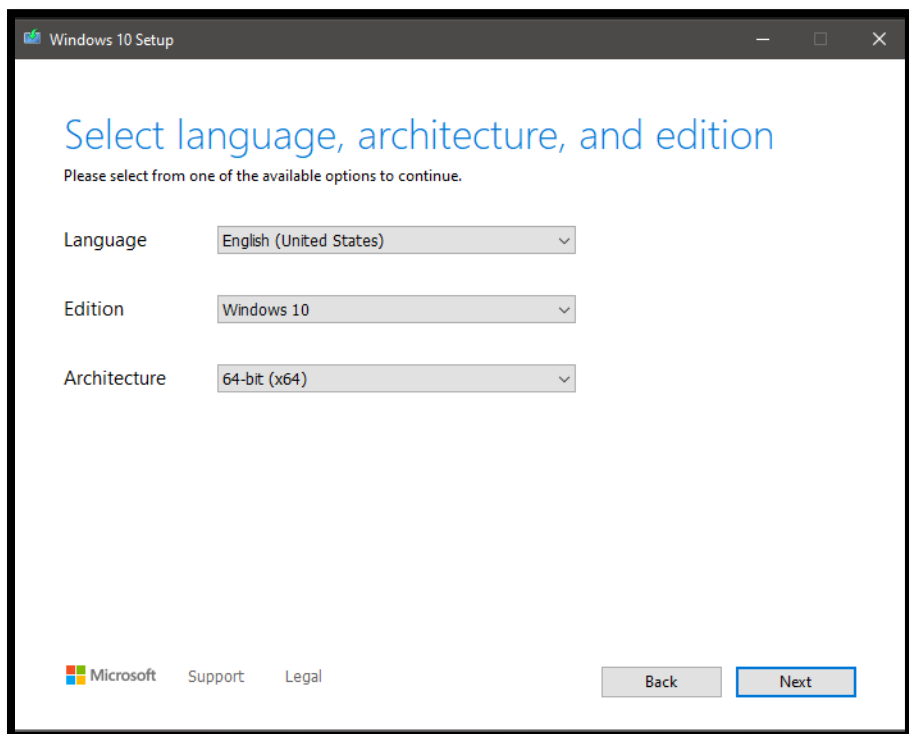
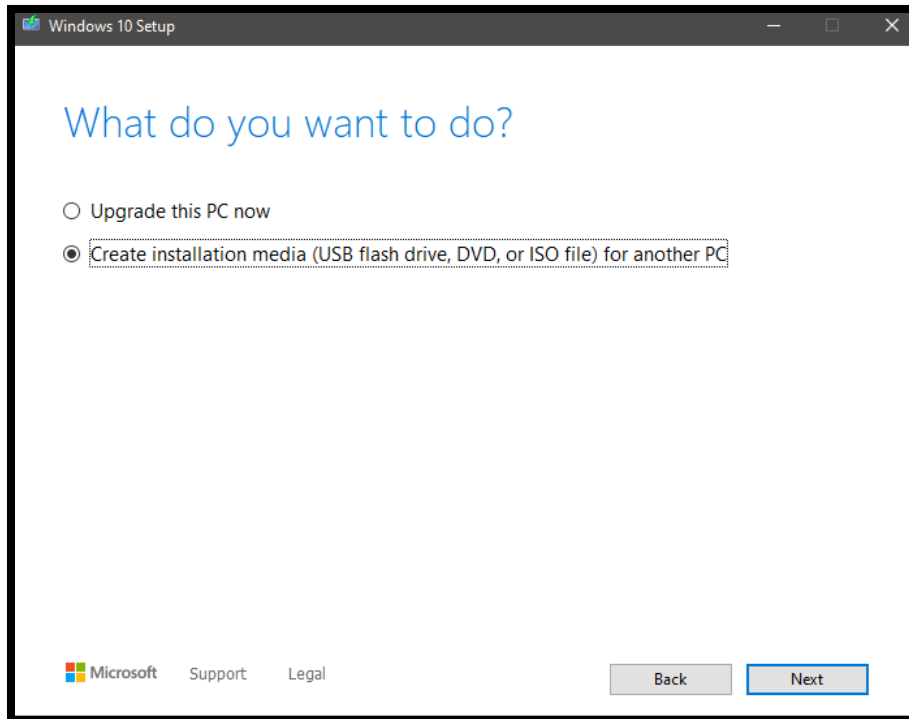




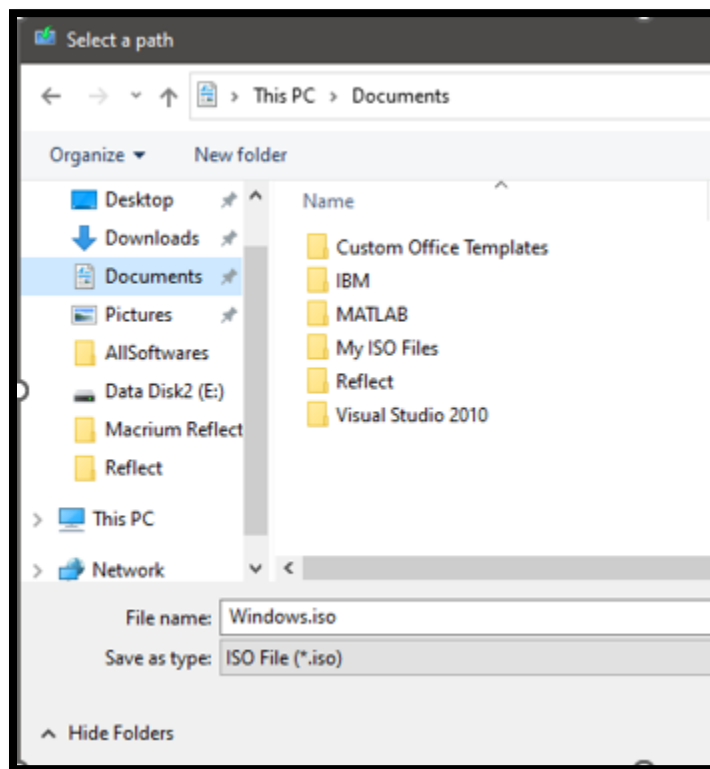
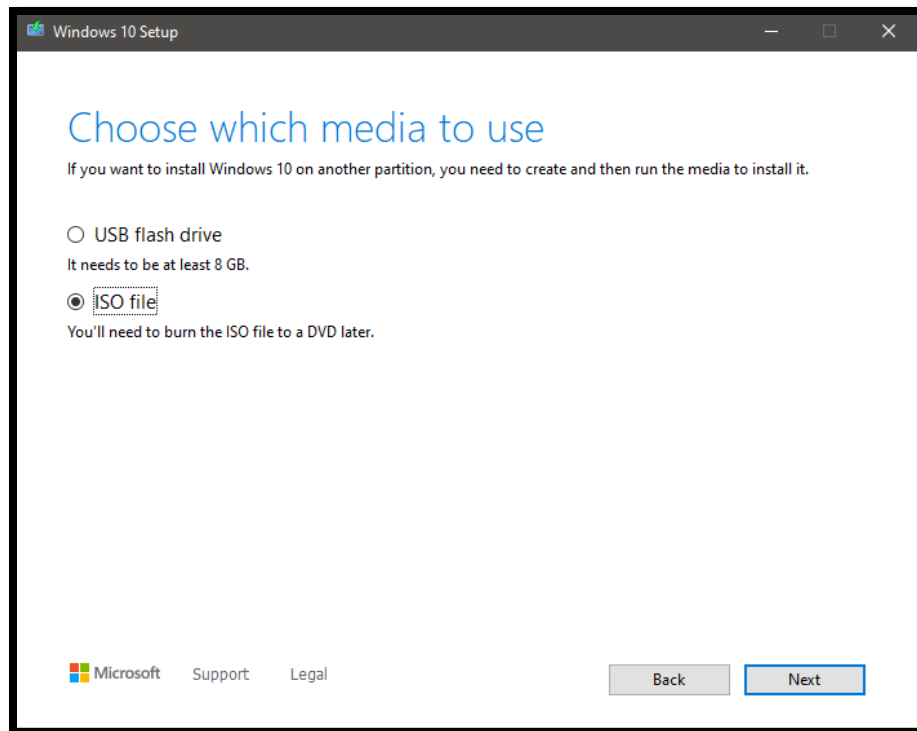
Click on accept

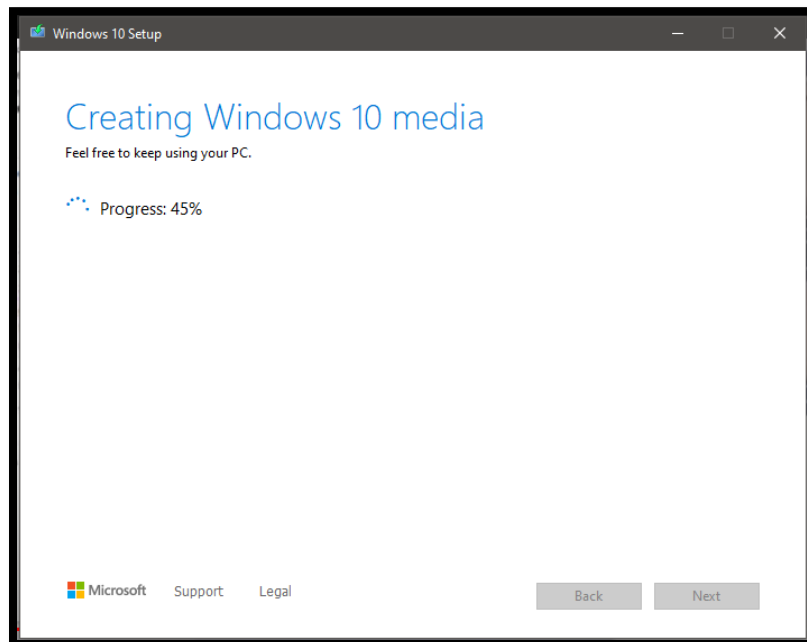
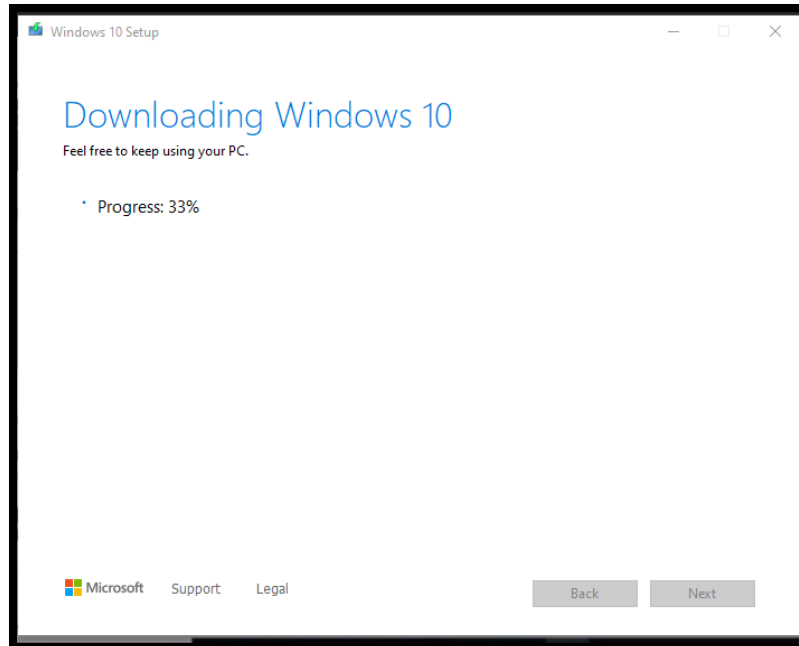


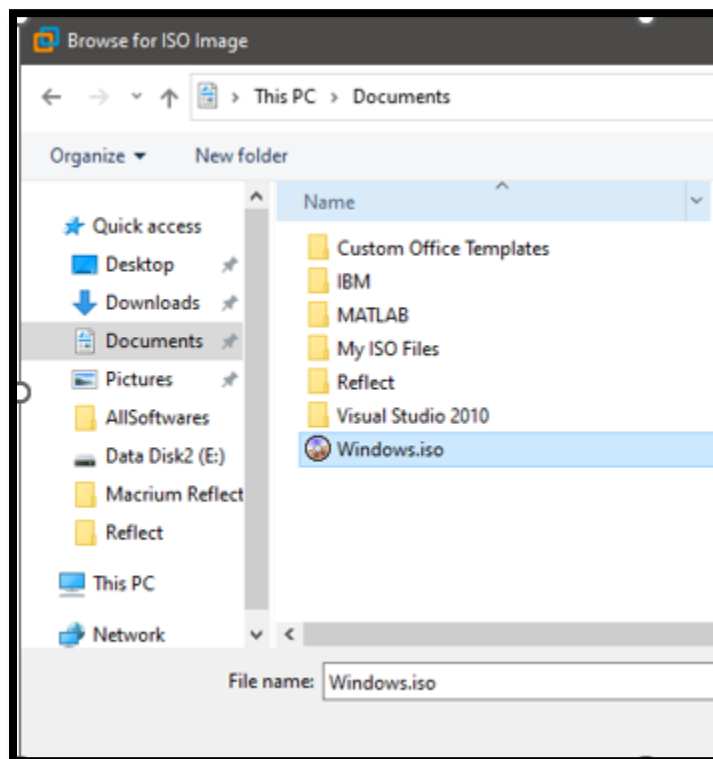
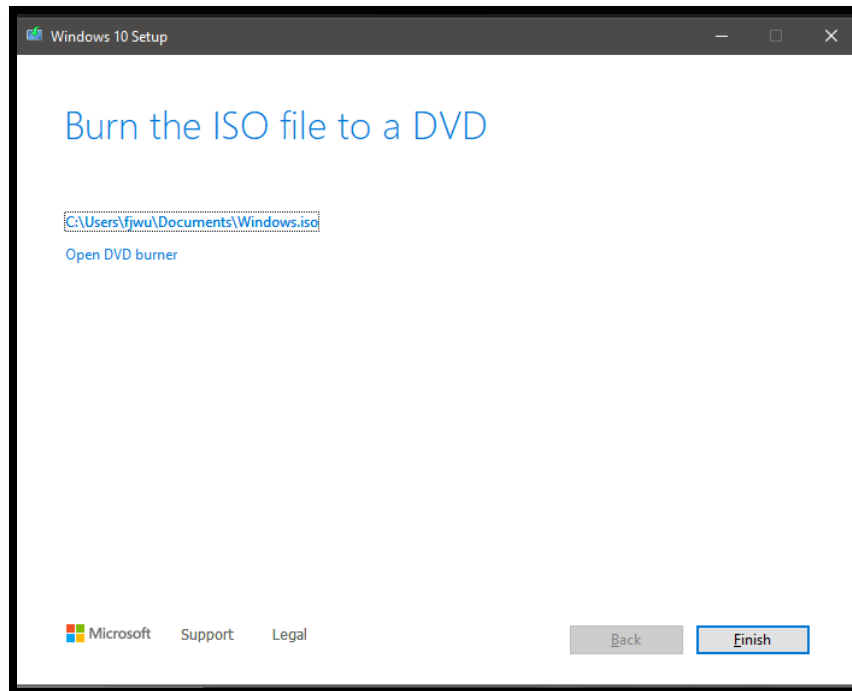
Select the second option and click on next.

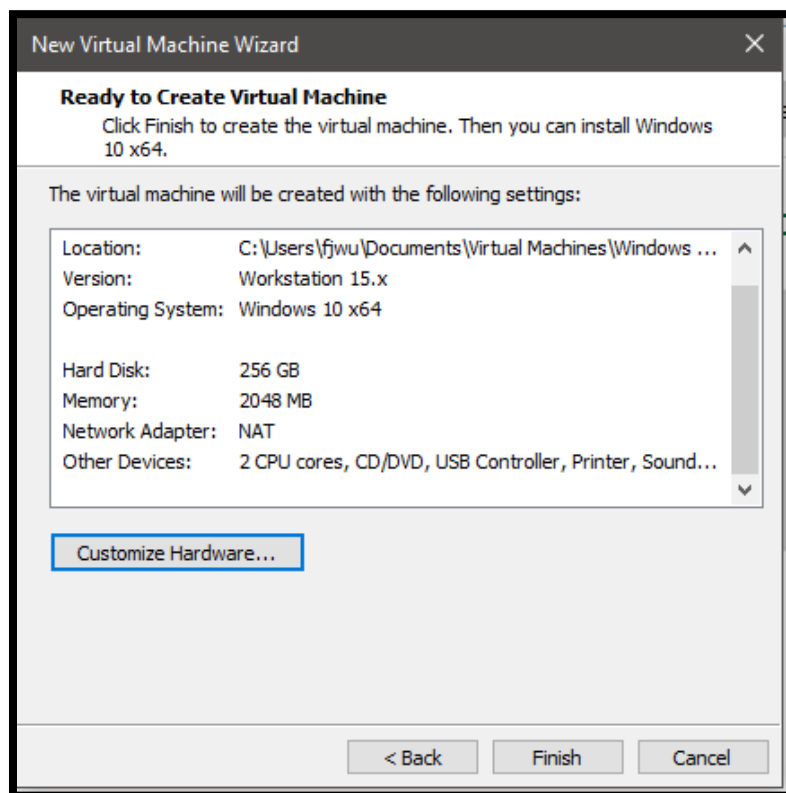
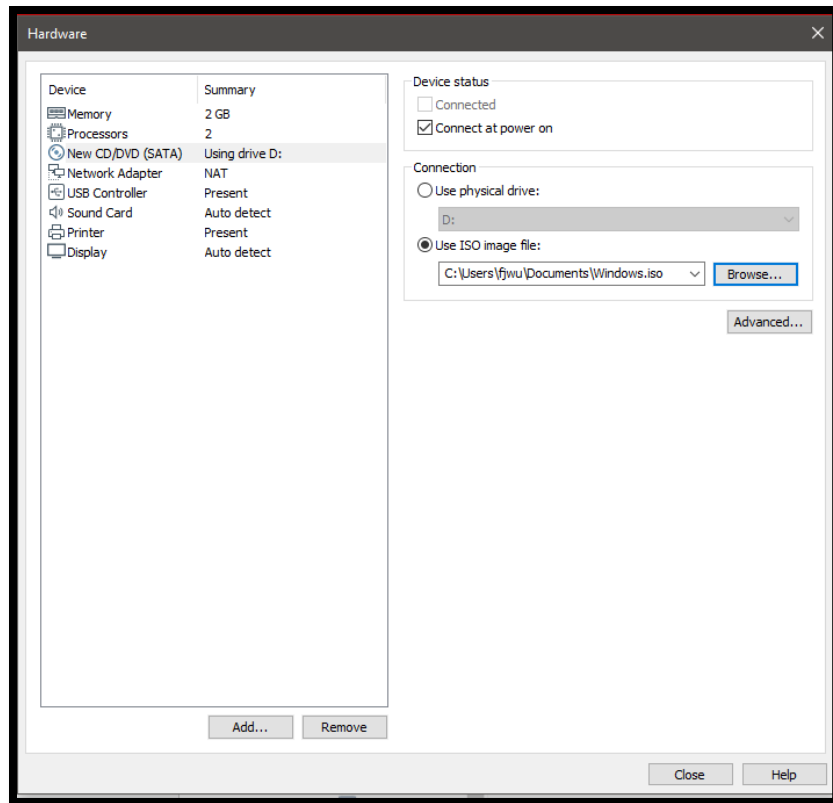


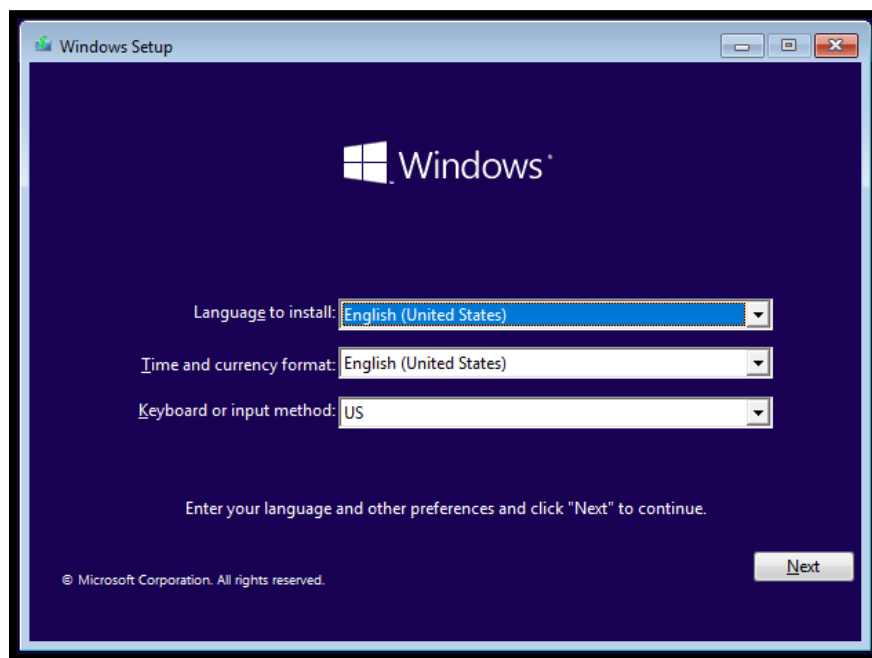
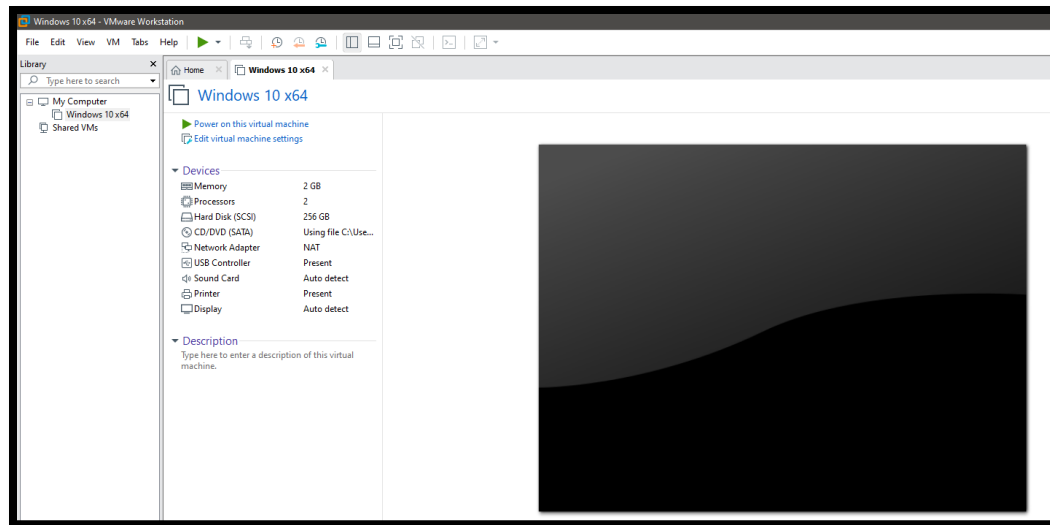
Choose ISO File.



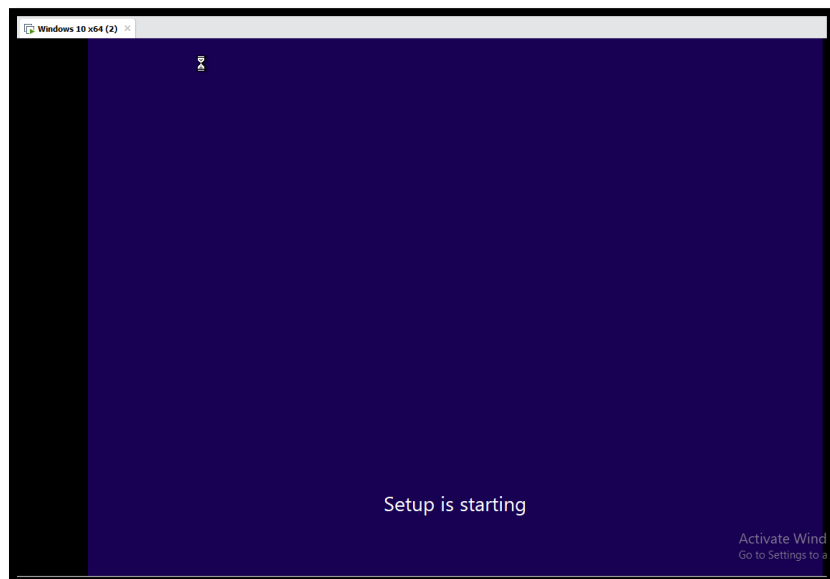
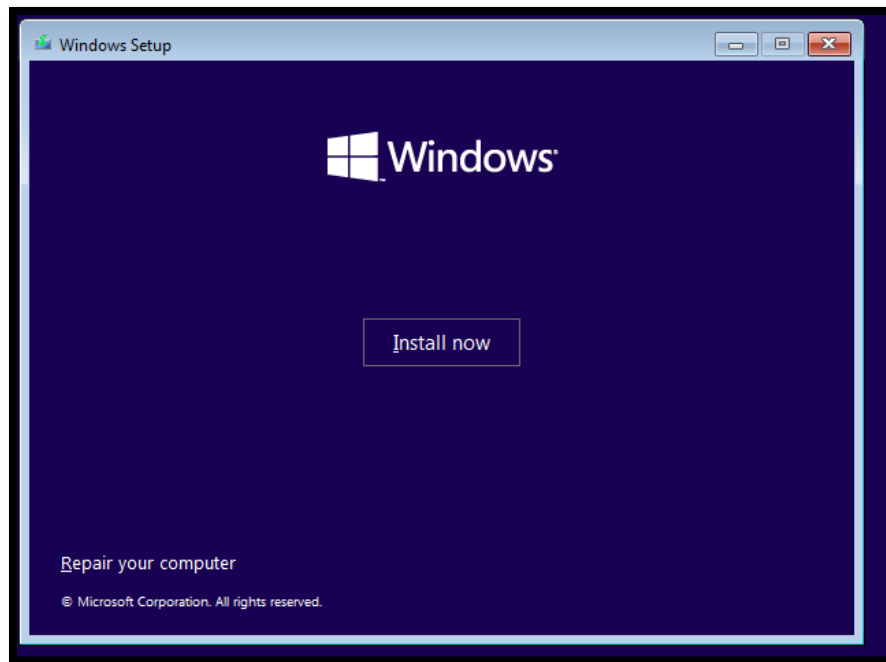


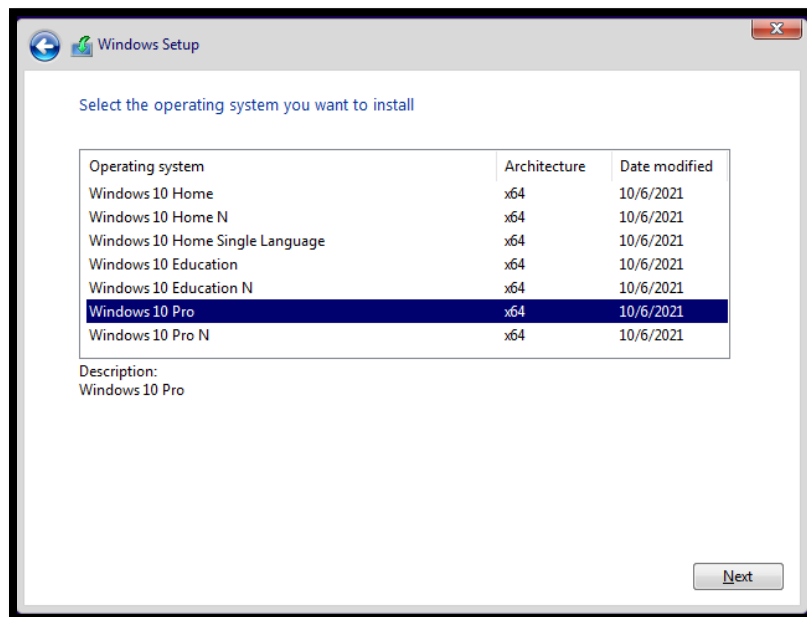
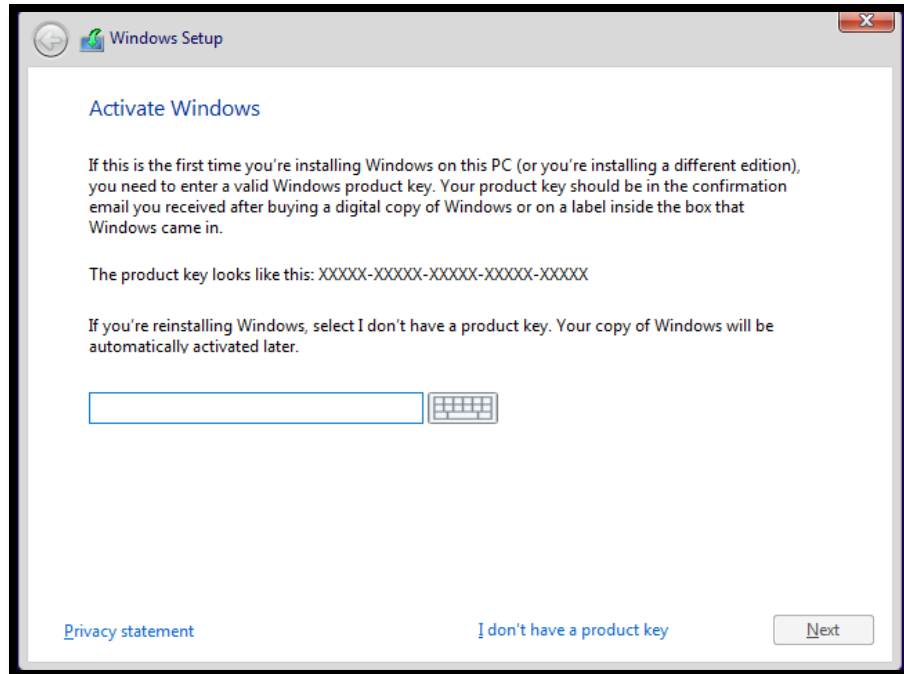


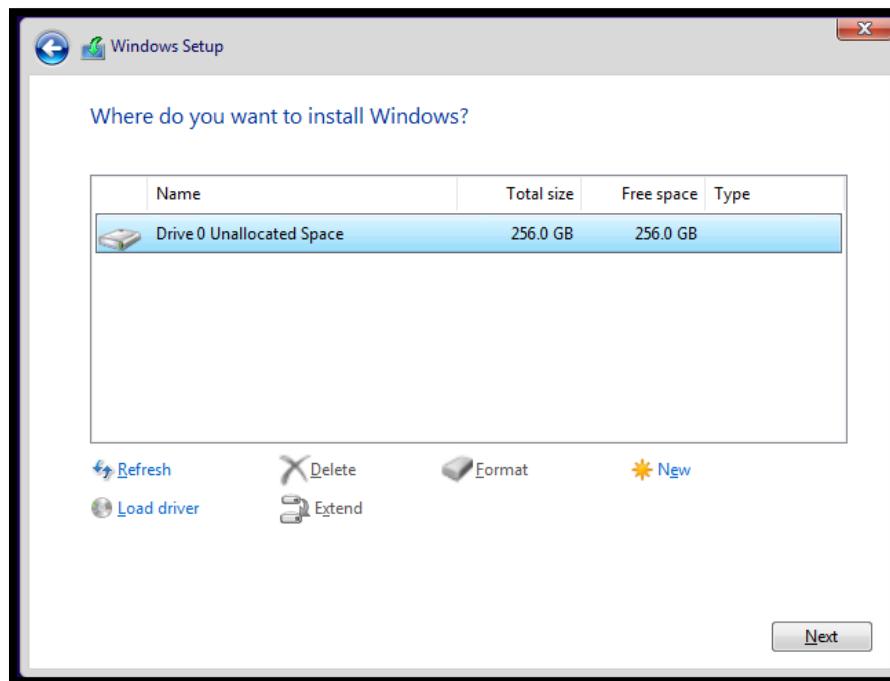
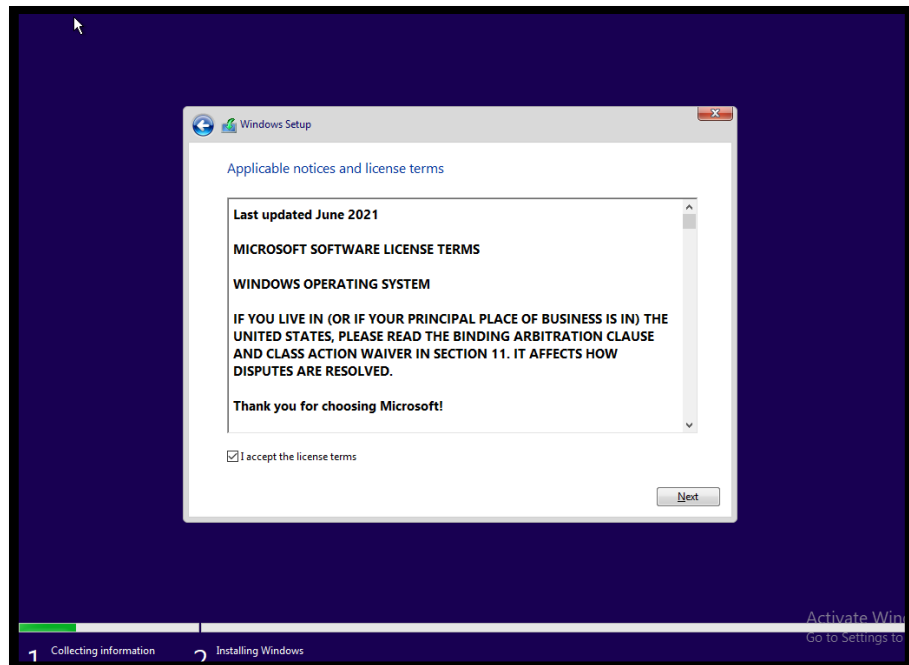


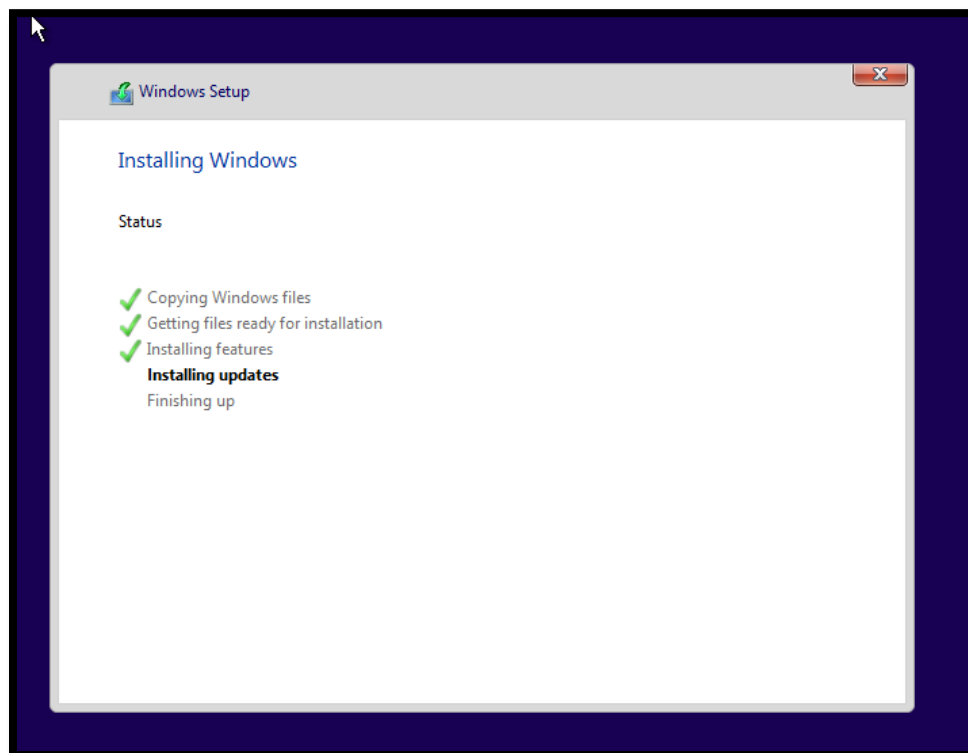
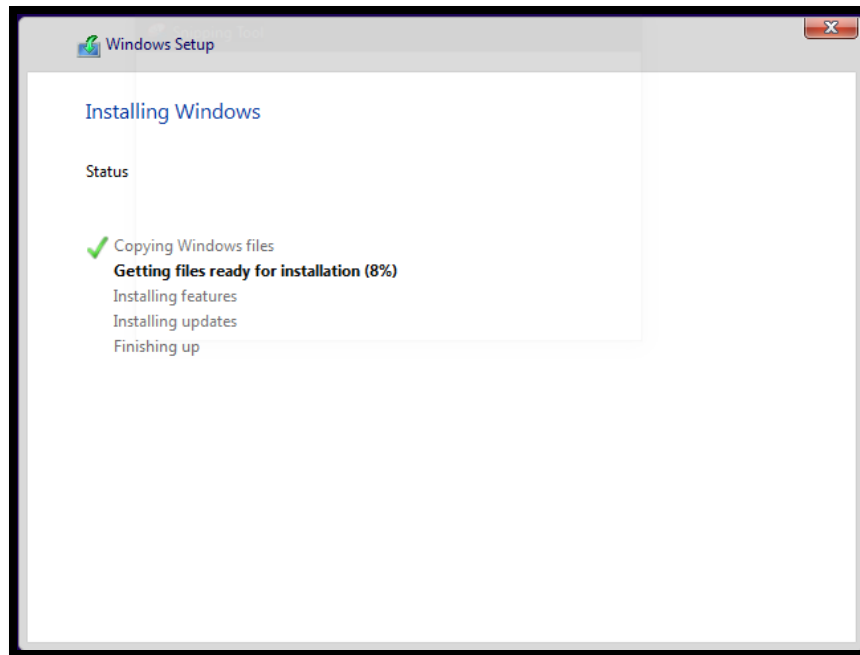


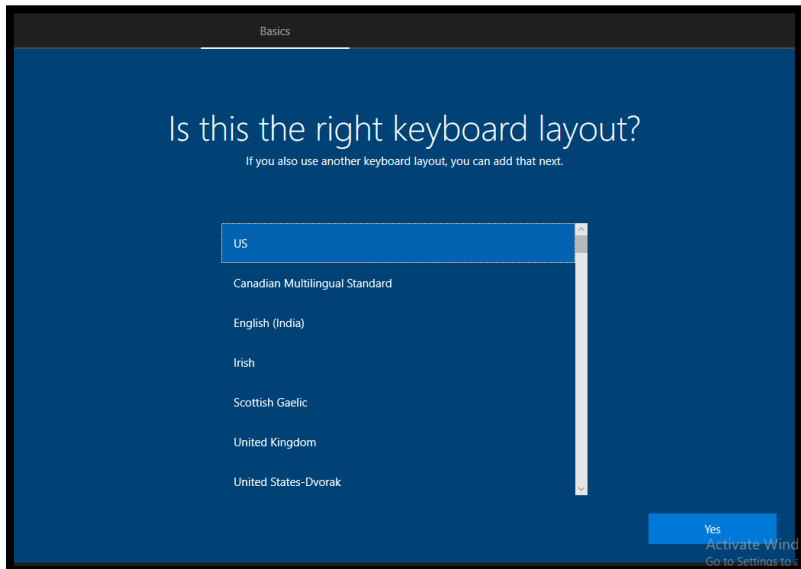
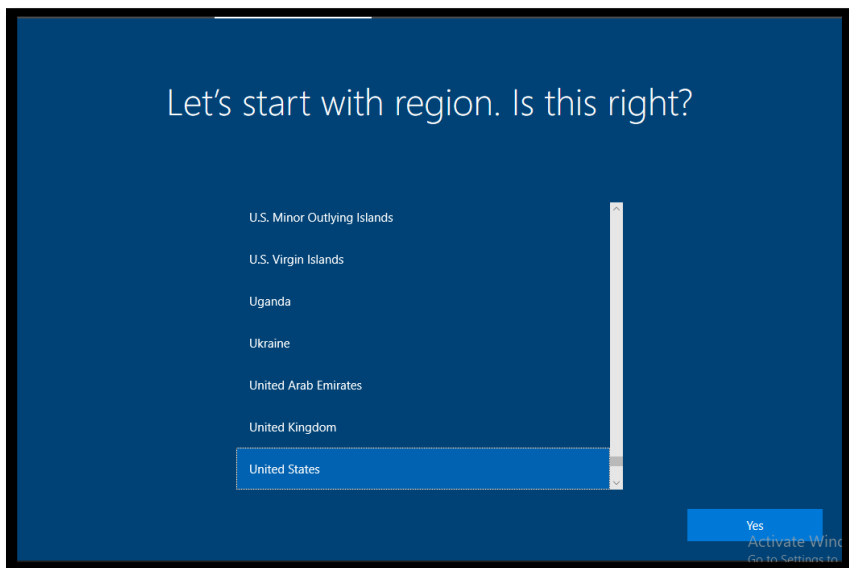
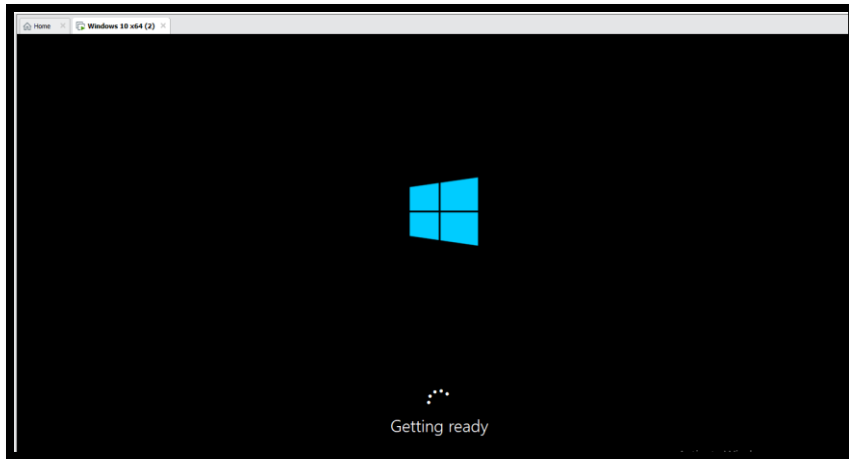
Click on install now

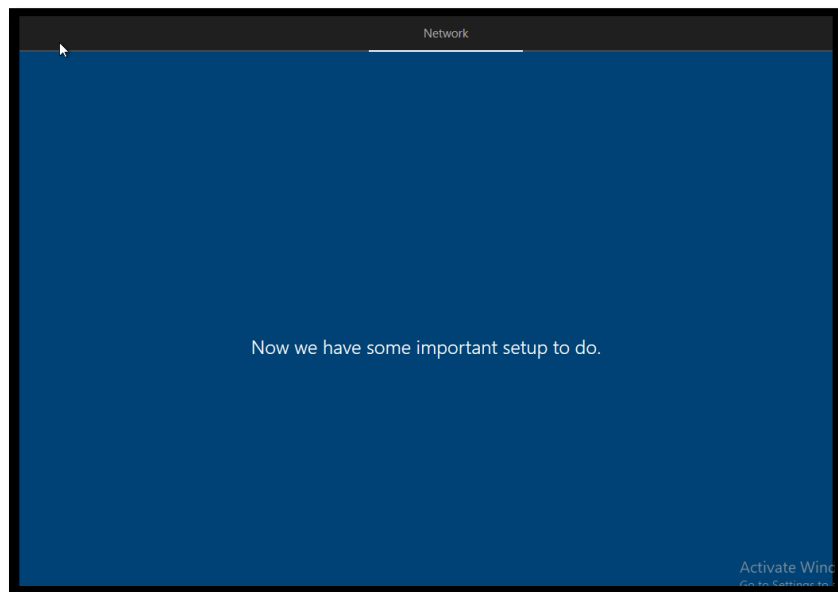
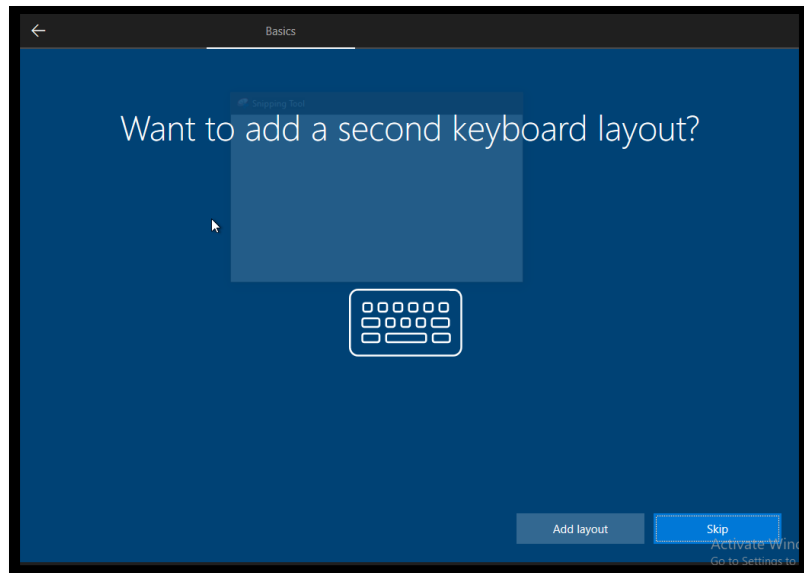


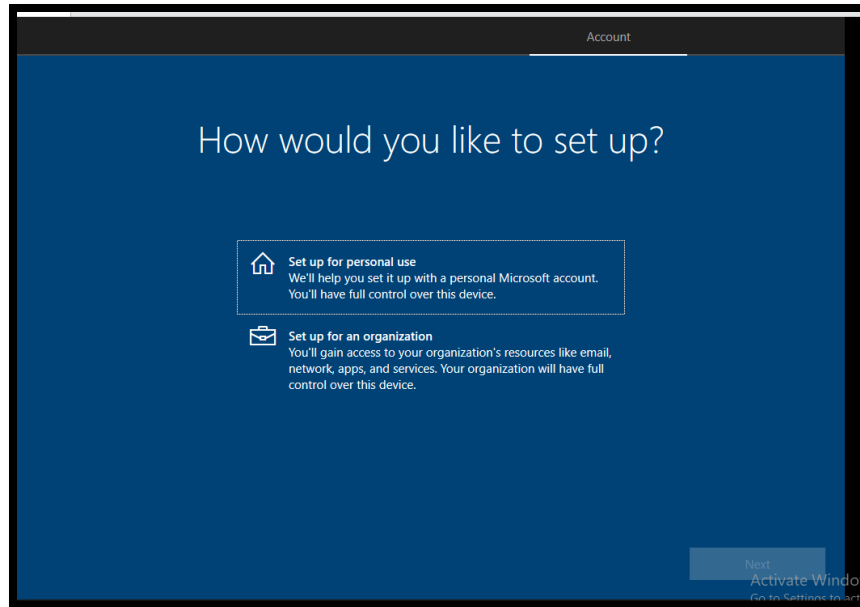




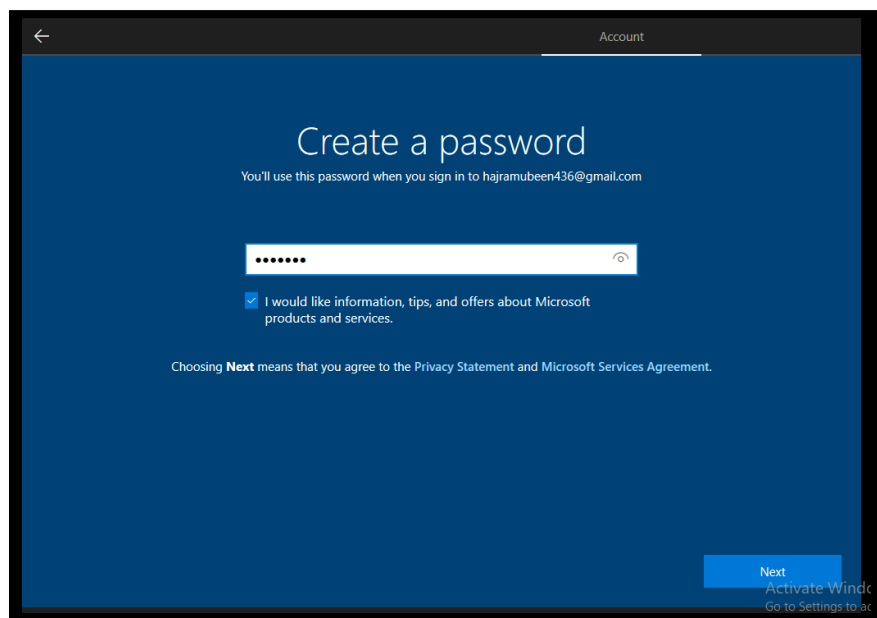


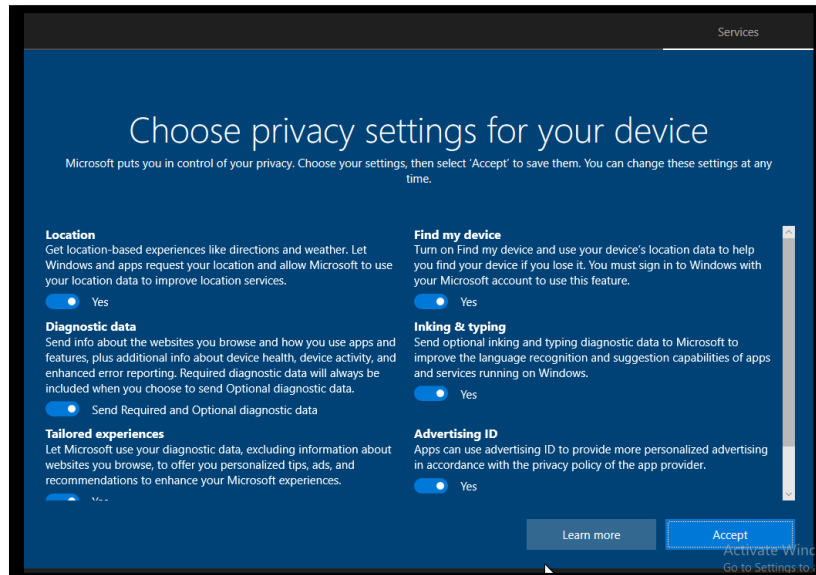
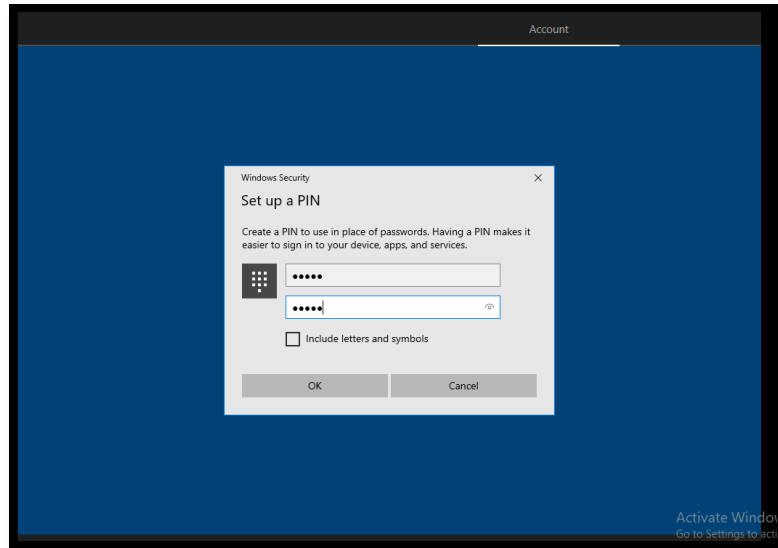


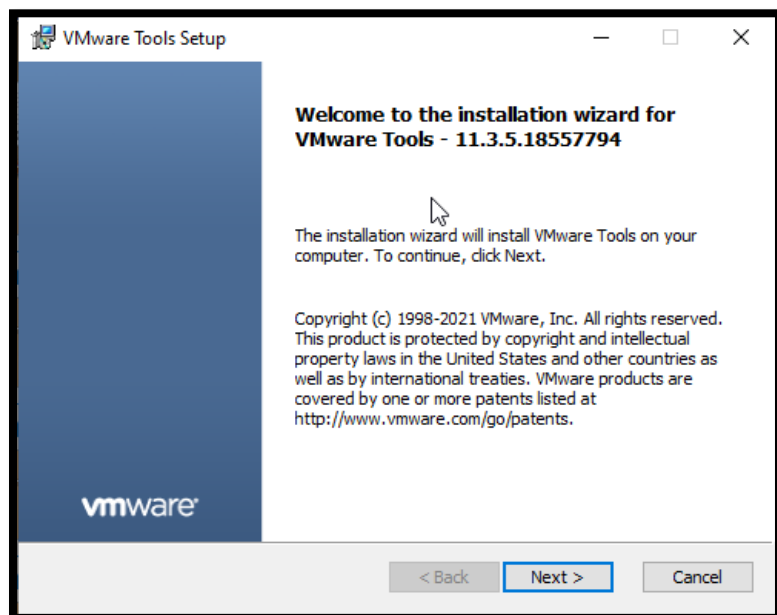
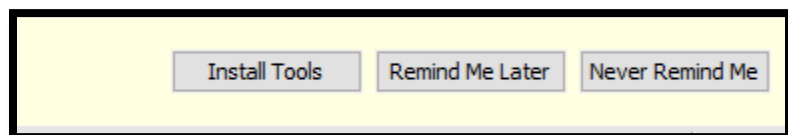
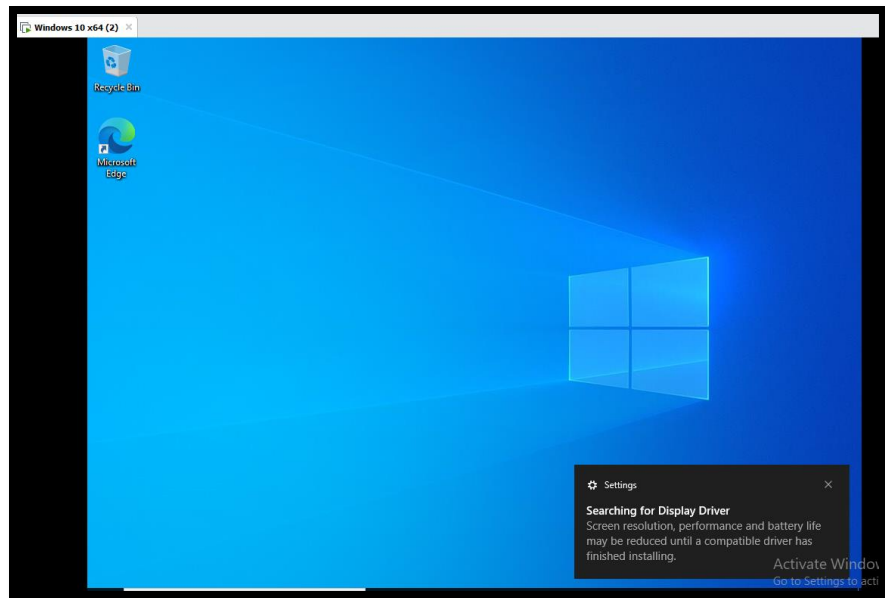


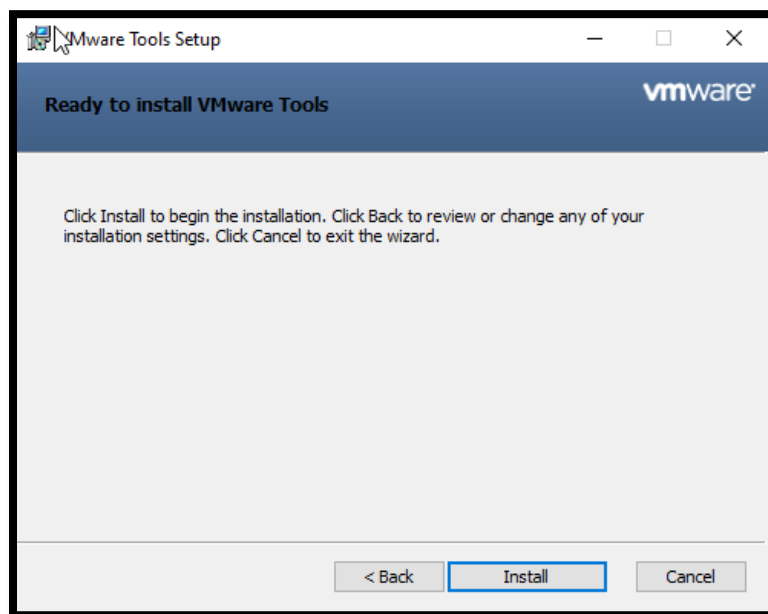
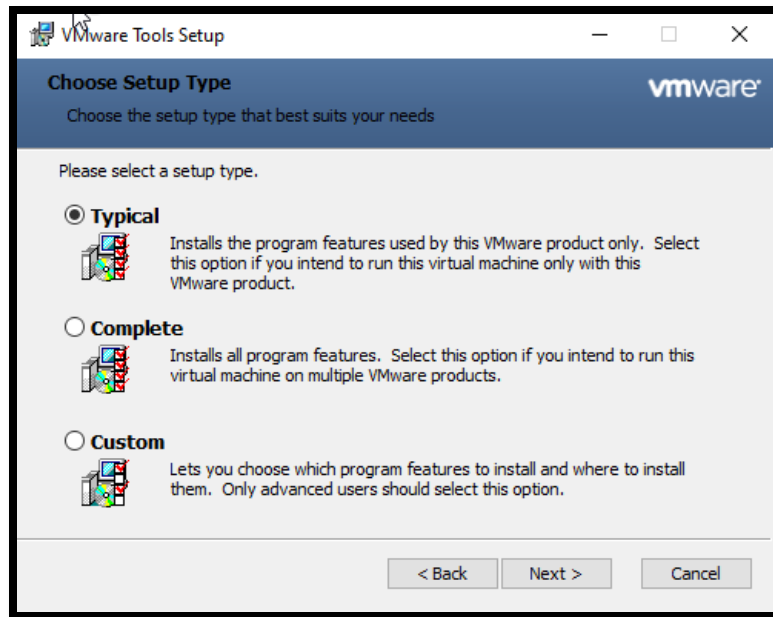


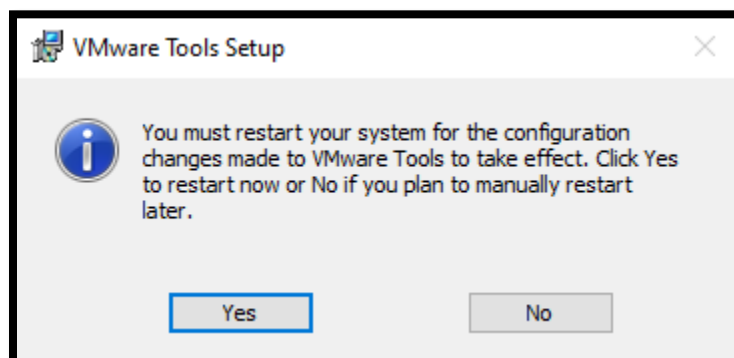
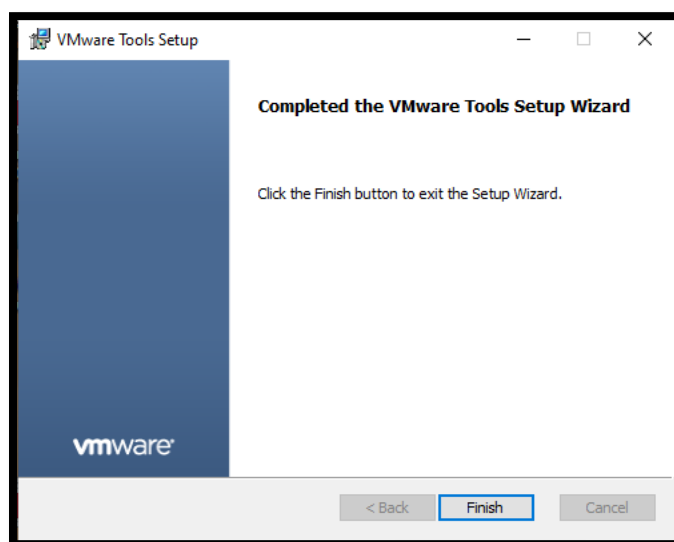
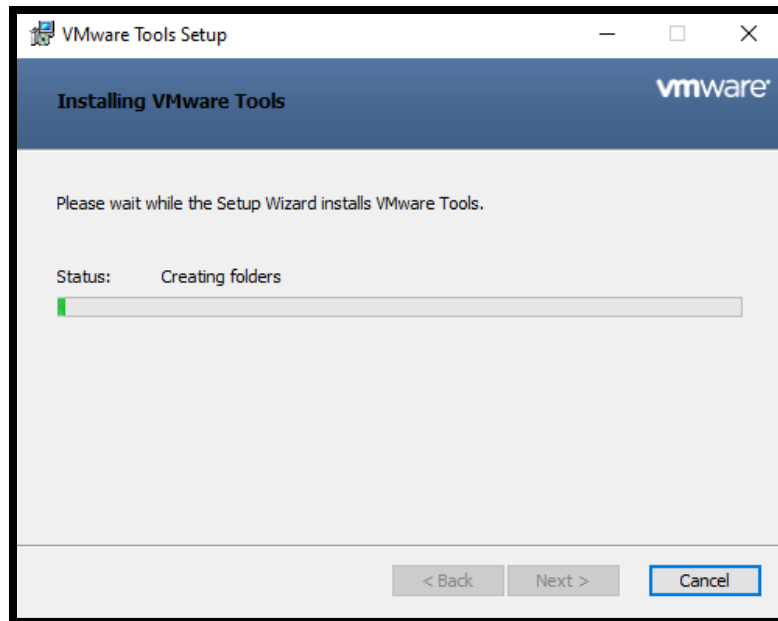
Set Password for the new window

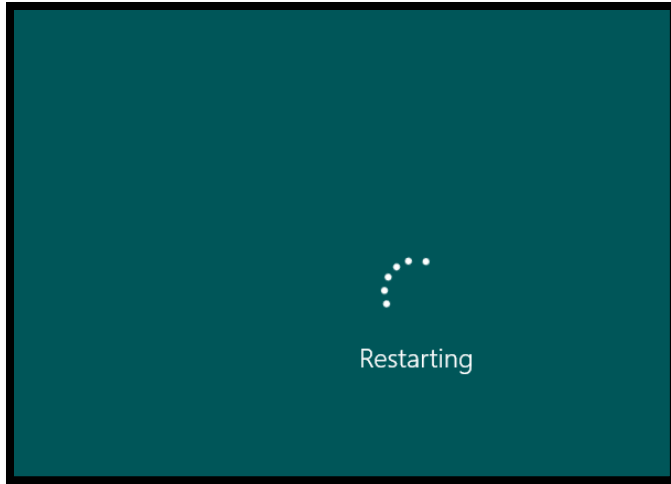




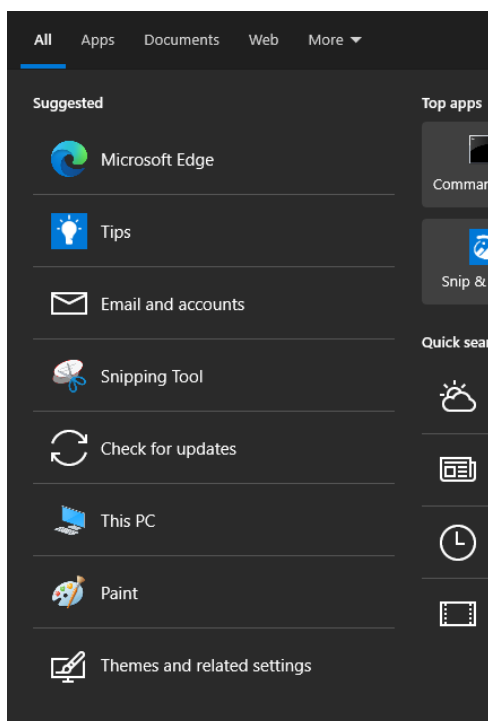


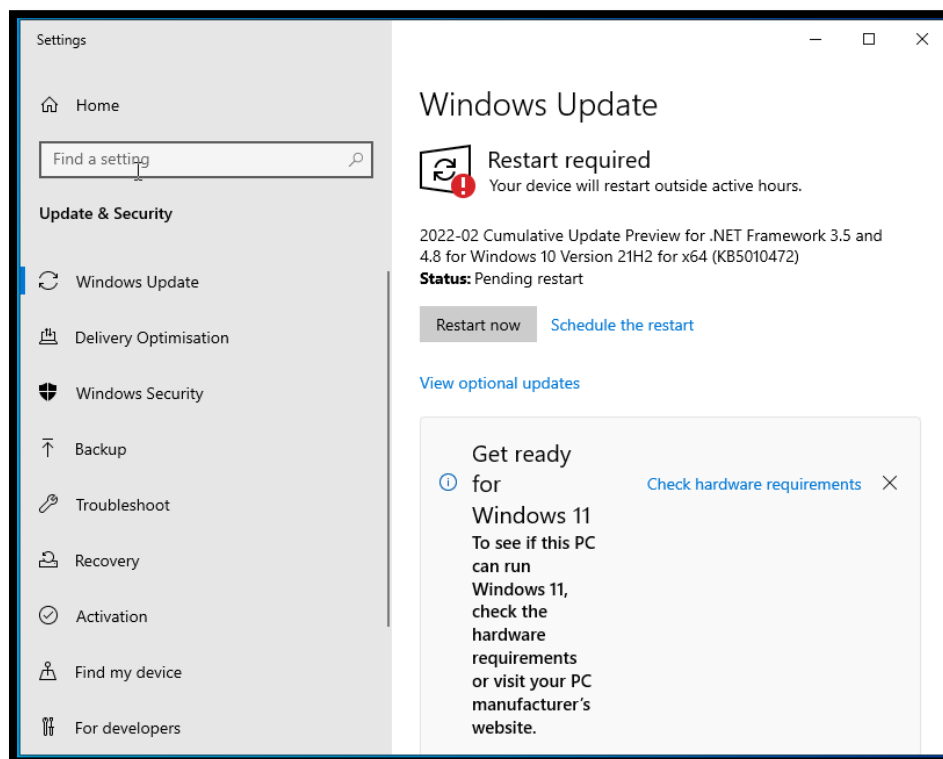
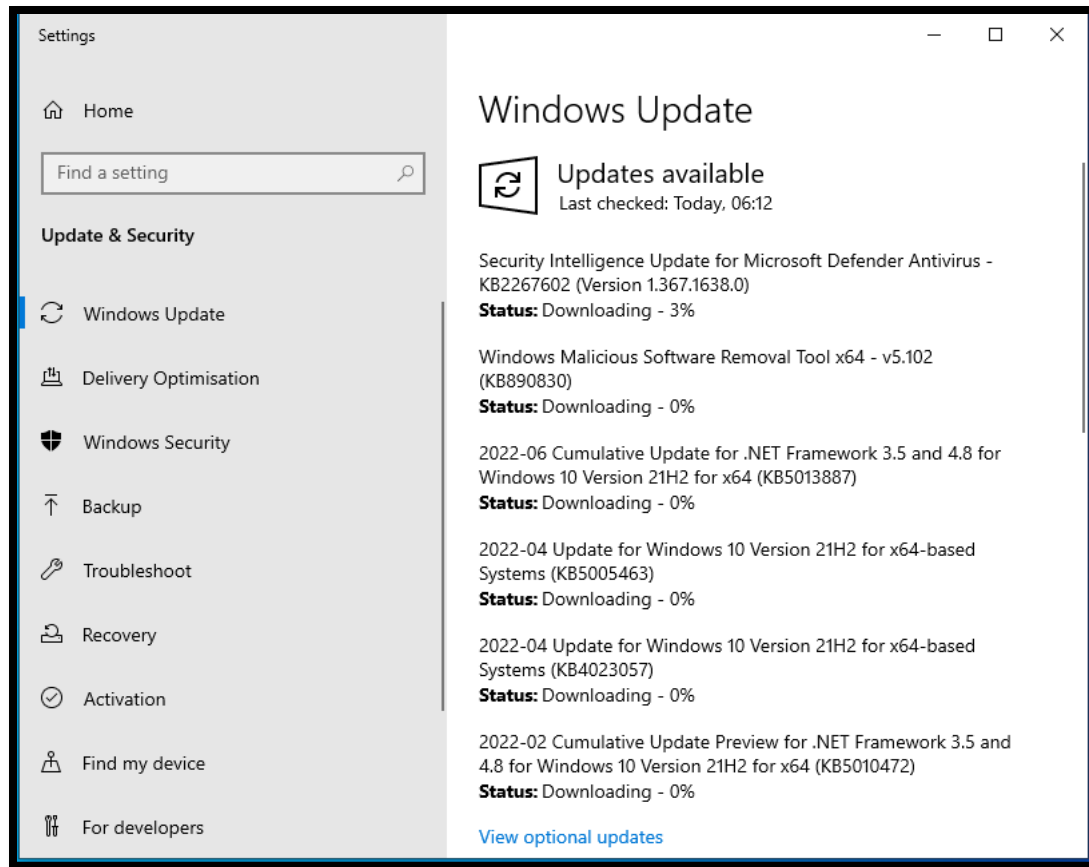


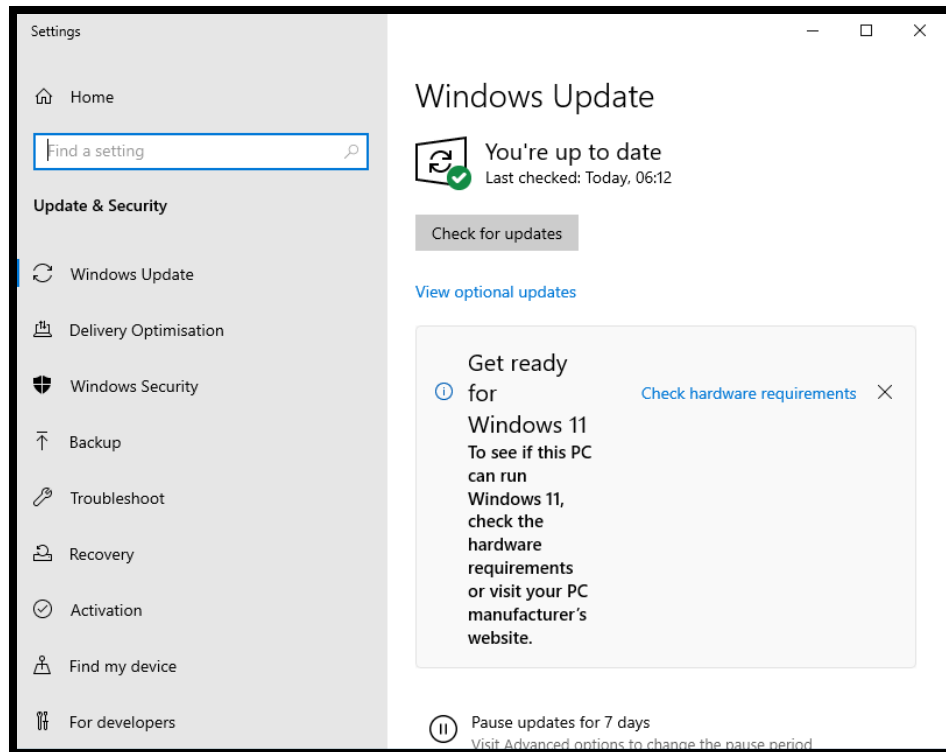




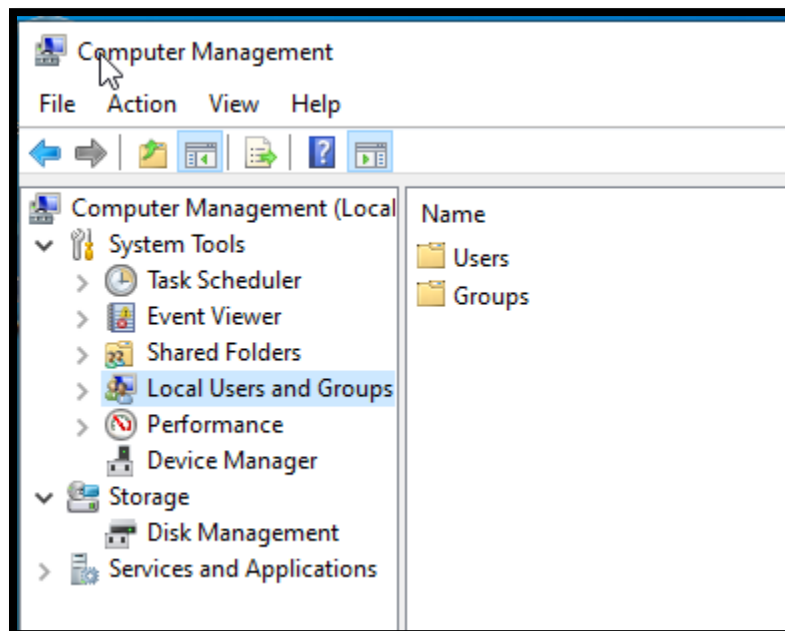
Create Backup Administrative Account

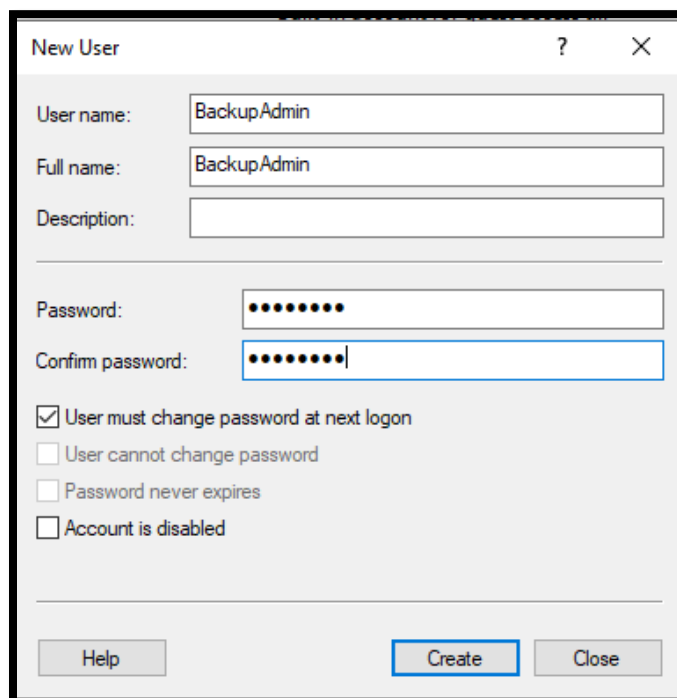
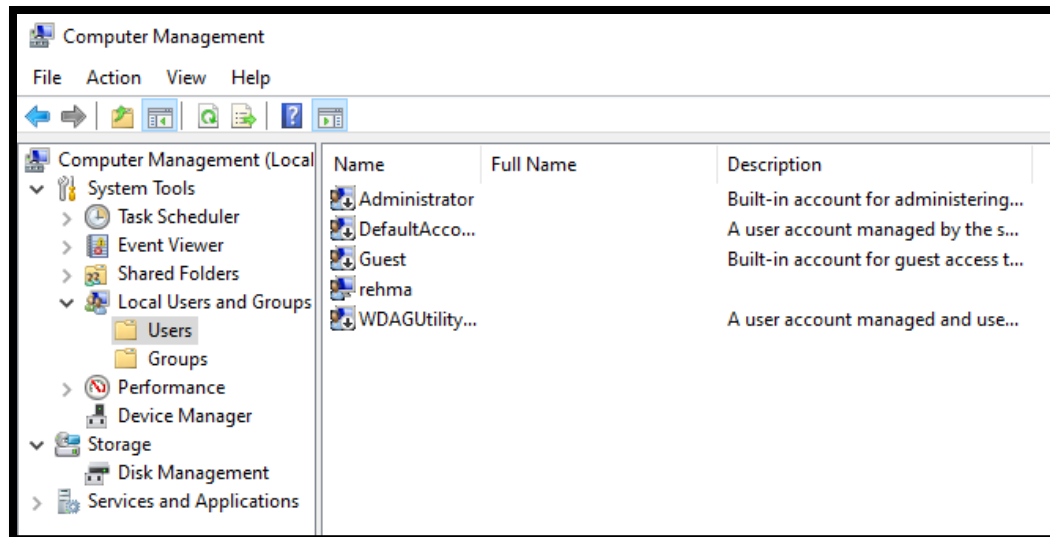


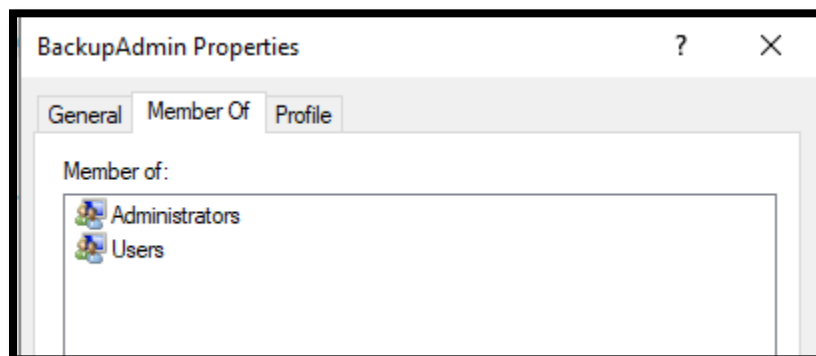
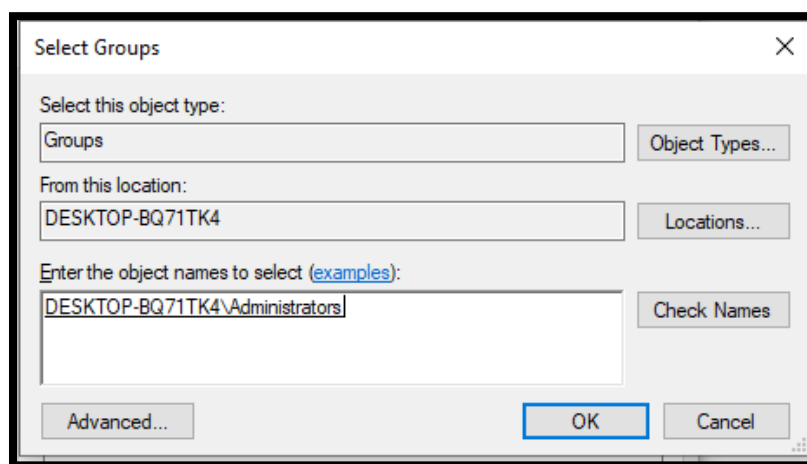
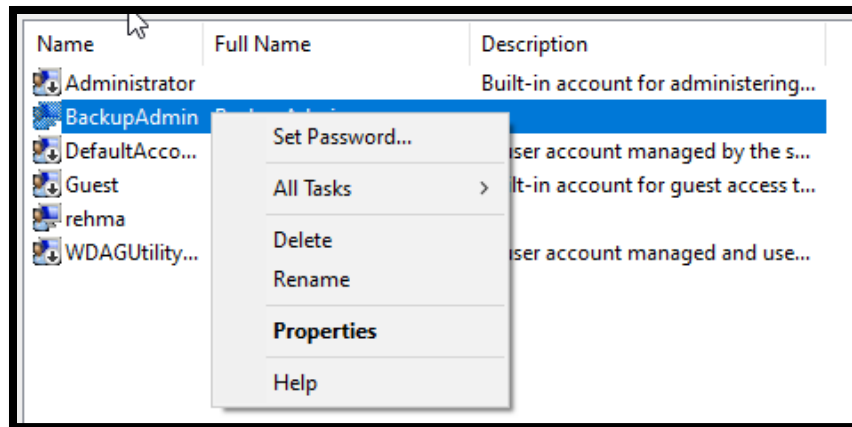


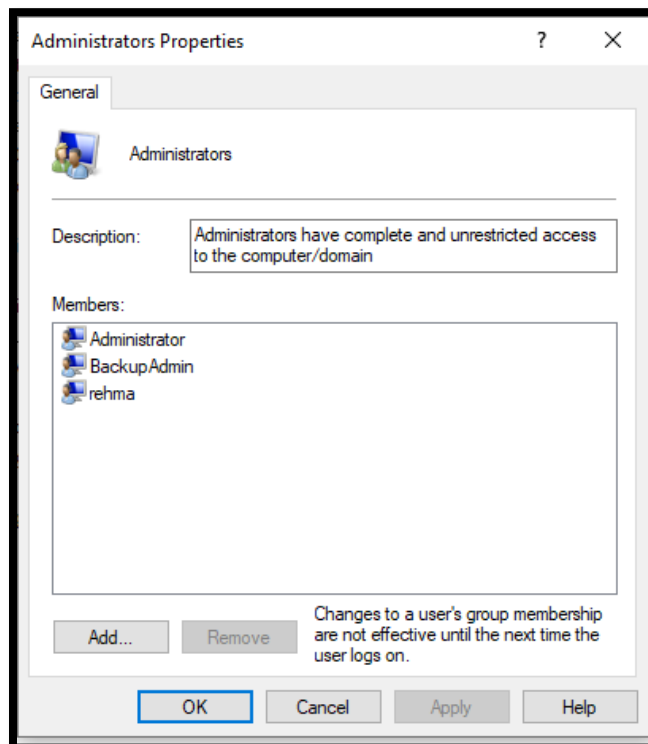
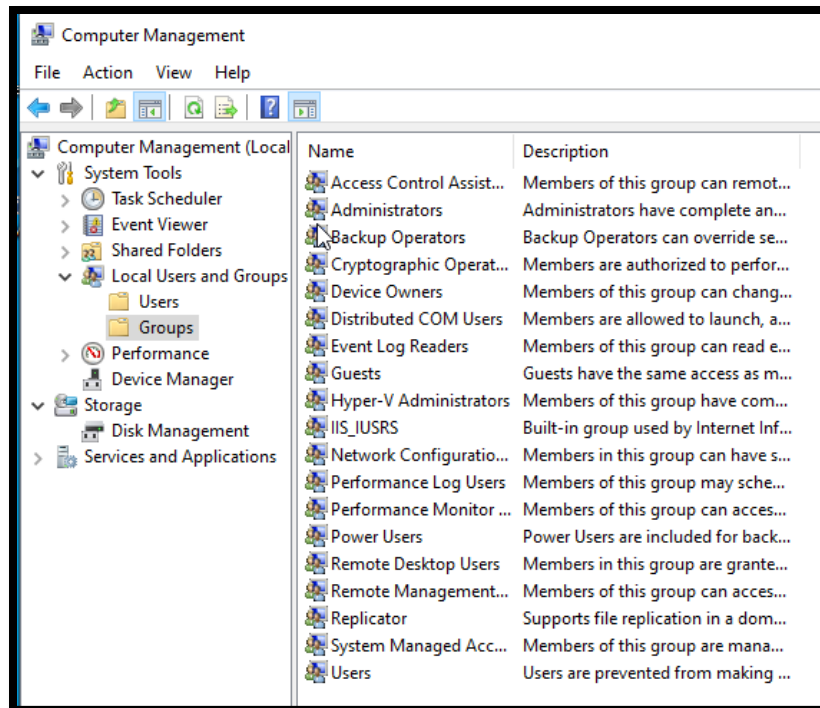


Create backup administrative account:

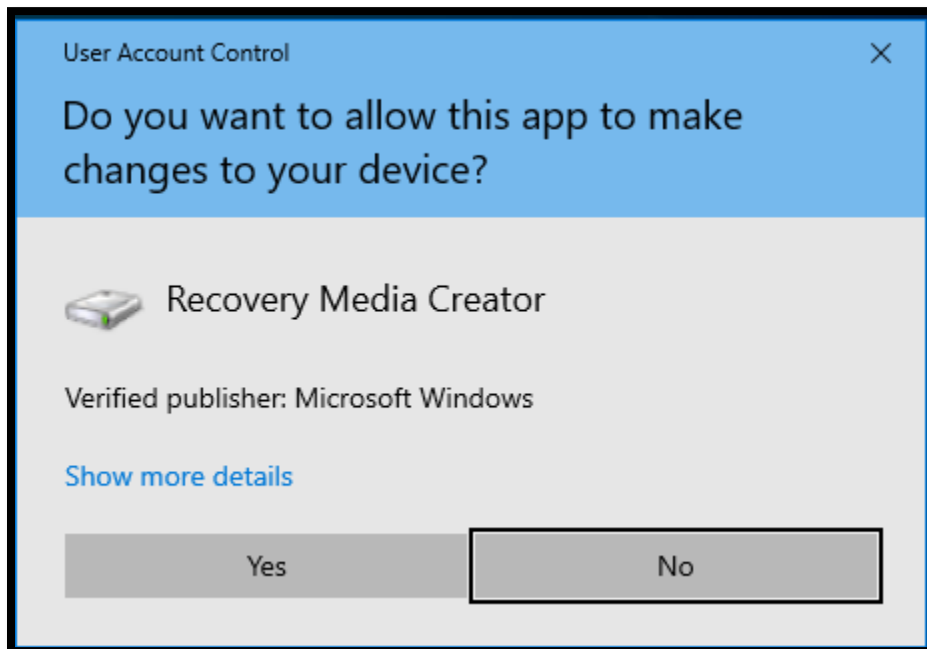
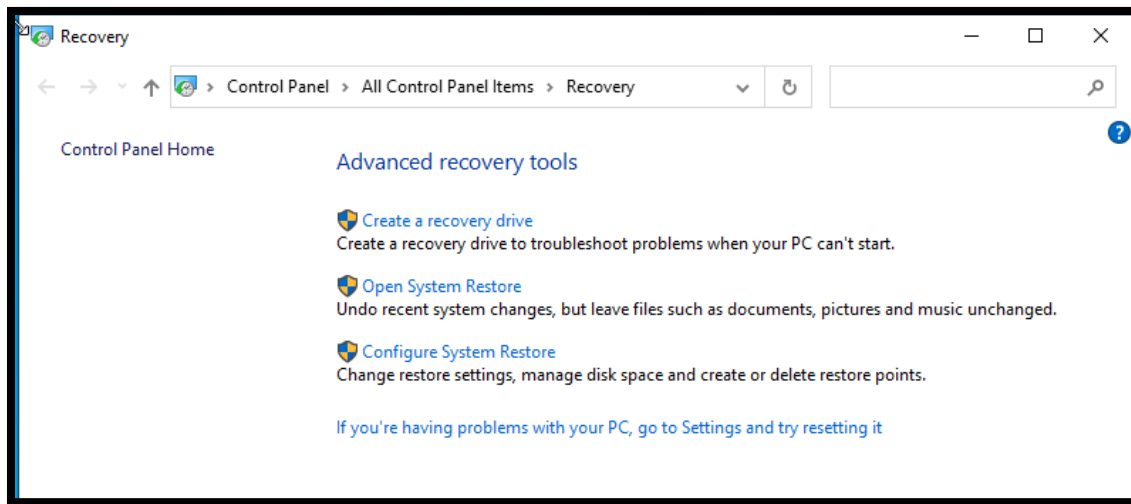


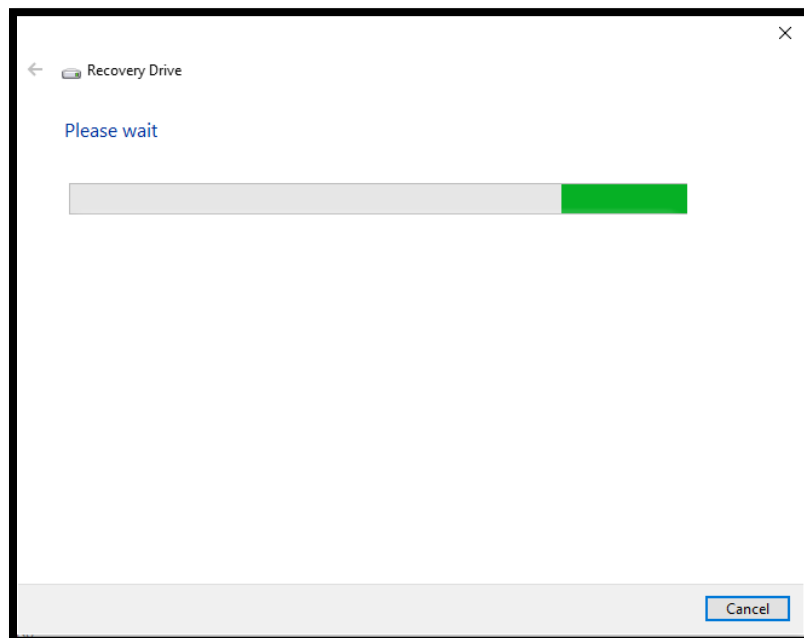
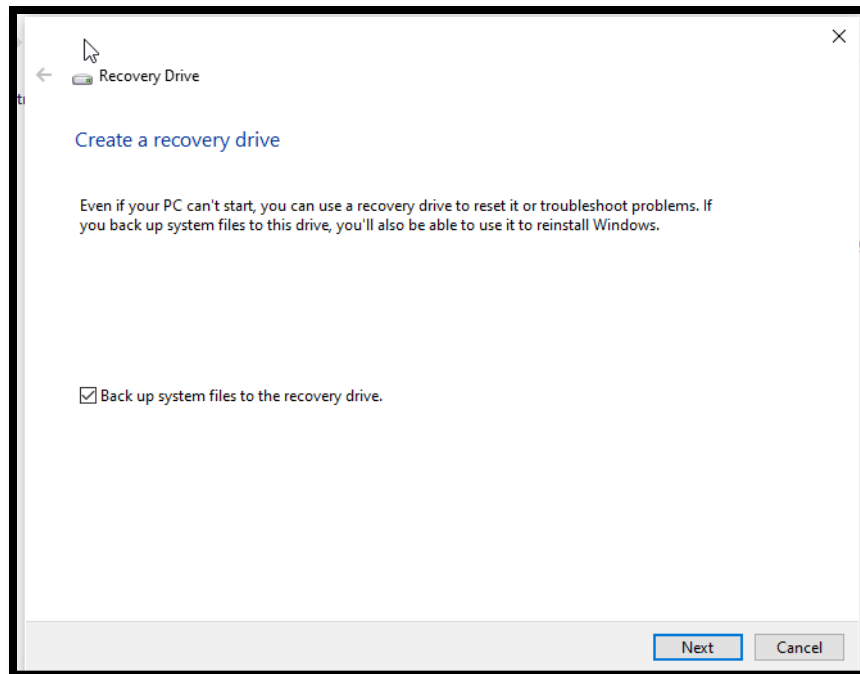




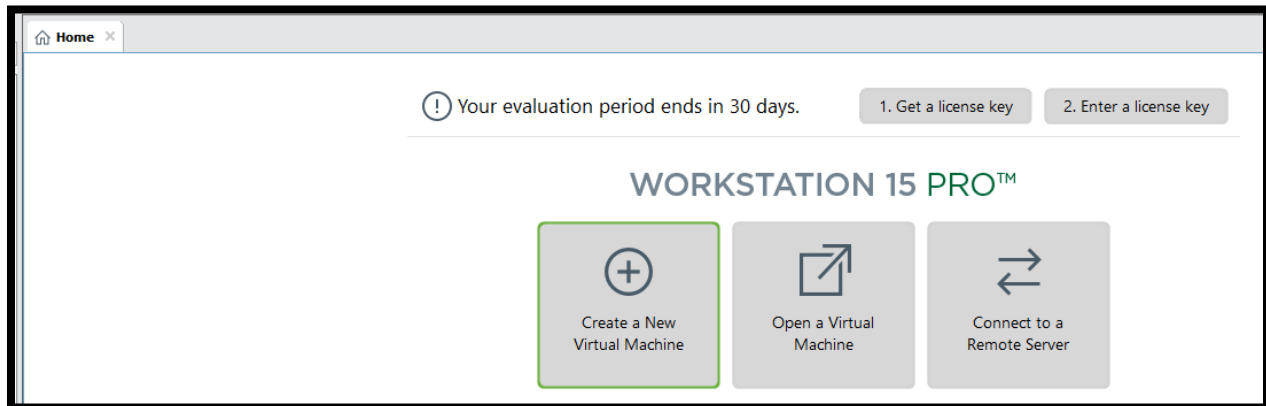


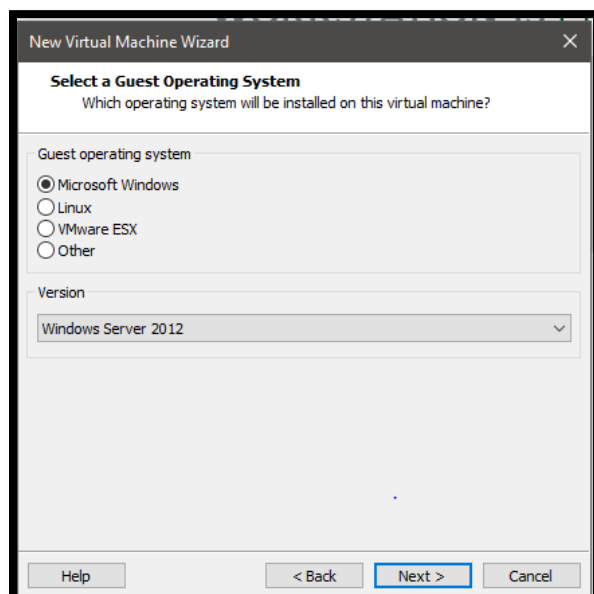
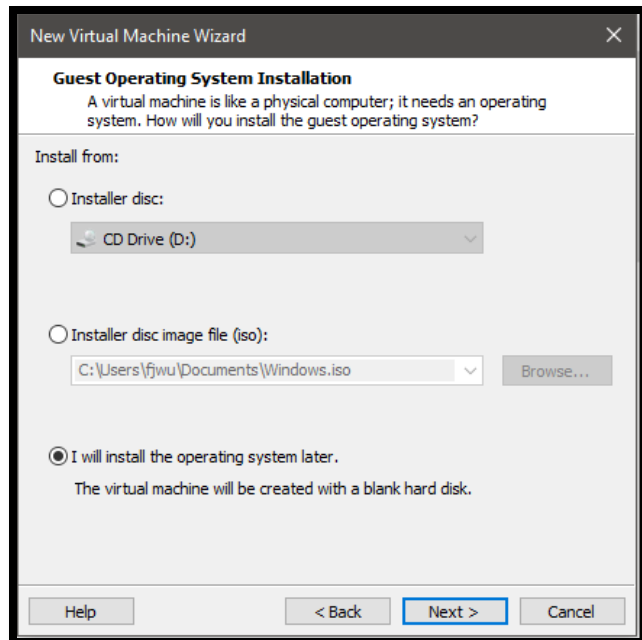
Create window Recover USB Drive





2. Installation of window server





New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

Location:

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

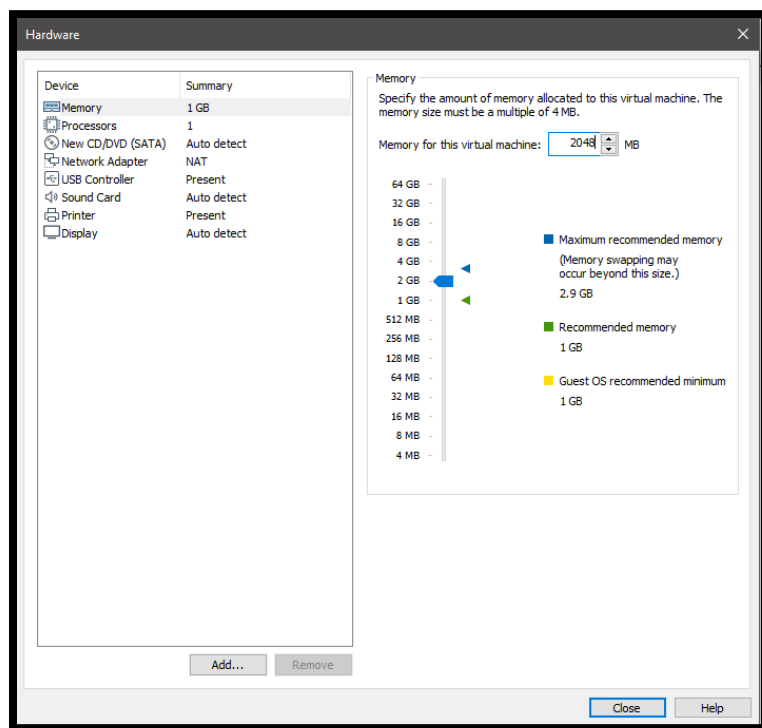
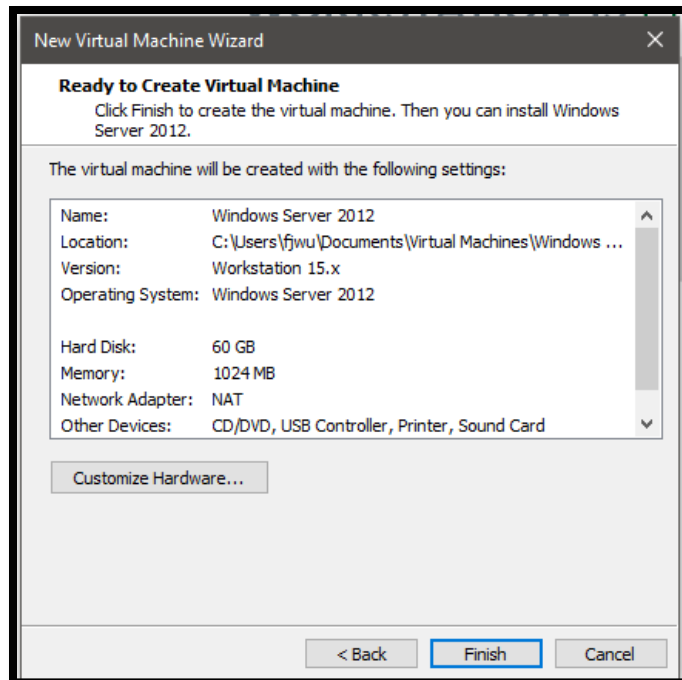
Maximum disk size (GB):

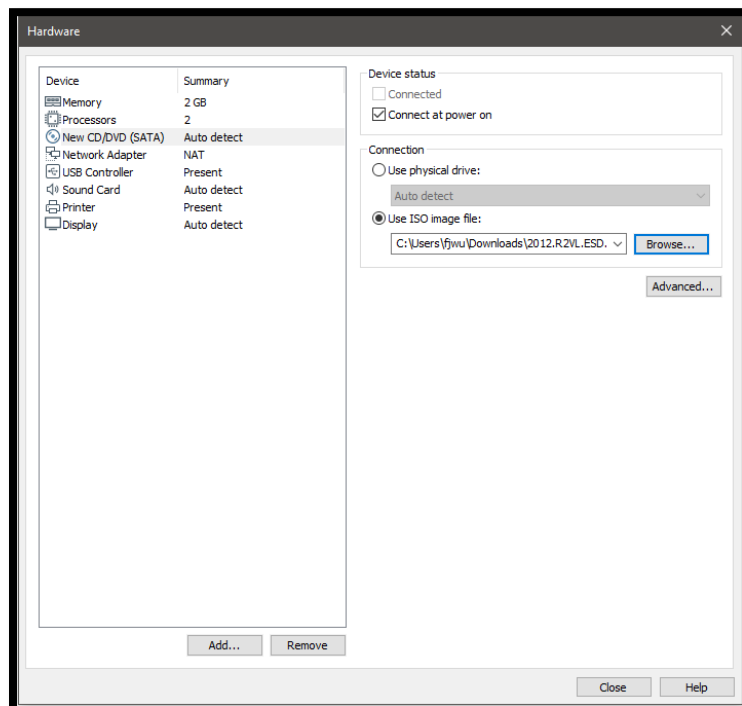
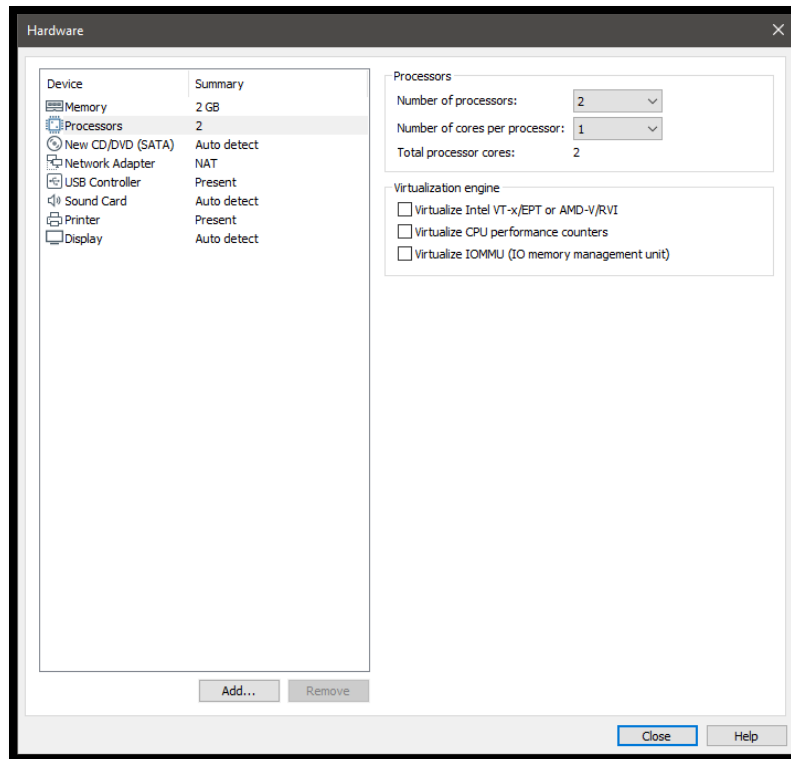
Recommended size for Windows Server 2012: 60 GB

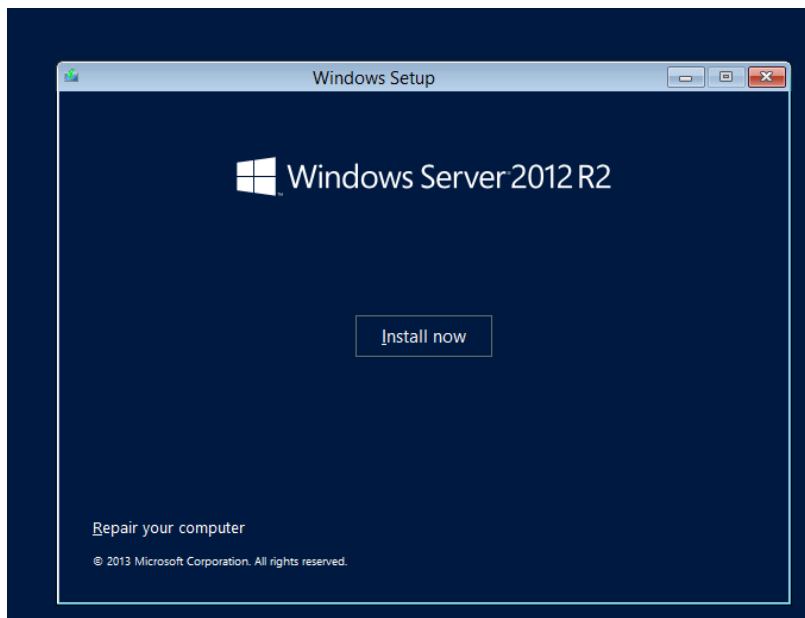
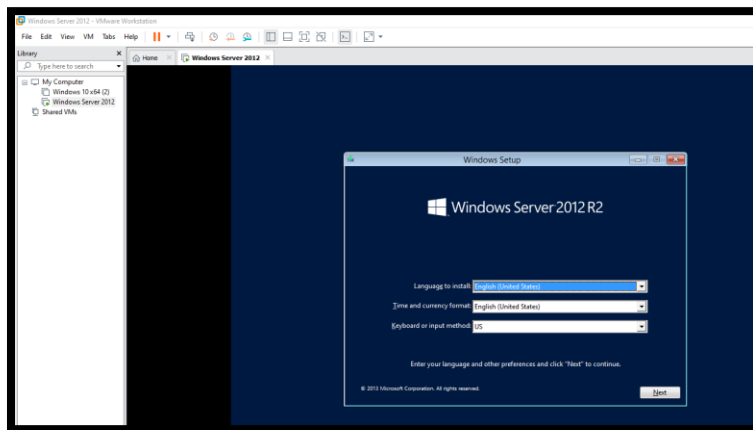
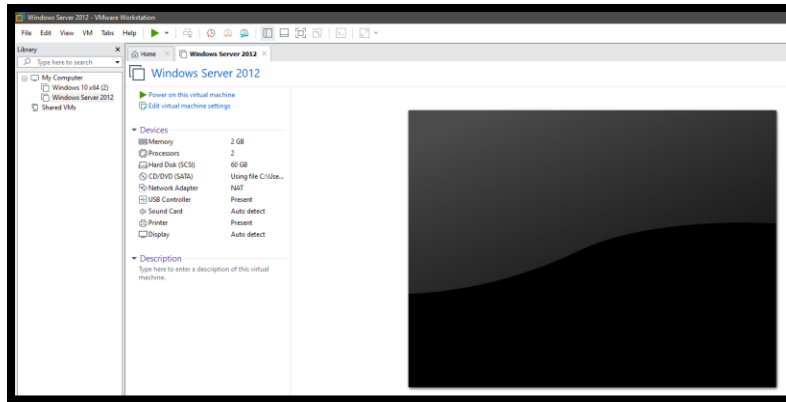
☒ Store virtual disk as a single file
☐ Split virtual disk into multiple files

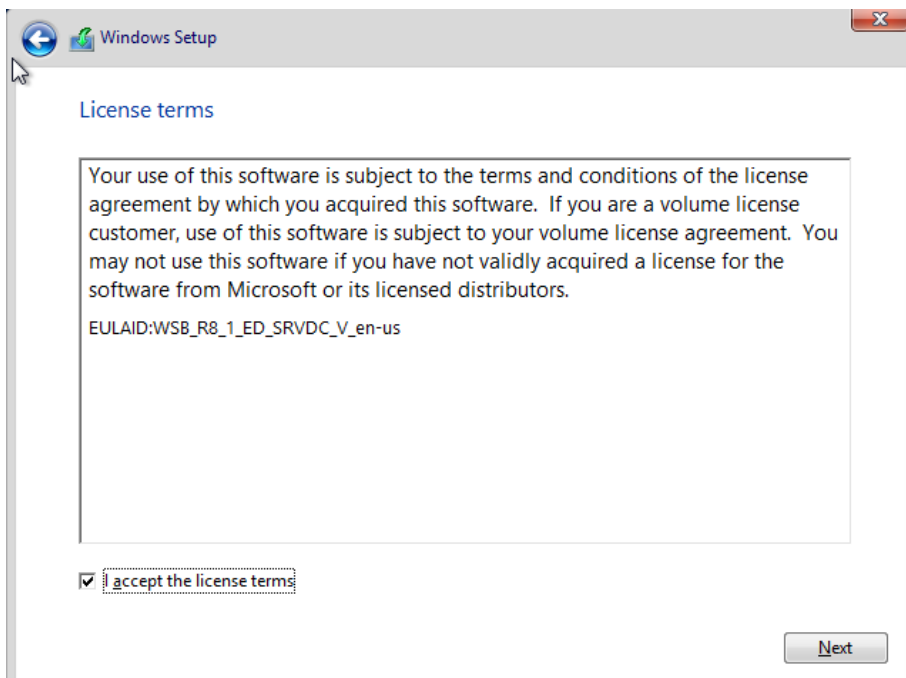
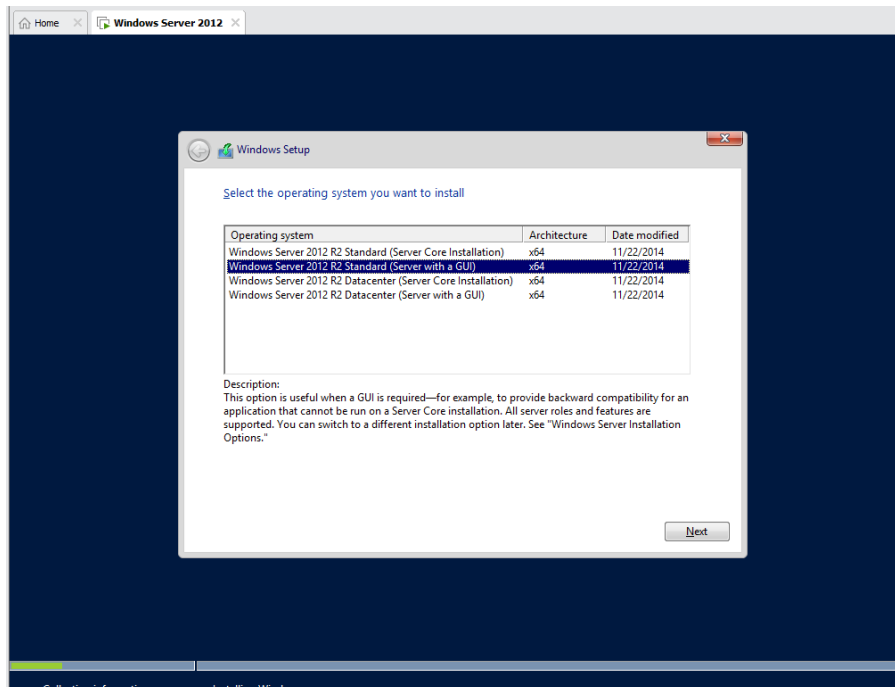
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

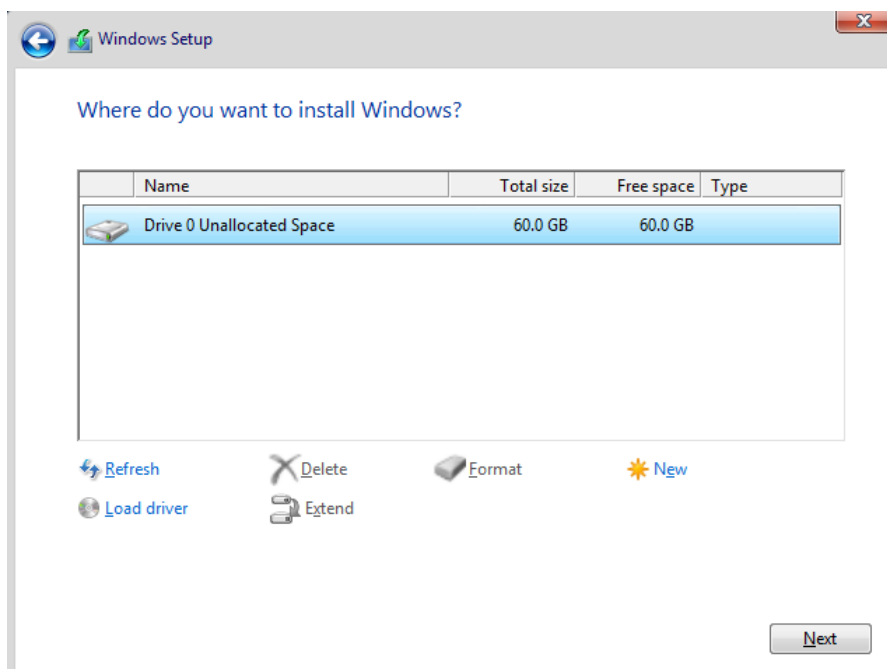
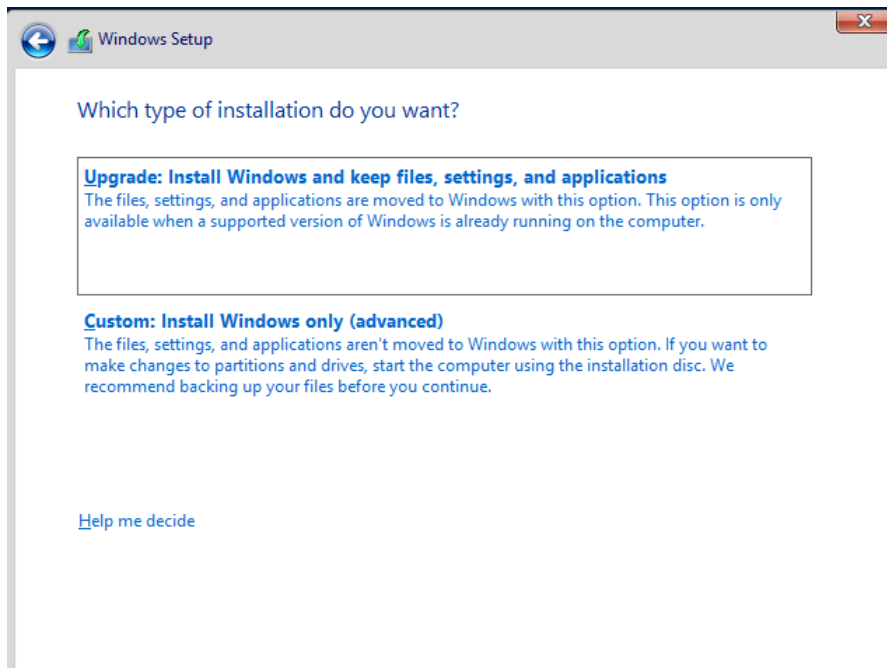
Help < Back Next > Cancel

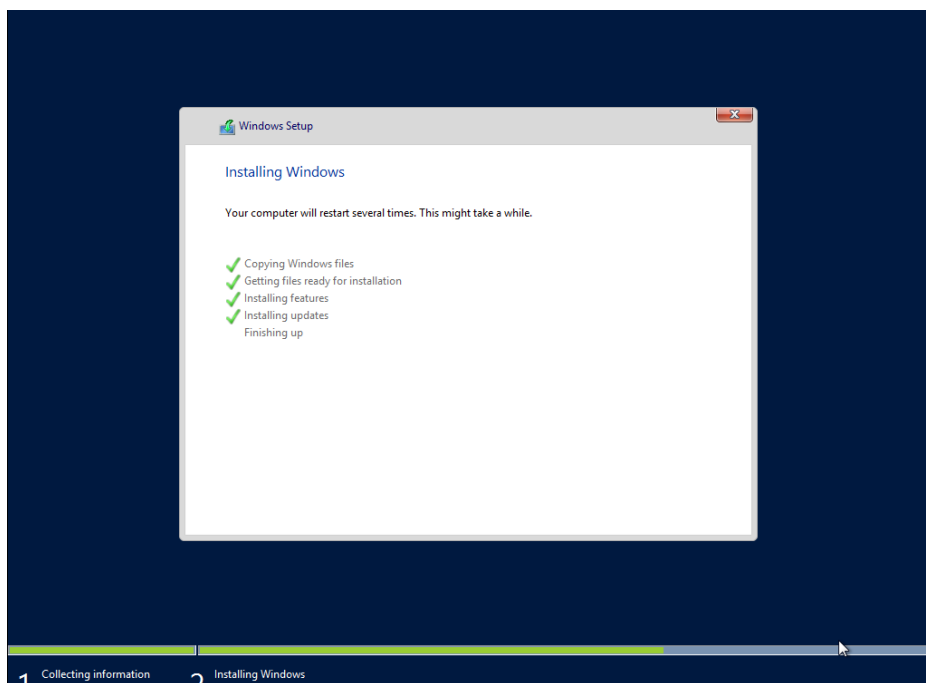
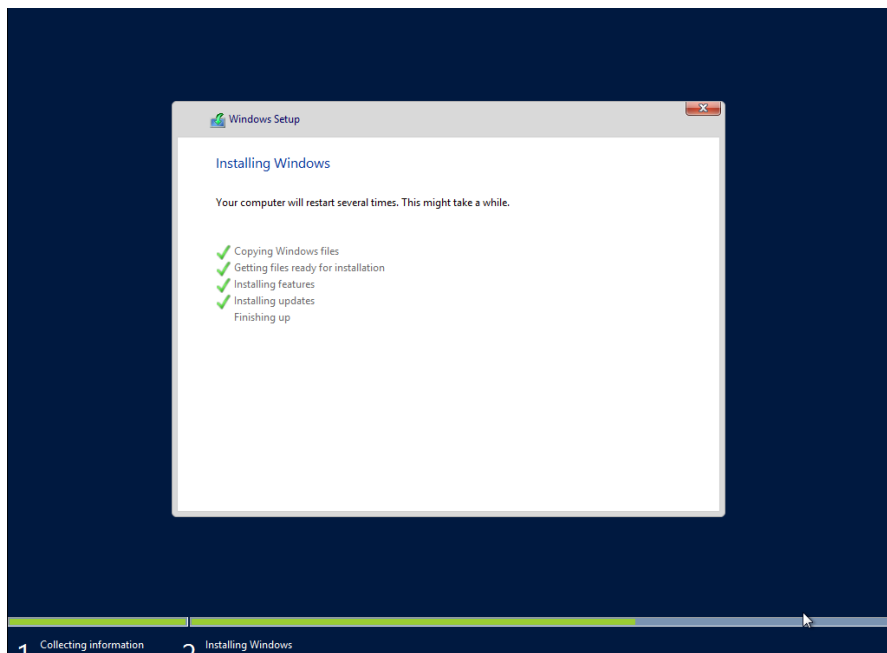


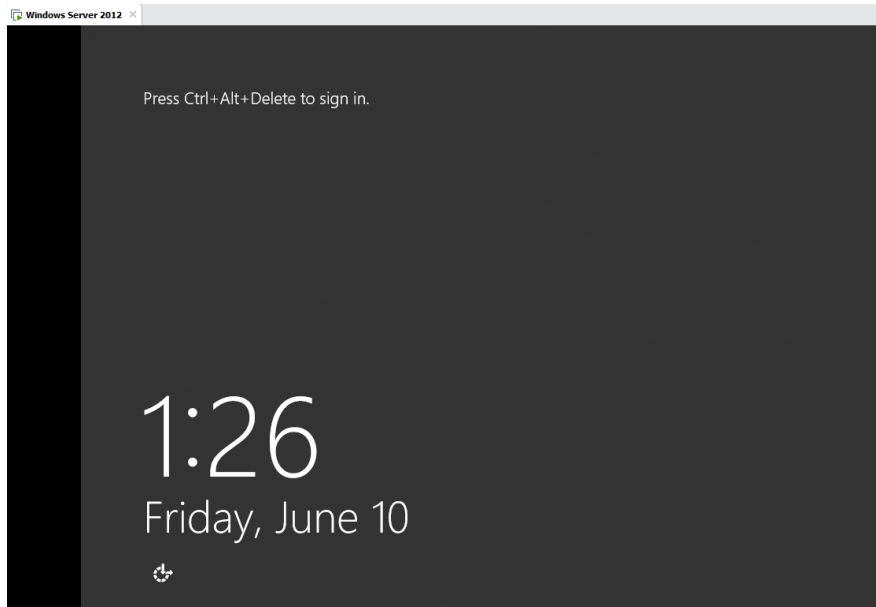




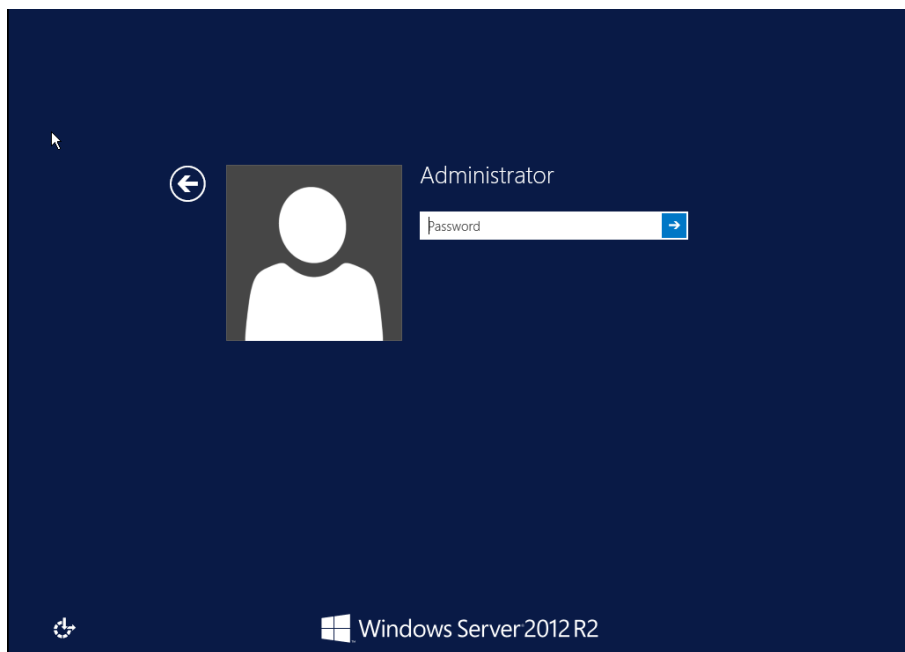


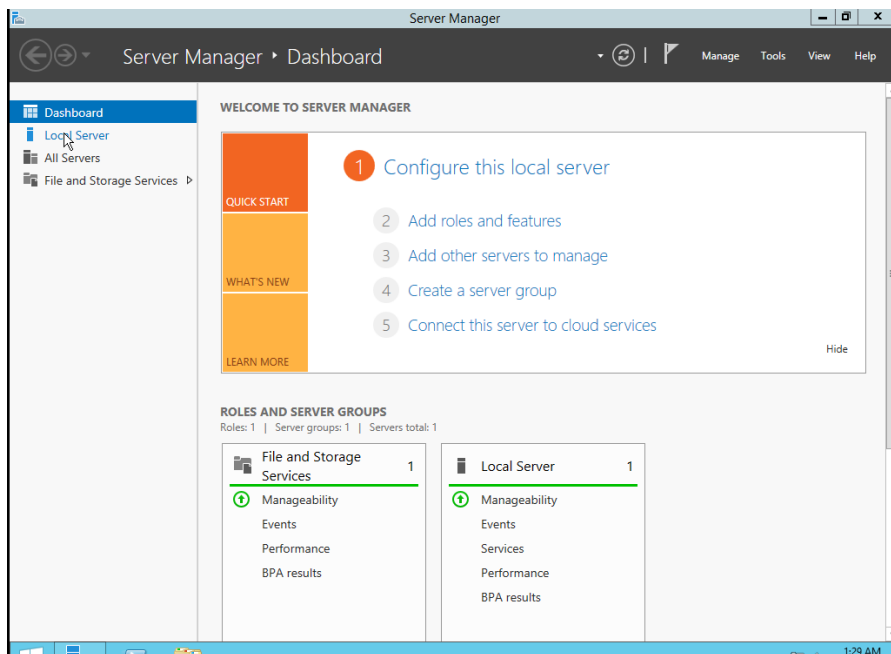
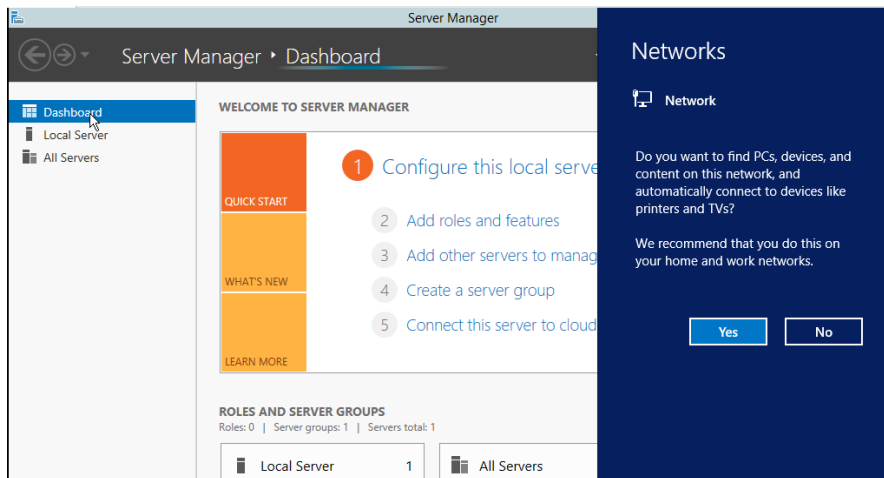


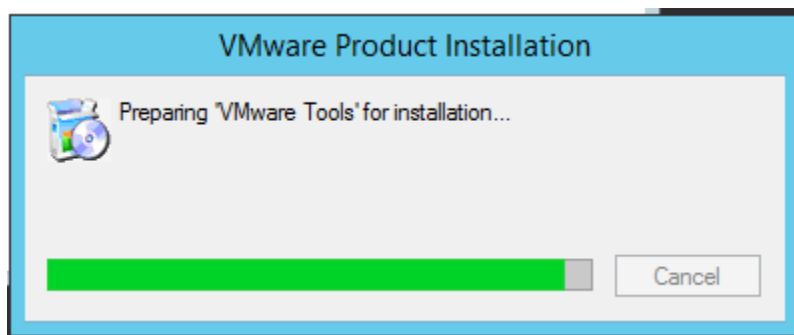
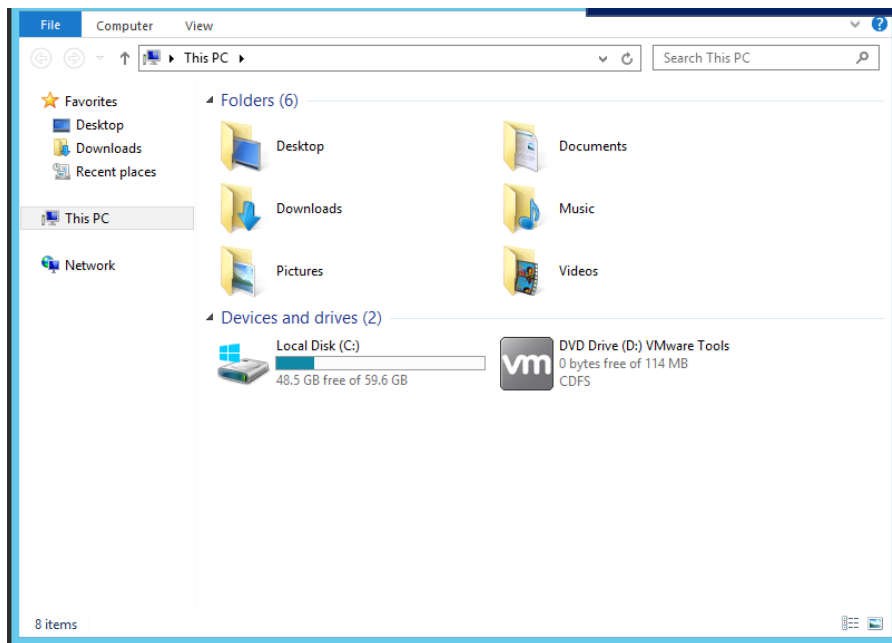


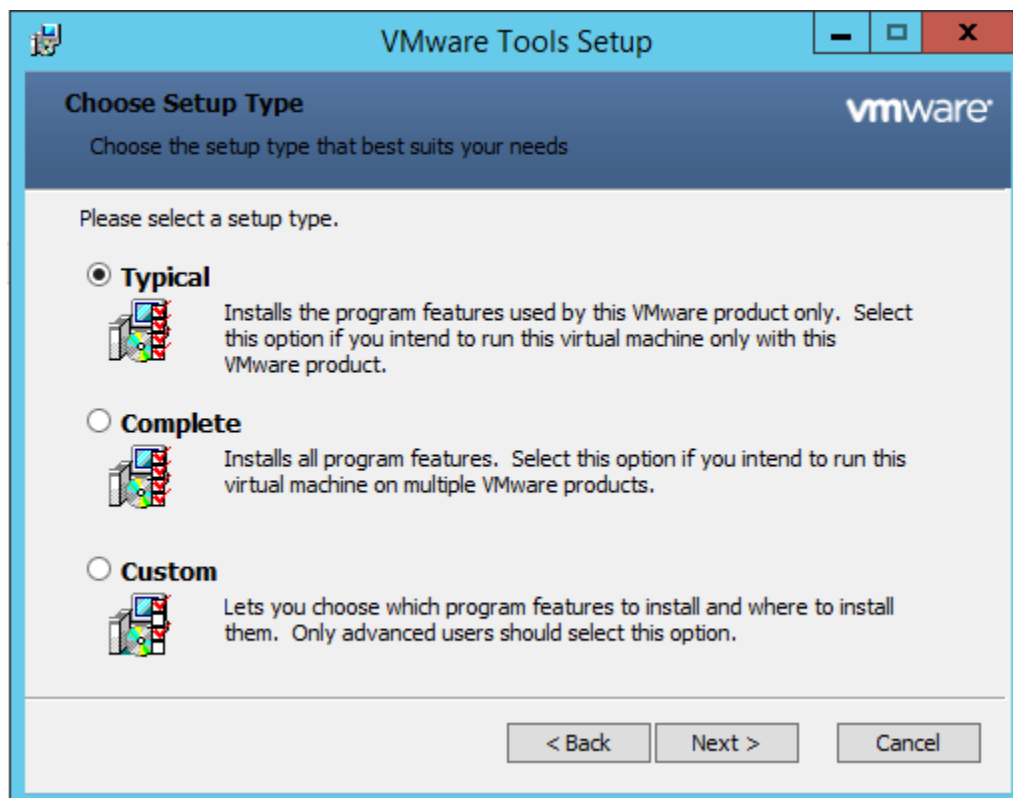
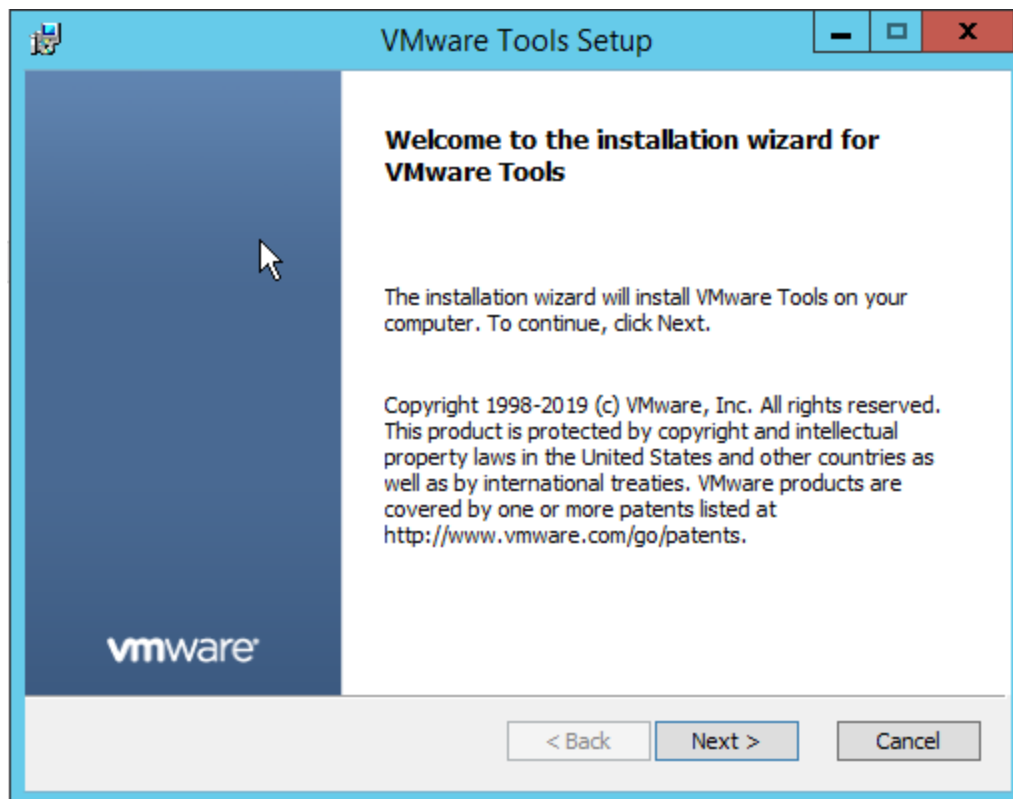


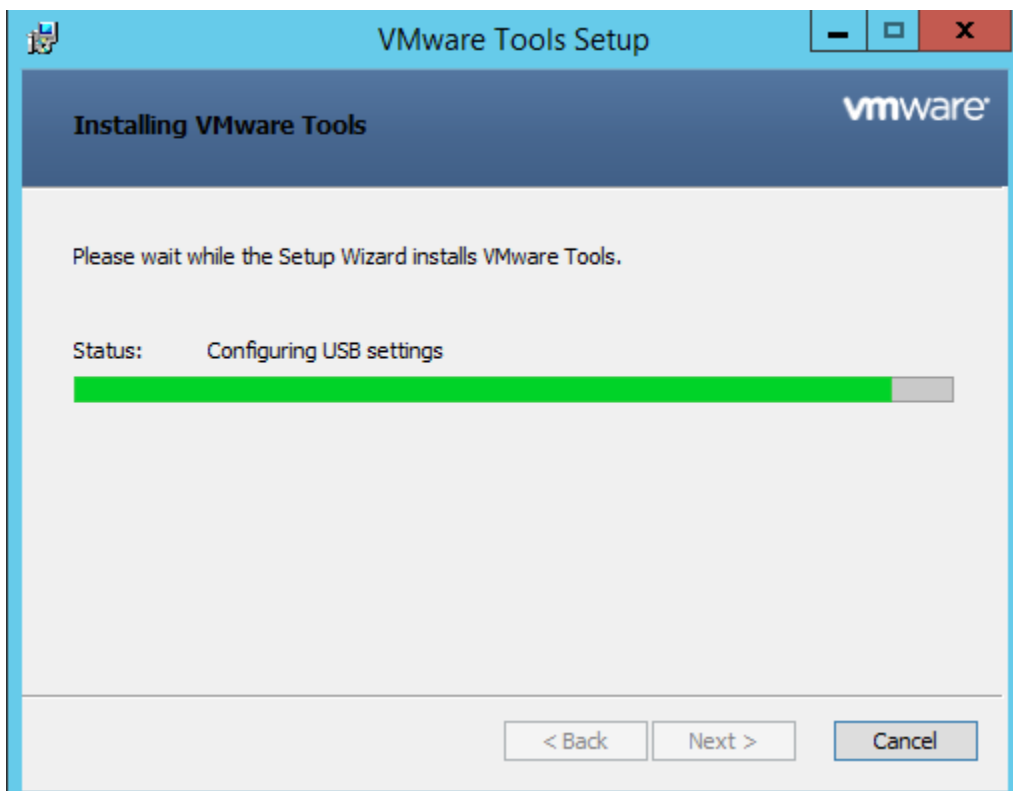
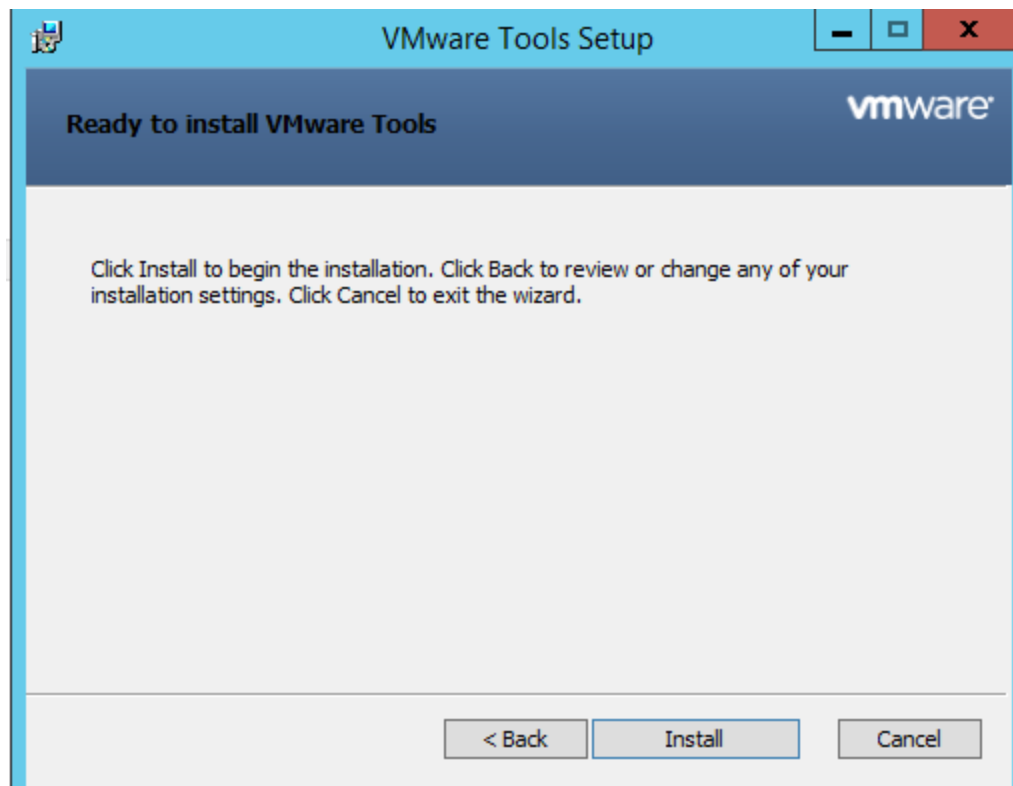
Enter the password that u have set before

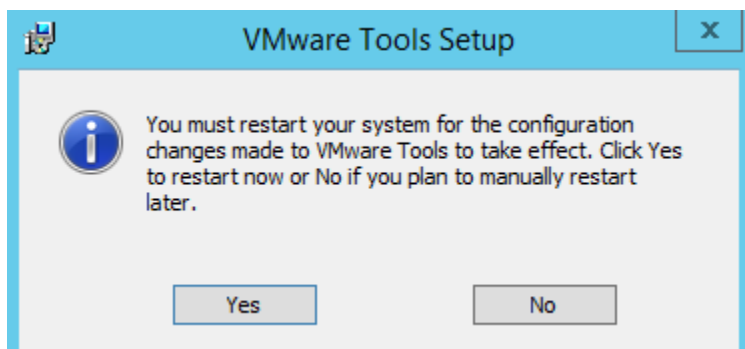
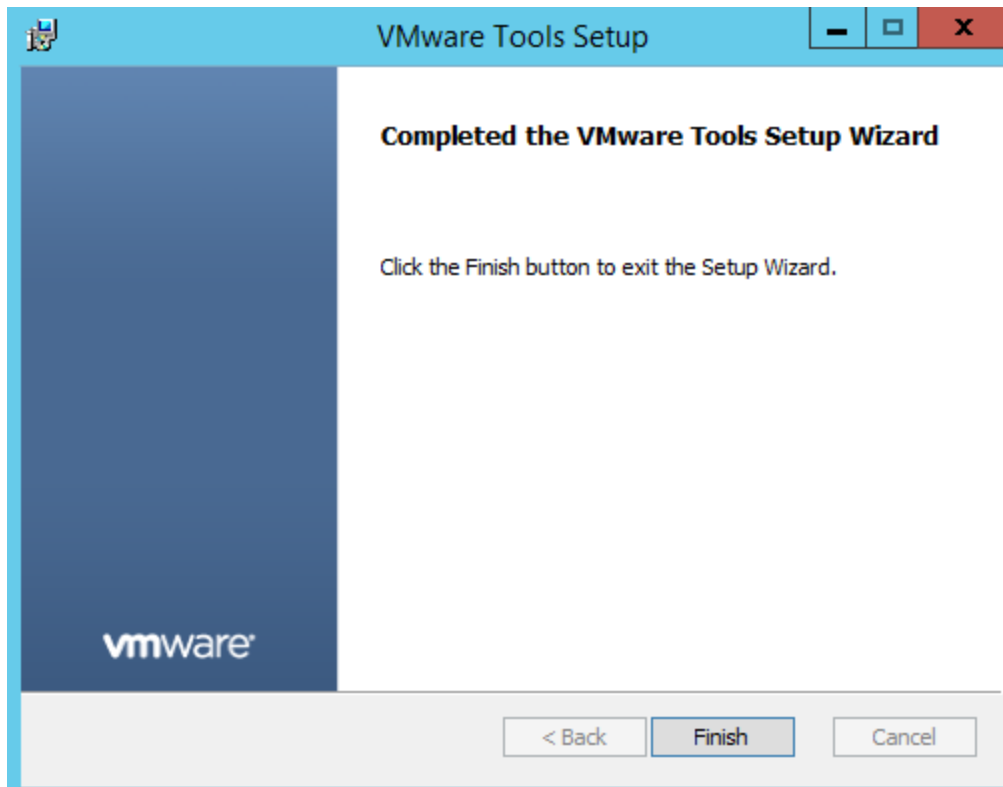












⋮ Stopping services

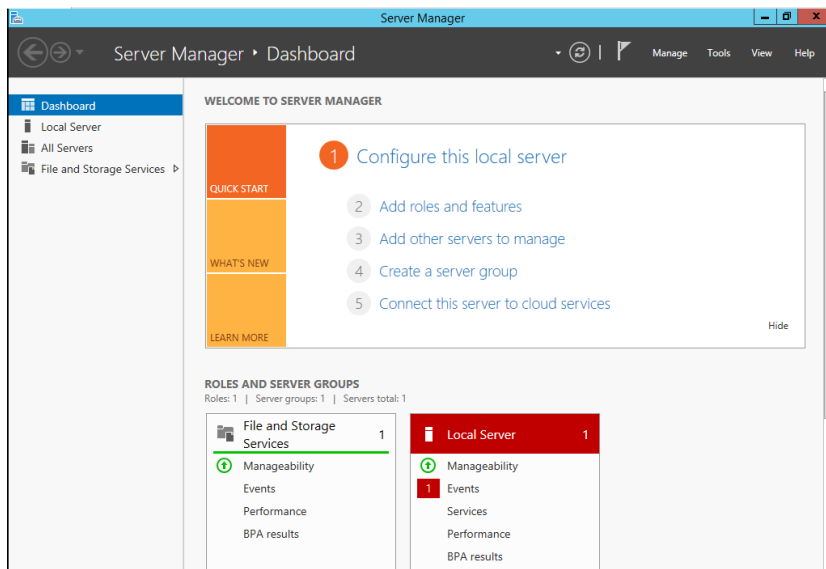
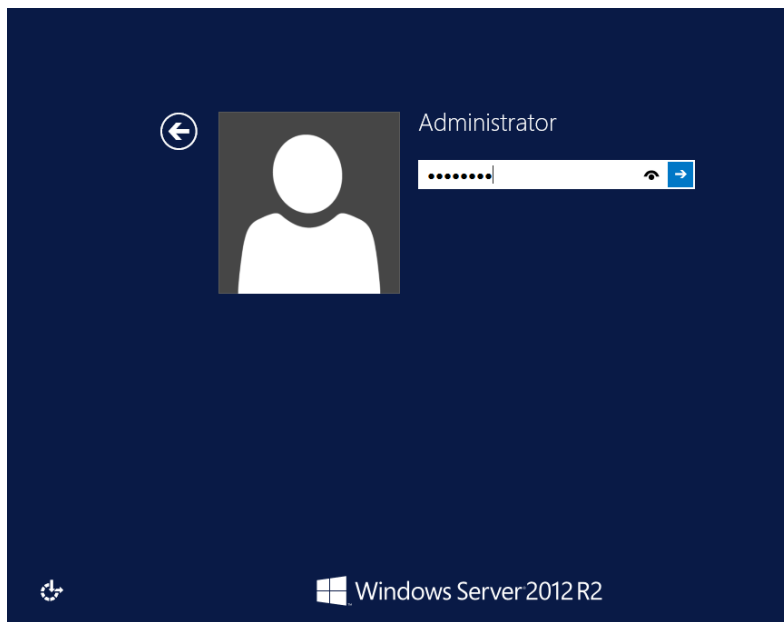
 Windows Server 2012 R2

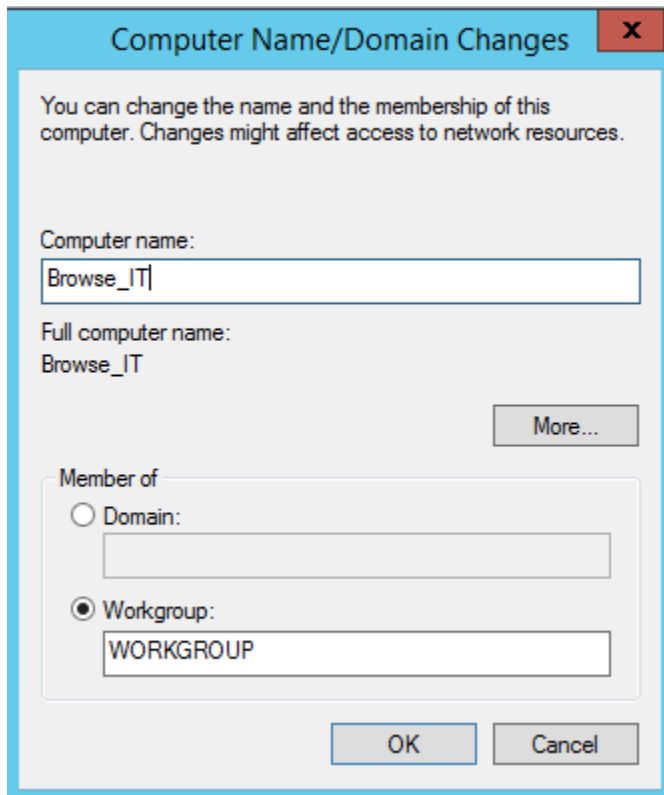
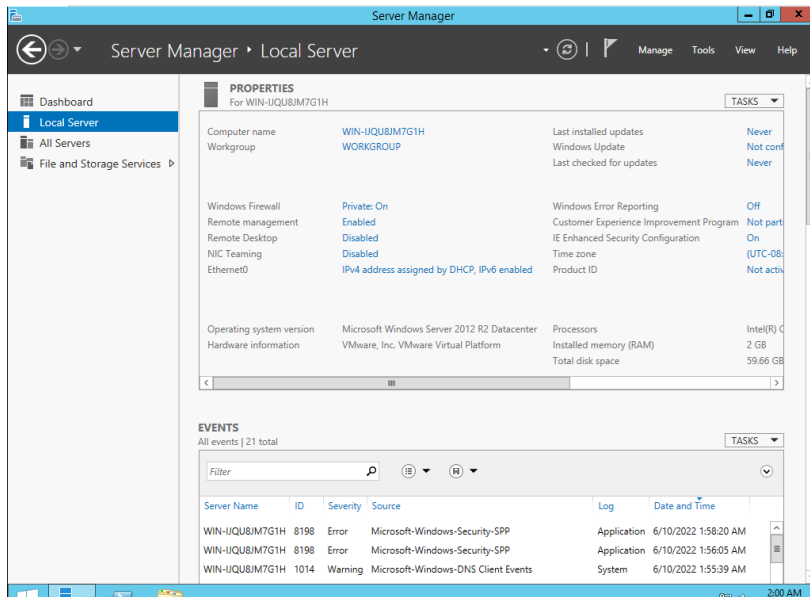
Press Ctrl+Alt+Delete to sign in.

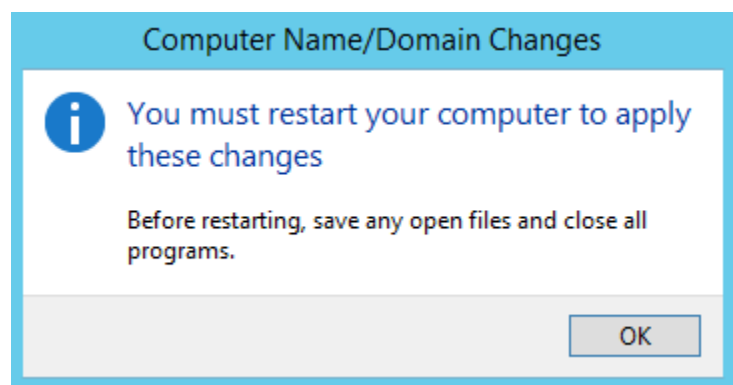
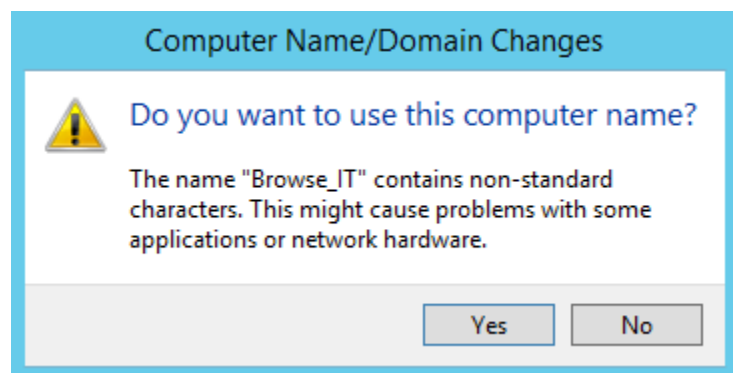
1:55

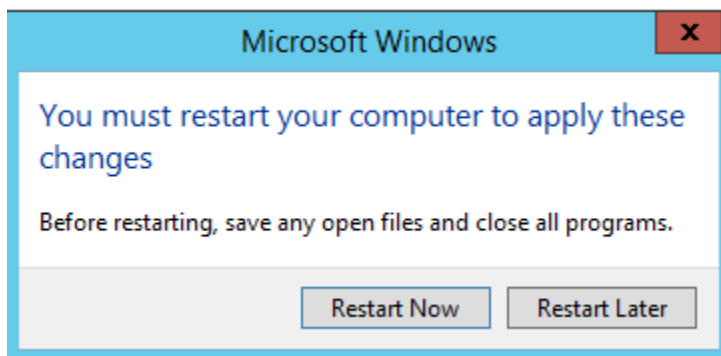
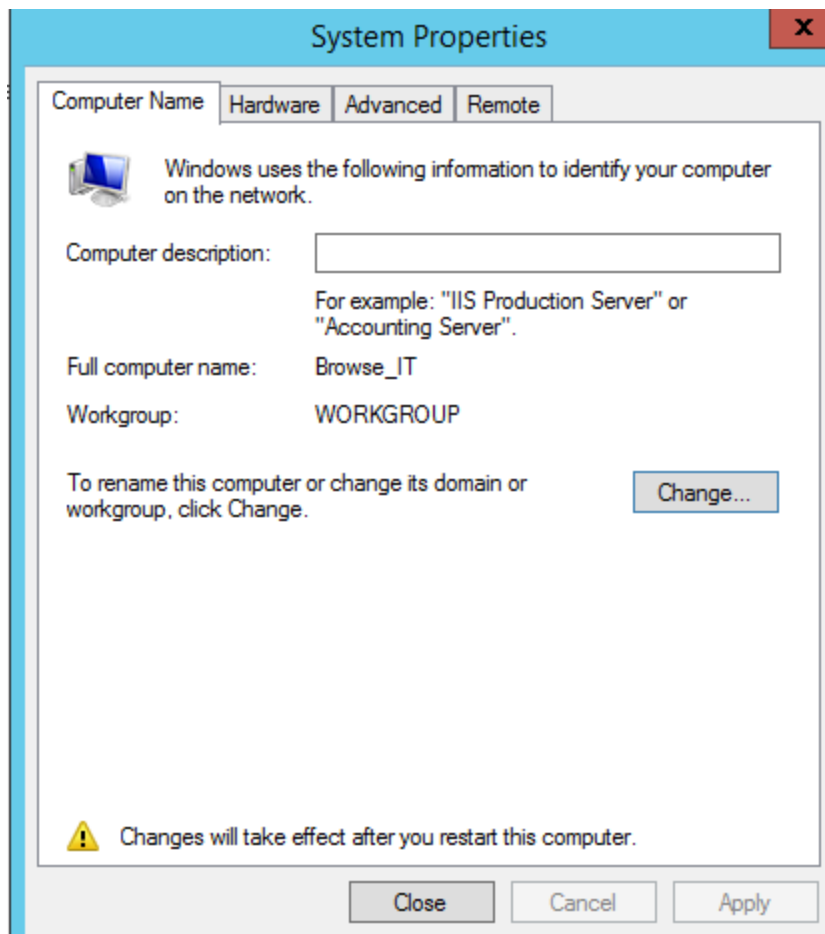
Friday, June 10











Shutting down service: VMware Tools.

Windows Server 2012 R2

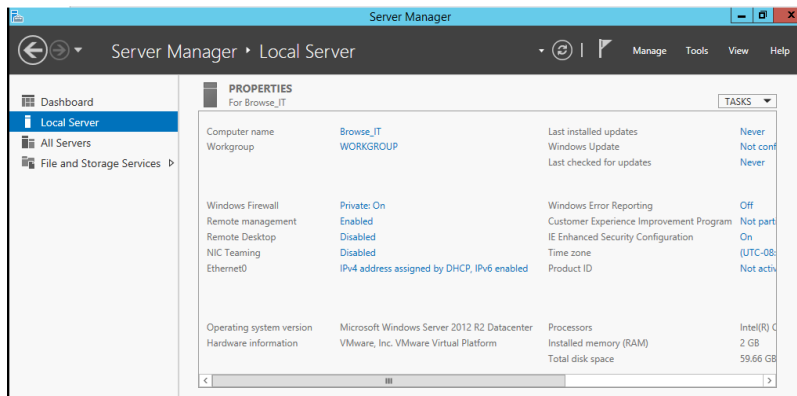


Administrator

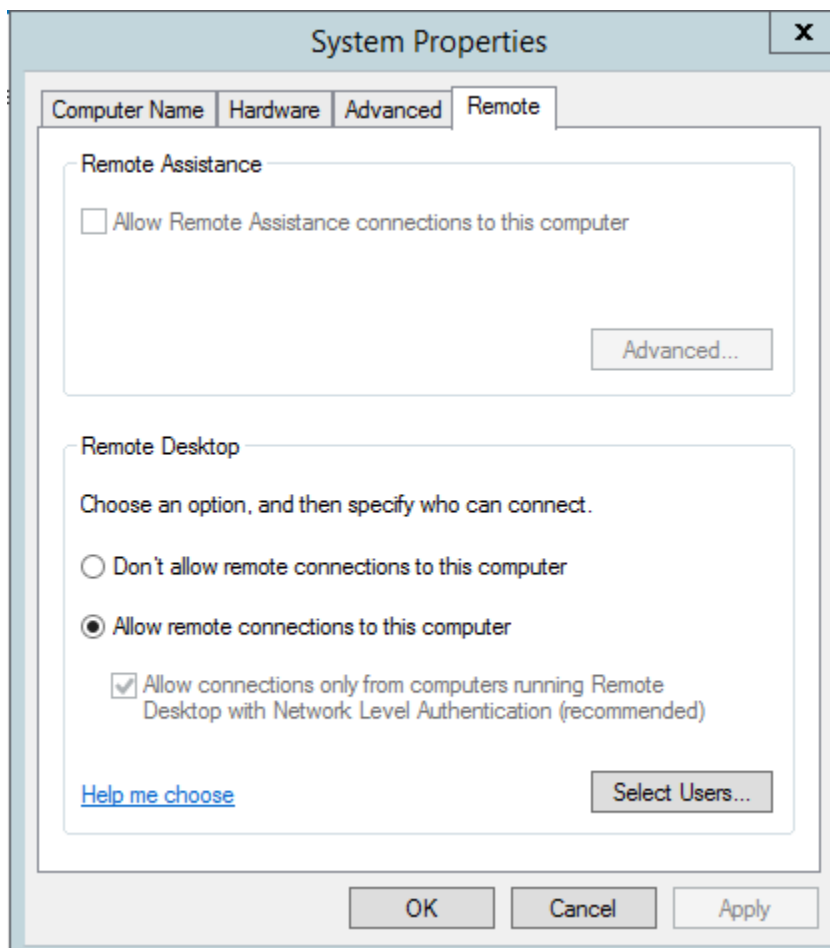
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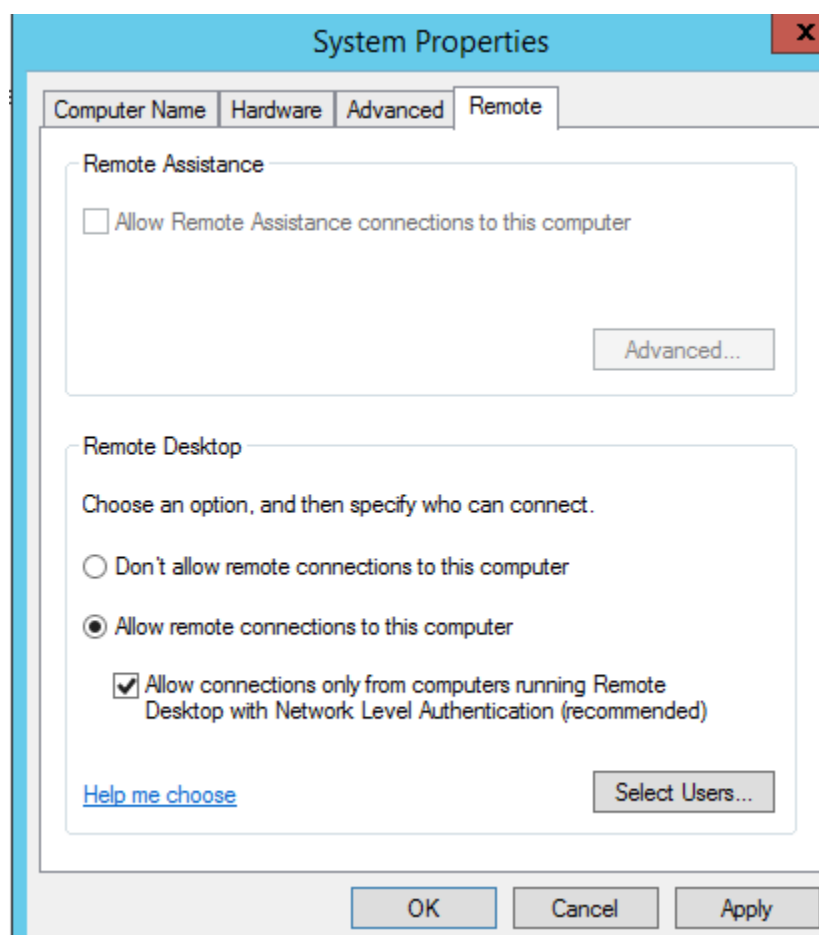
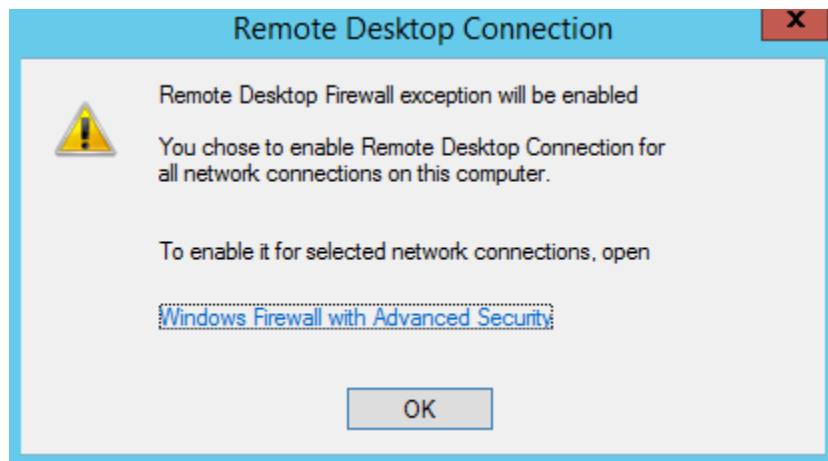


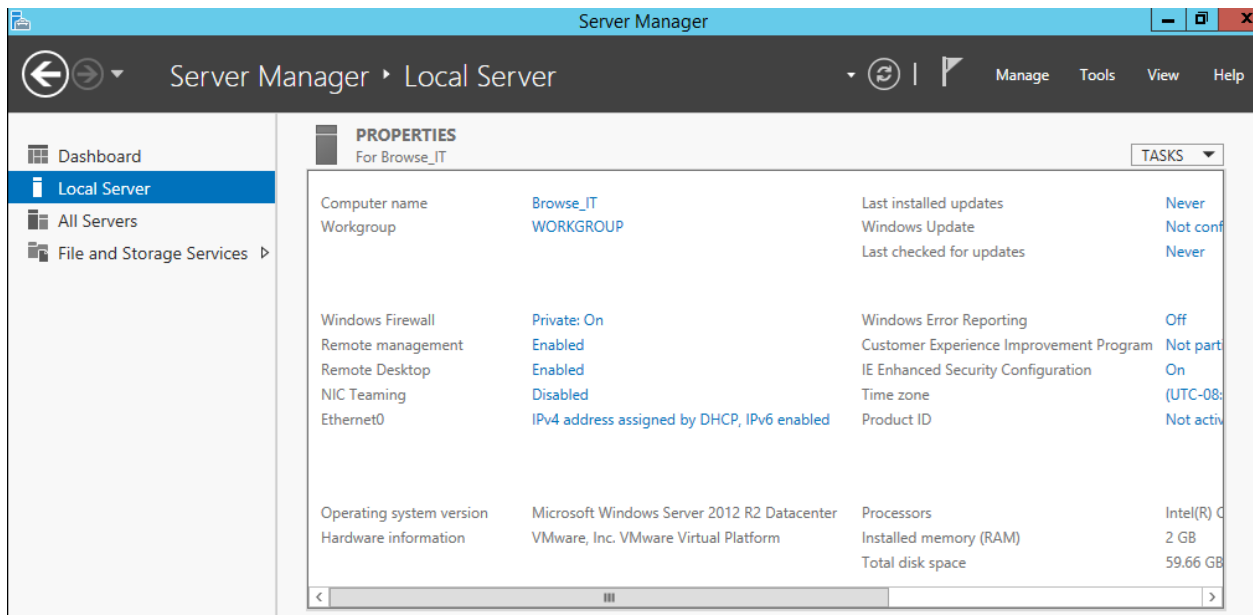
Windows Server 2012 R2



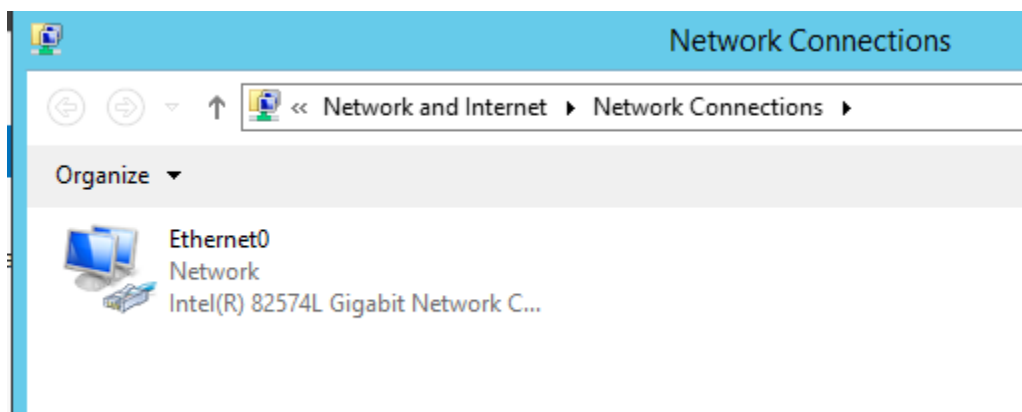
Enable remote desktop connection:

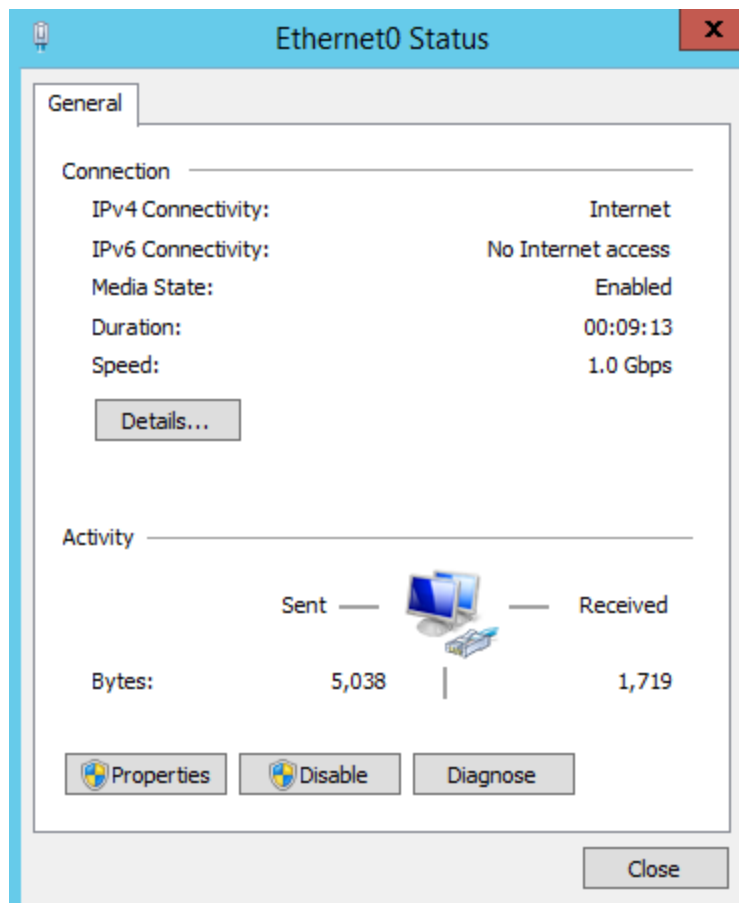


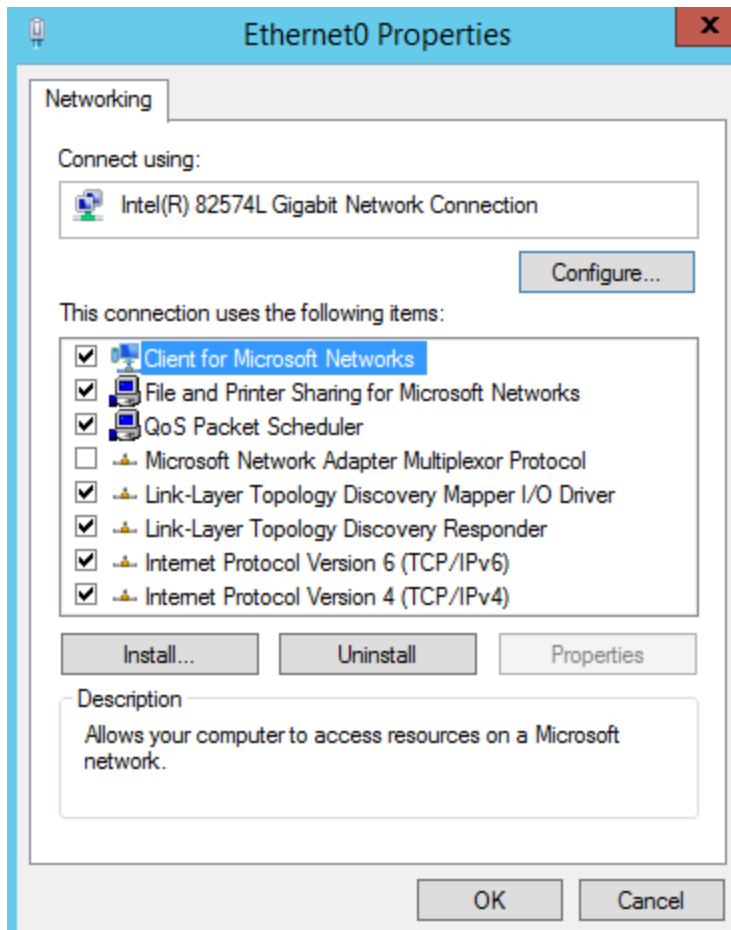




Configure Server's IP Settings







Internet Protocol Version 4 (TCP/IPv4) Properties ✕

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically
☒ Use the following IP address:

IP address:
 Subnet mask:
 Default gateway:

☐ Obtain DNS server address automatically
☒ Use the following DNS server addresses:

Preferred DNS server:
 Alternate DNS server:

☐ Validate settings upon exit
 Advanced...

OK
Cancel

PROPERTIES			
For Browse_IT		TASKS ▼	
Computer name	Browse_IT	Last installed updates	Never
Workgroup	WORKGROUP	Windows Update	Not conf
		Last checked for updates	Never
Windows Firewall	Public: On	Windows Error Reporting	Off
Remote management	Enabled	Customer Experience Improvement Program	Not part
Remote Desktop	Enabled	IE Enhanced Security Configuration	On
NIC Teaming	Disabled	Time zone	(UTC-08:
Ethernet0	10.10.10.25, IPv6 enabled	Product ID	Not activ

Configure Windows update:

TASKS ▼	
Last installed updates	Never
Windows Update	Not configured
Last checked for updates	Never
Windows Error Reporting	Off
Customer Experience Improvement Program	Not participating
IE Enhanced Security Configuration	On
Time zone	(UTC-08:00) Pacific Time (US & Canada)
Product ID	Not activated

Windows Update

Control Panel
System and Security
Windows Update

Search Control Panel

Control Panel Home
Check for updates
Change settings
View update history
Restore hidden updates

Windows Update

Turn on automatic updating

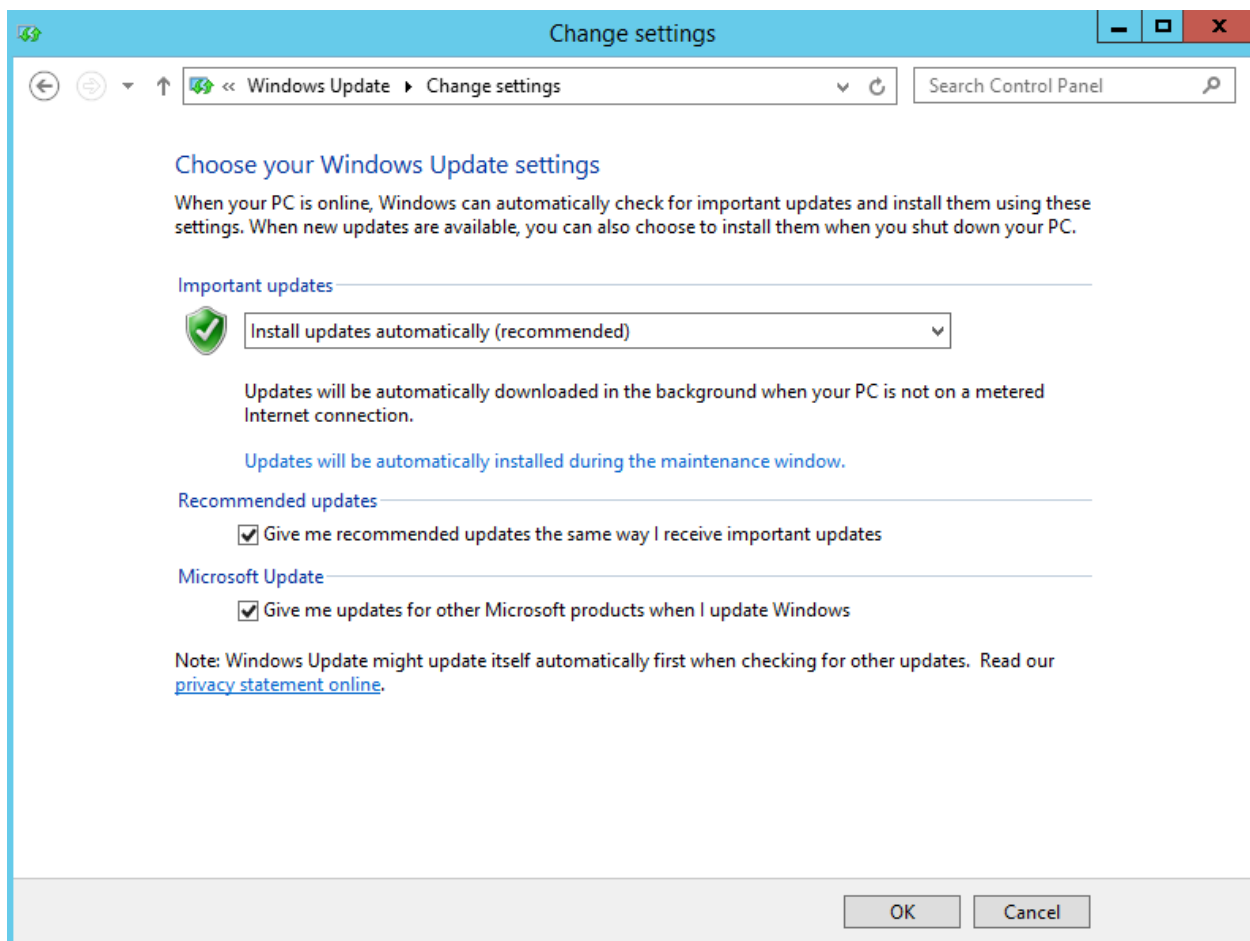
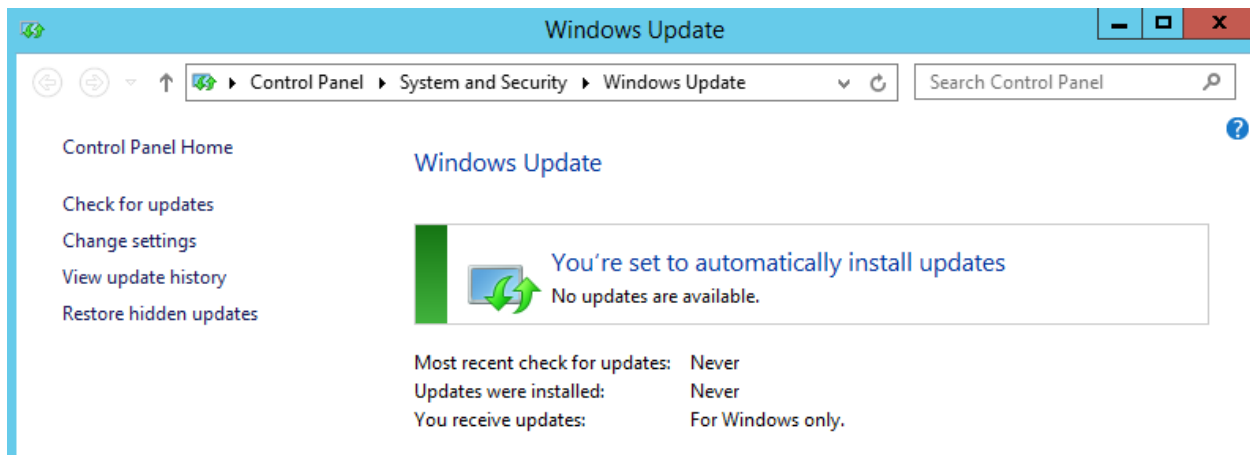
Updates are not being installed automatically

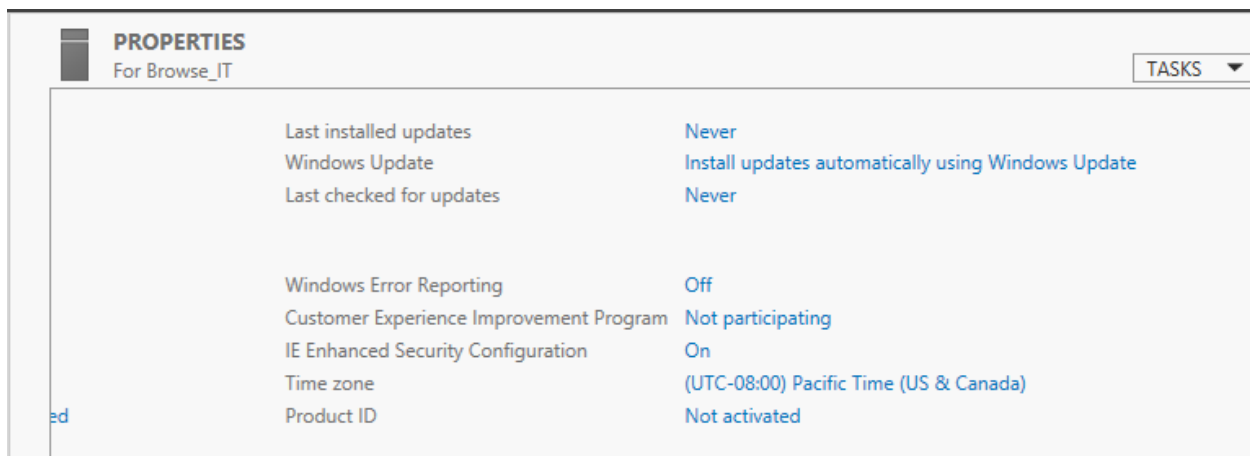
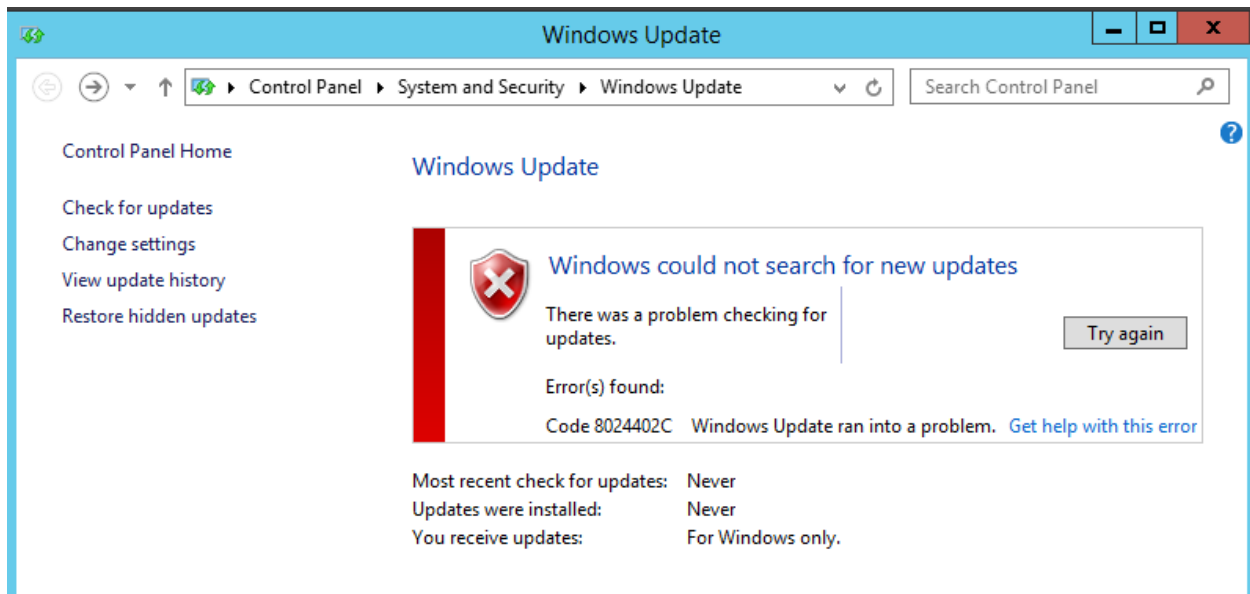
Turn on automatic updating to help improve the security and performance of your PC and allow standard users to install updates on this PC.

Turn on automatic updates
Let me choose my settings

Most recent check for updates:
Updates were installed:
You receive updates:

Never
Never
For Windows only.

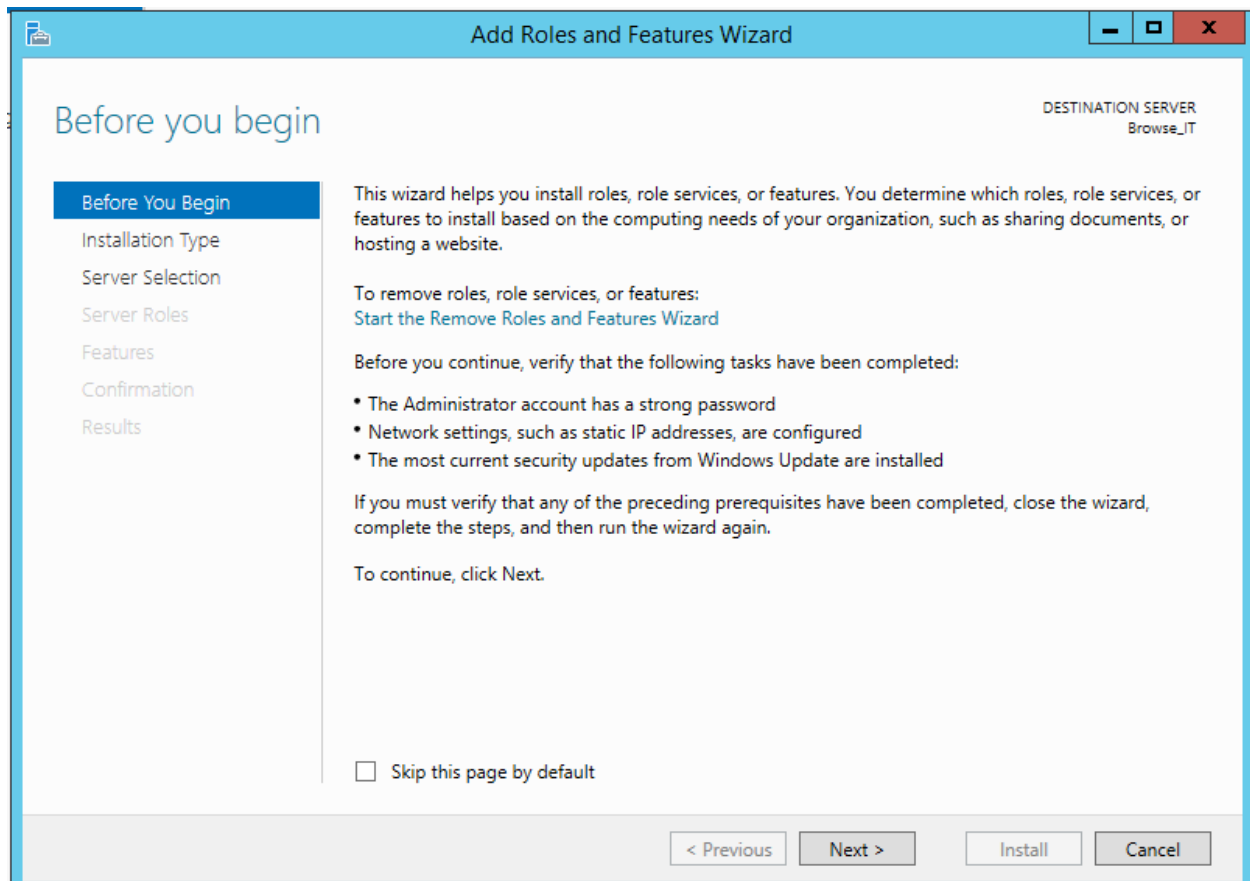
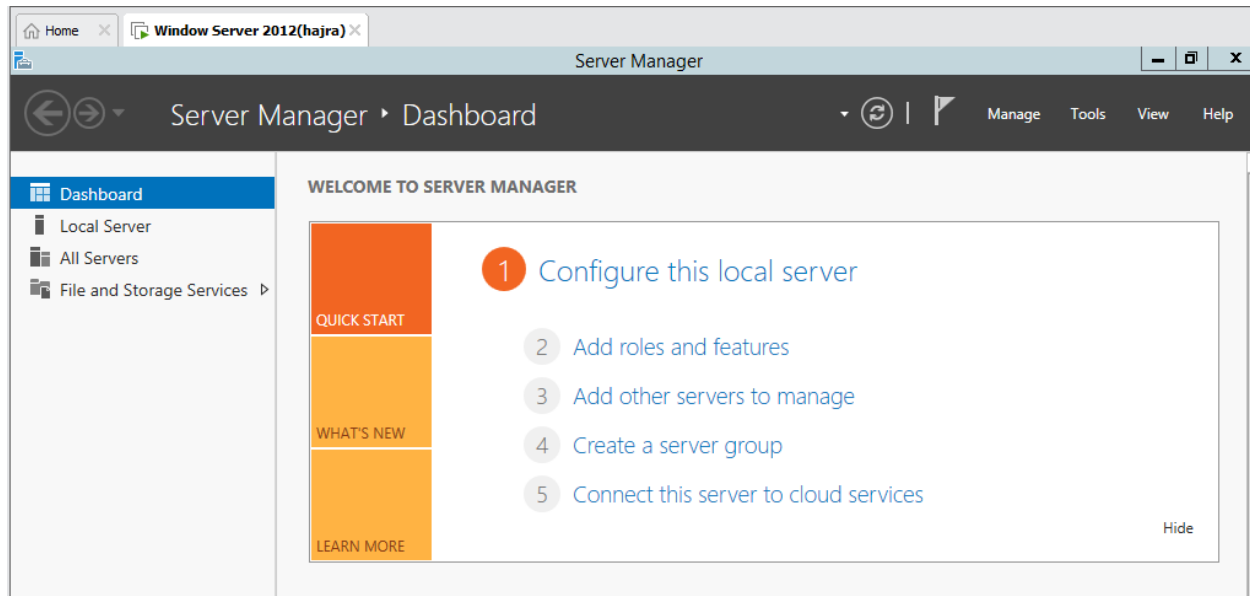


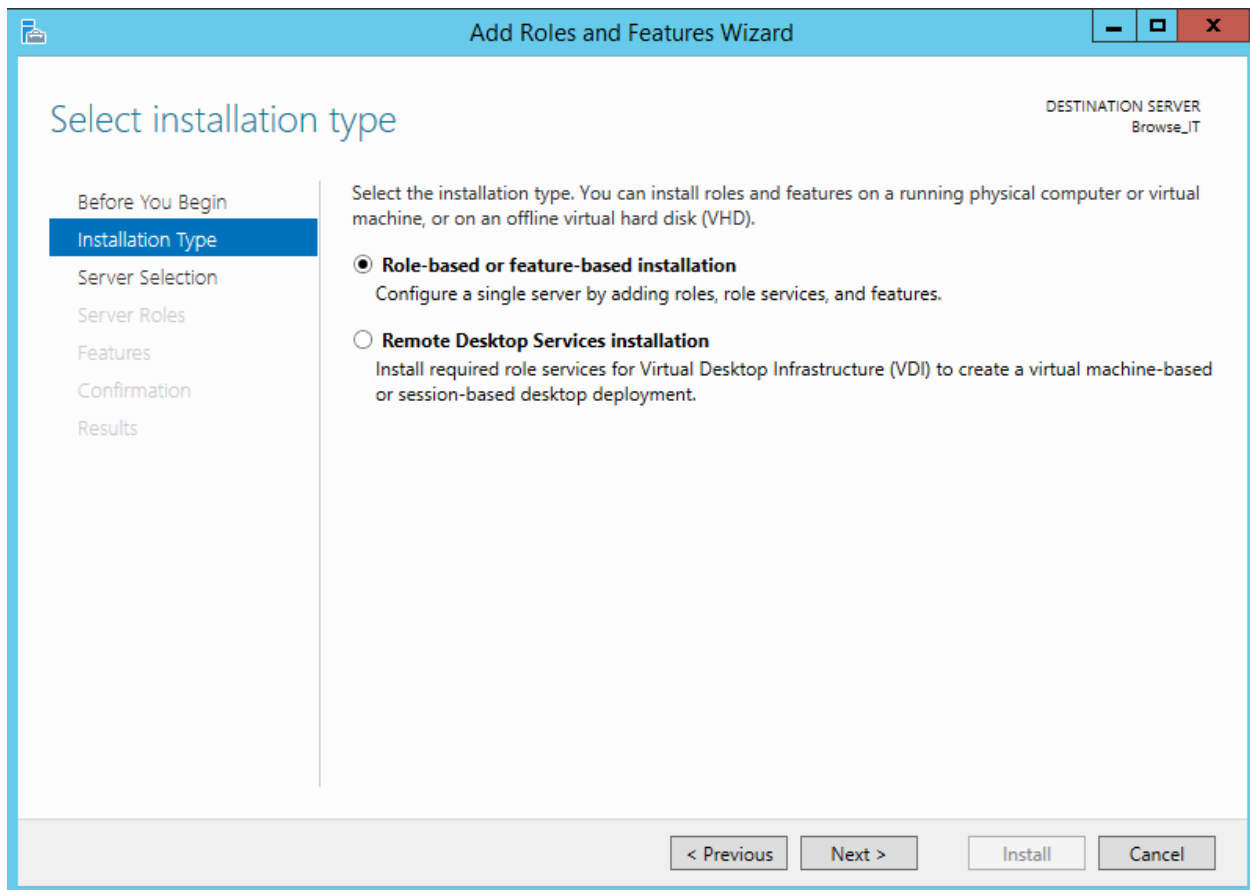


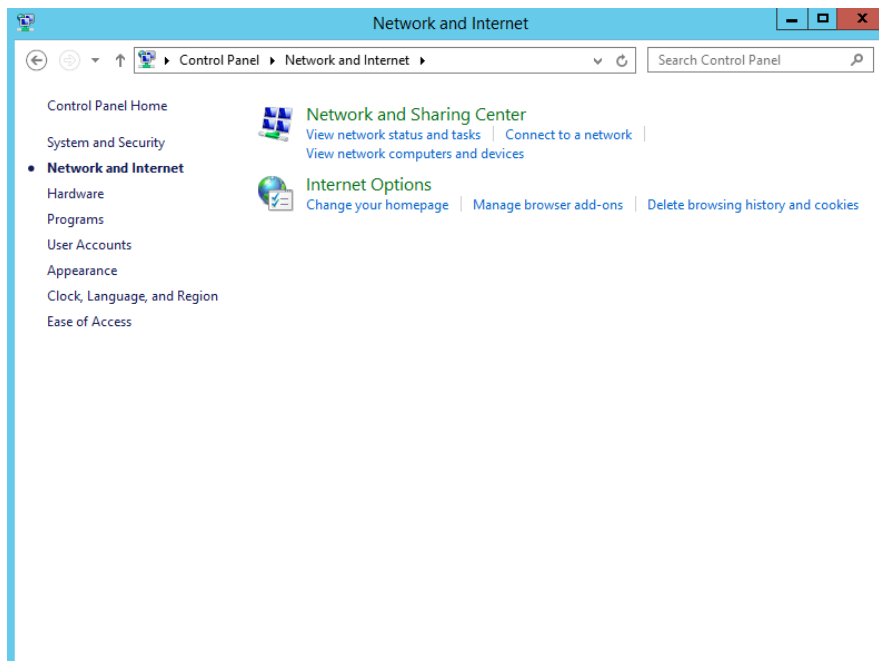
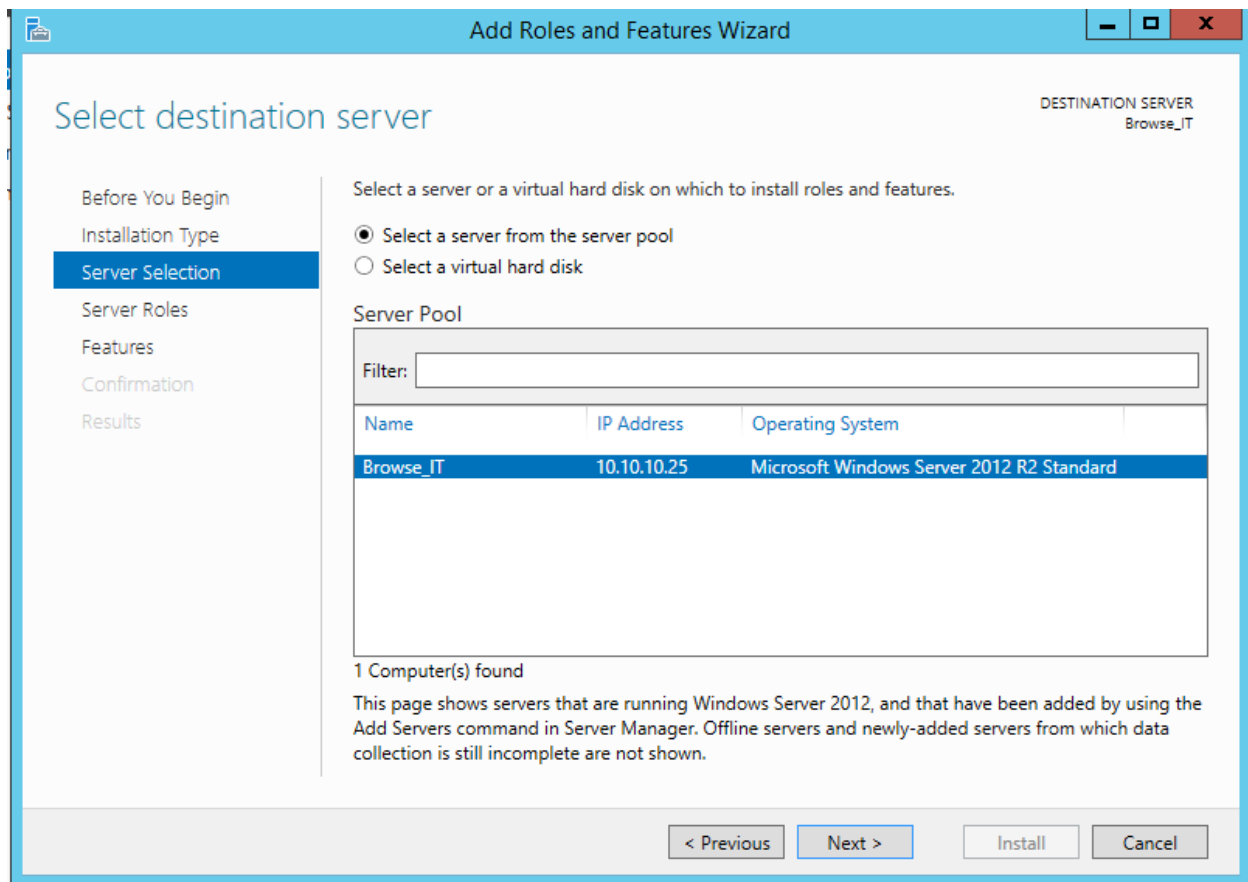
3.Create an active directory:

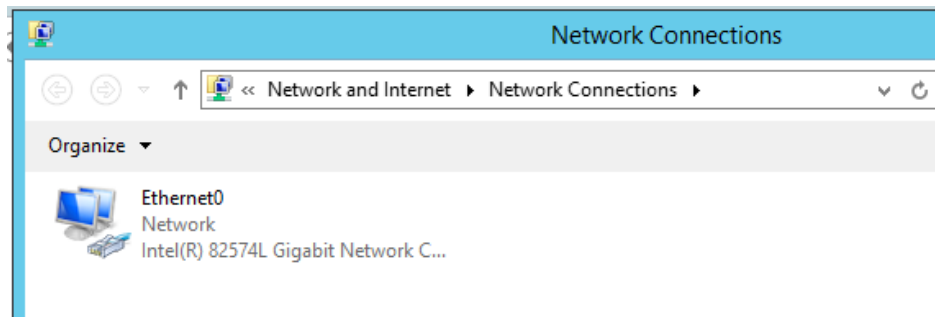
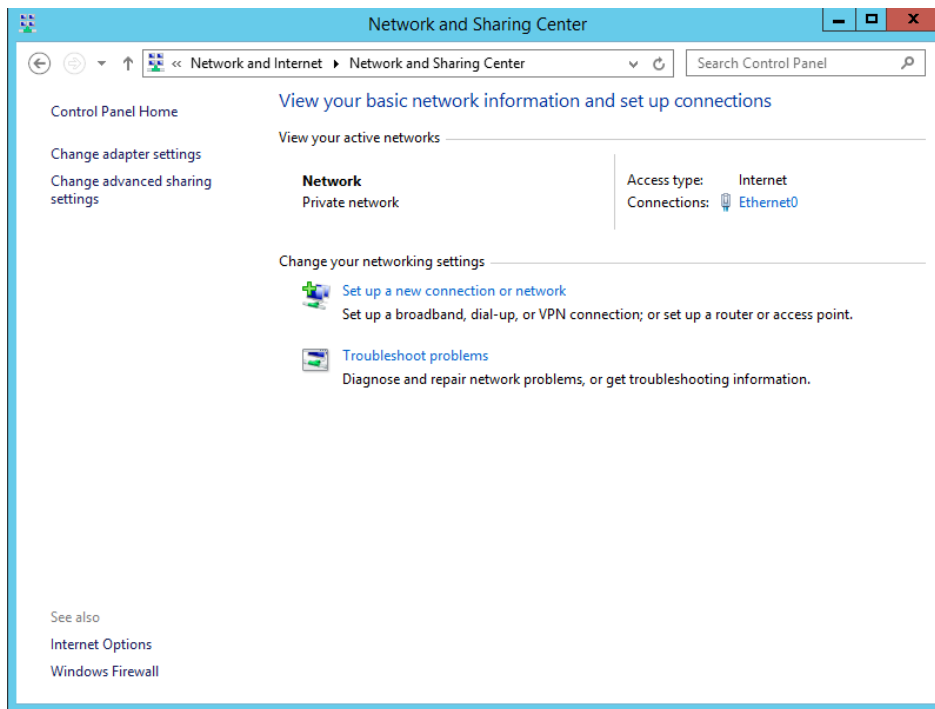
Open virtual machine that install server in it.

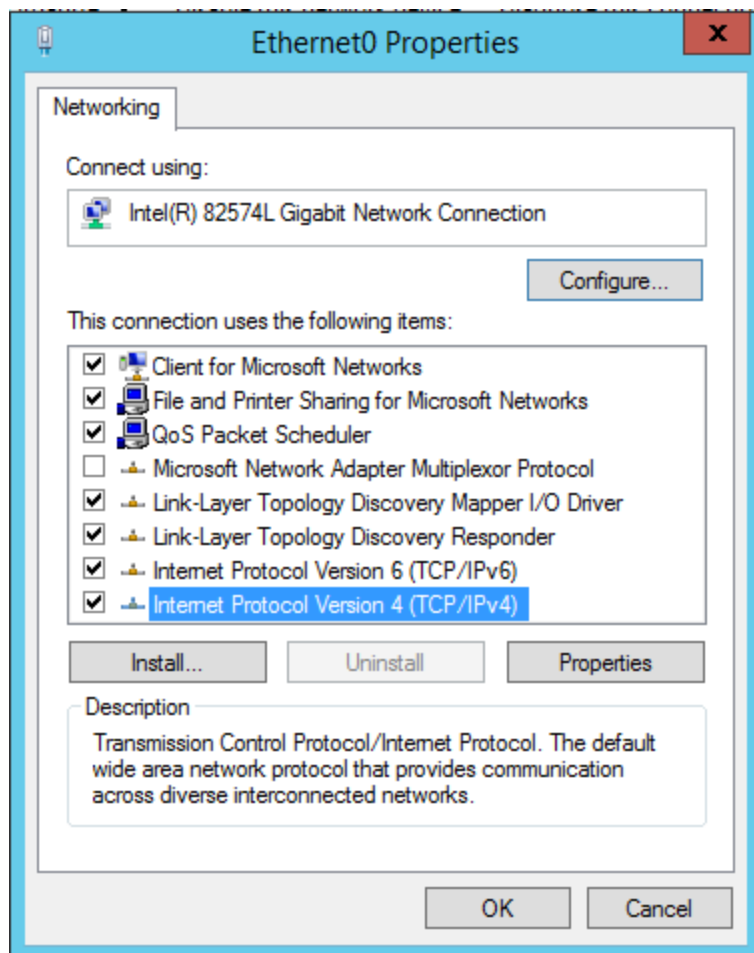
Then perform the following steps.

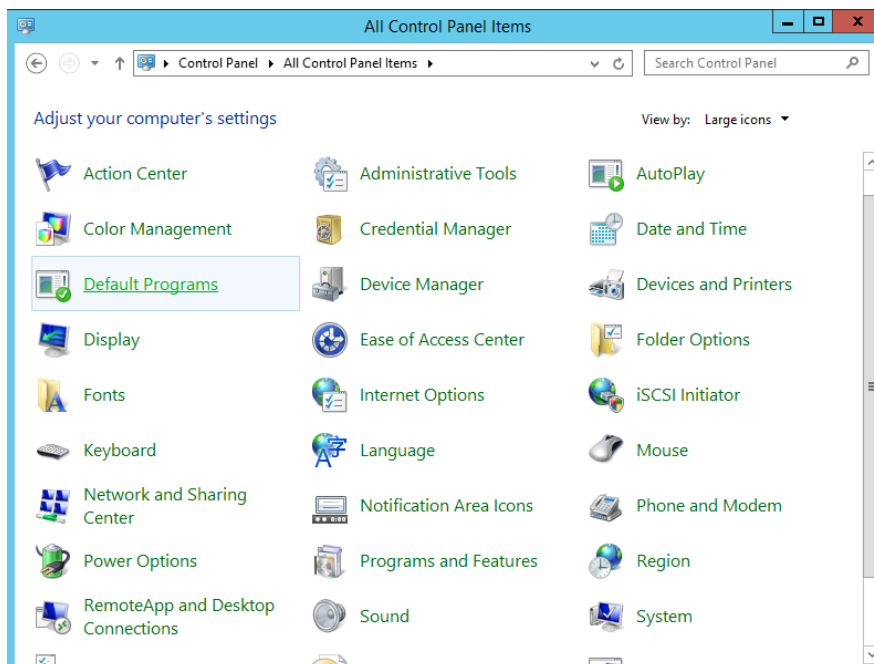
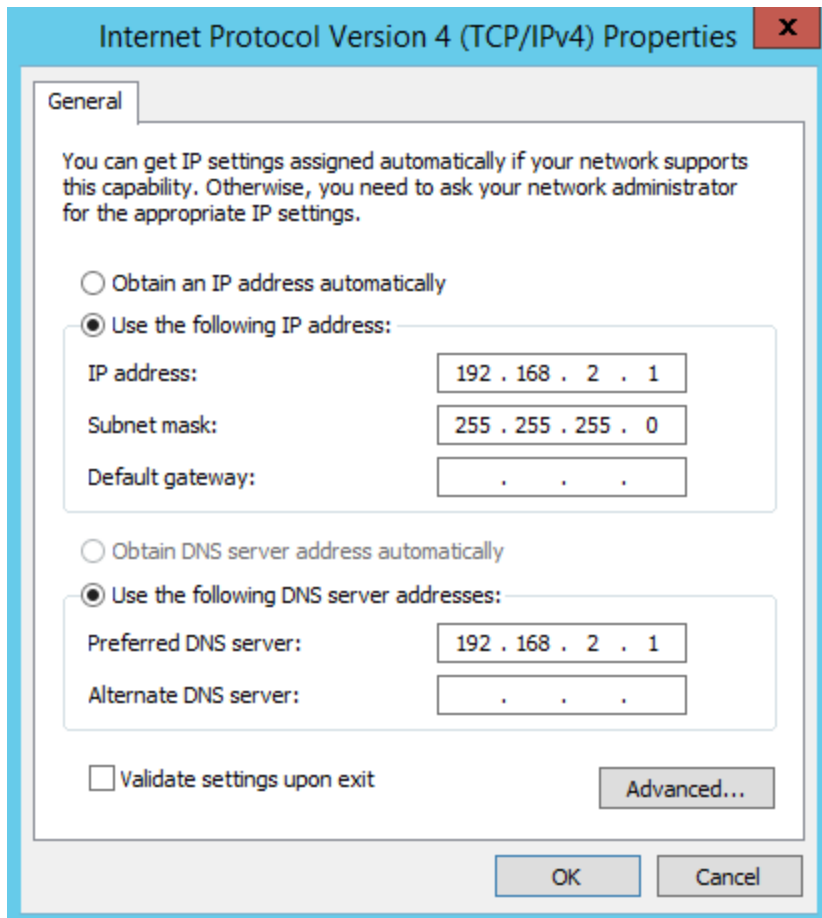


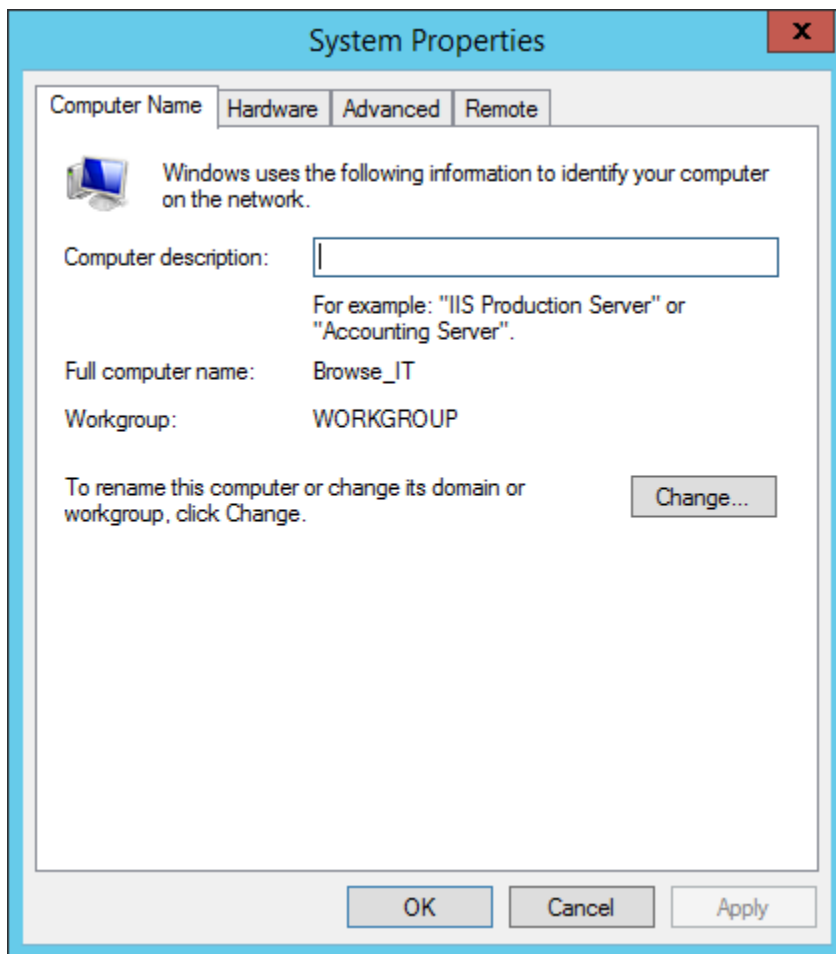
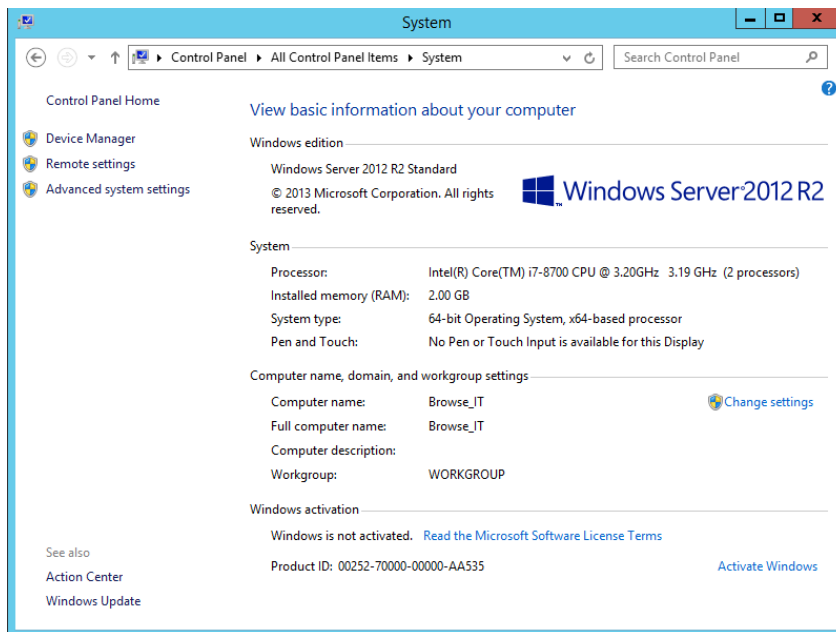


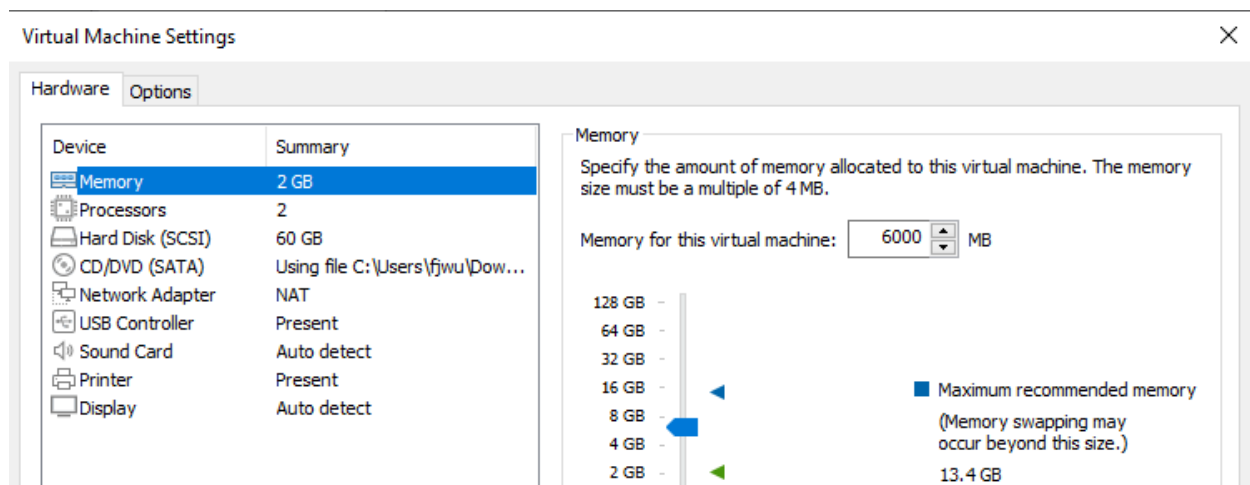
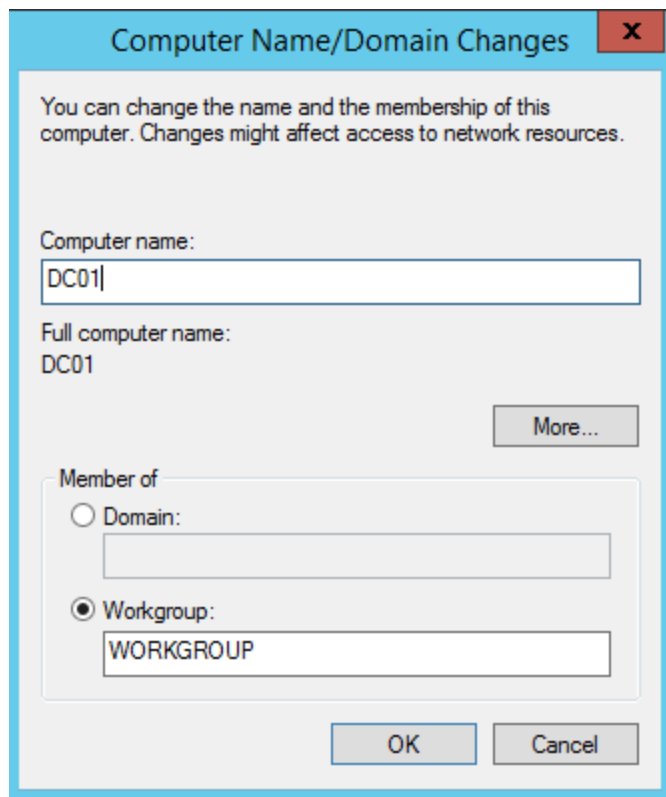


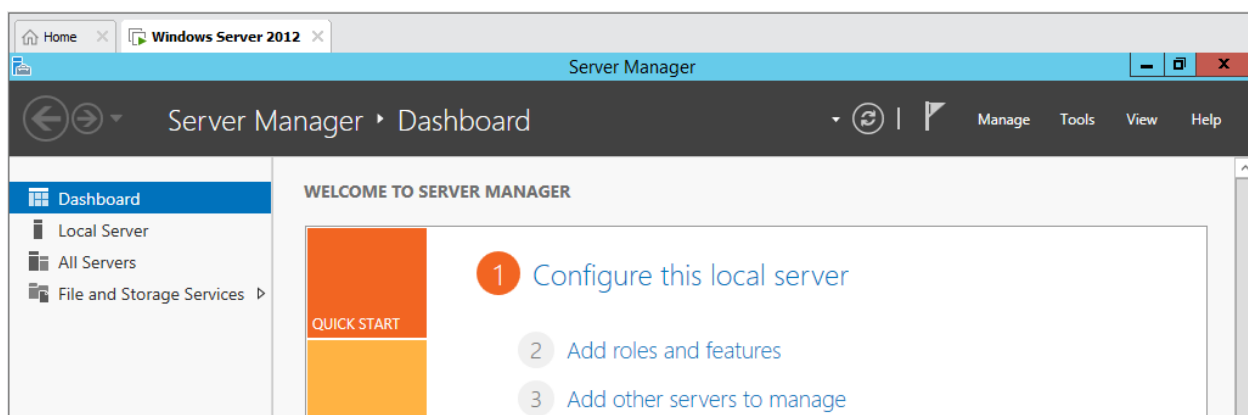
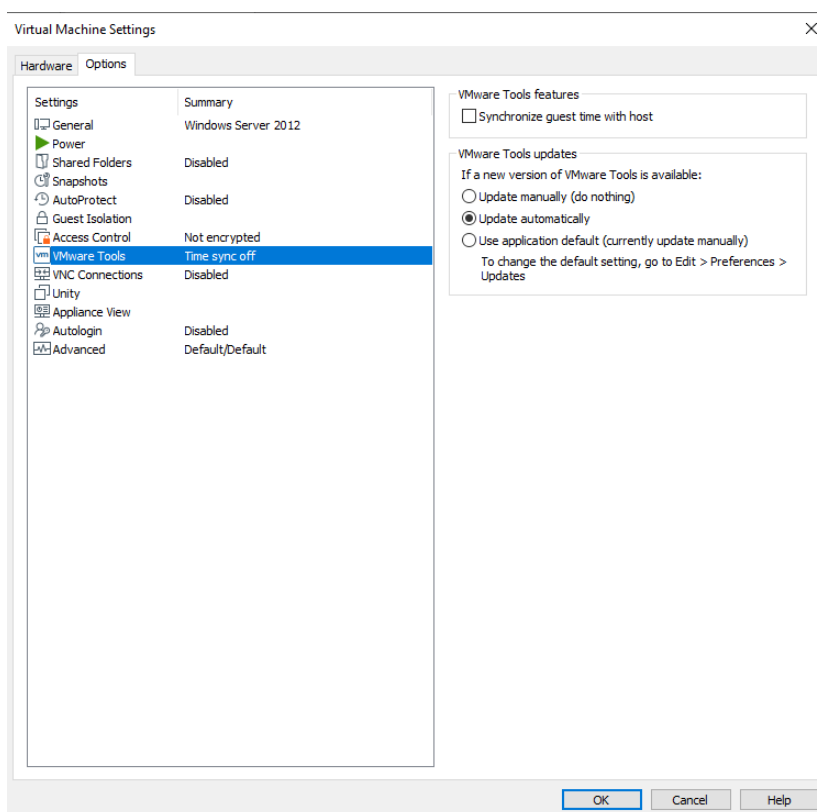
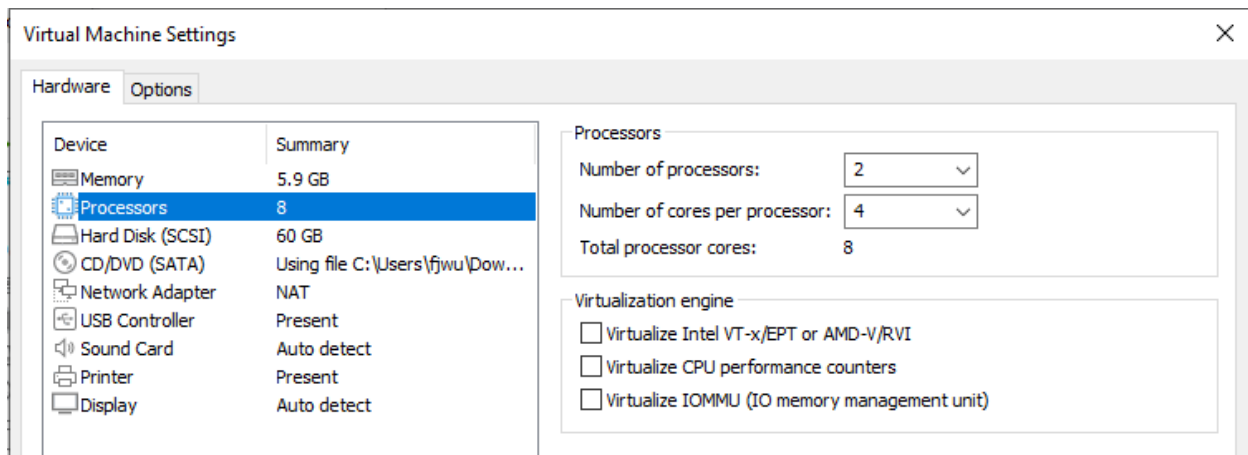












Add Roles and Features Wizard

Before you begin

DESTINATION SERVER
DC01

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

This wizard helps you install roles, role services, or features. You determine which roles, role services, or features to install based on the computing needs of your organization, such as sharing documents, or hosting a website.

To remove roles, role services, or features:
[Start the Remove Roles and Features Wizard](#)

Before you continue, verify that the following tasks have been completed:

- The Administrator account has a strong password
- Network settings, such as static IP addresses, are configured
- The most current security updates from Windows Update are installed

If you must verify that any of the preceding prerequisites have been completed, close the wizard, complete the steps, and then run the wizard again.

To continue, click Next.

☐ Skip this page by default

< Previous

Next >

Install

Cancel

Add Roles and Features Wizard

Select installation type

DESTINATION SERVER
DC01

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select the installation type. You can install roles and features on a running physical computer or virtual machine, or on an offline virtual hard disk (VHD).

☒ **Role-based or feature-based installation**
Configure a single server by adding roles, role services, and features.

☐ **Remote Desktop Services installation**
Install required role services for Virtual Desktop Infrastructure (VDI) to create a virtual machine-based or session-based desktop deployment.

Add Roles and Features Wizard

Select destination server

DESTINATION SERVER
DC01

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool

☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
DC01	192.168.0.1	Microsoft Windows Server 2012 R2 Standard

1 Computer(s) found

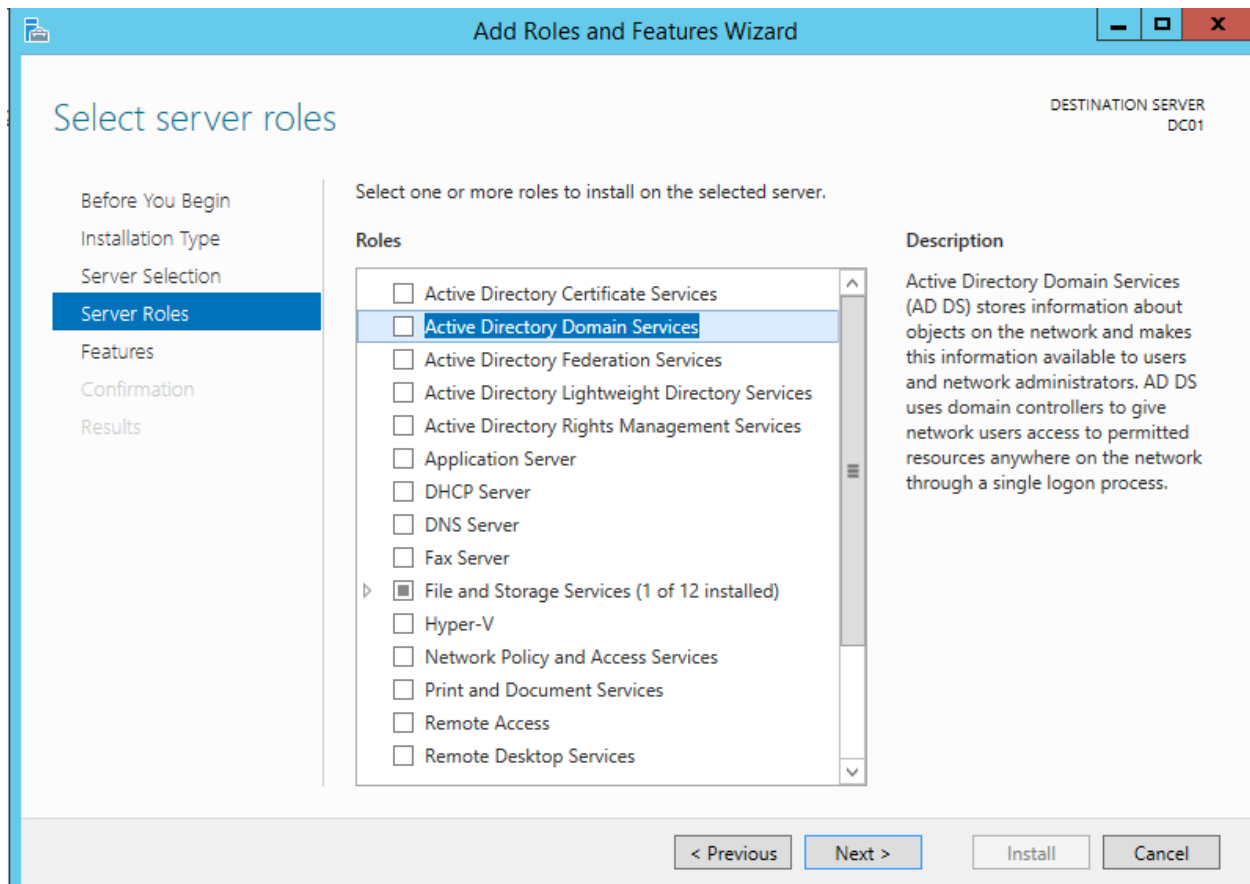
This page shows servers that are running Windows Server 2012, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

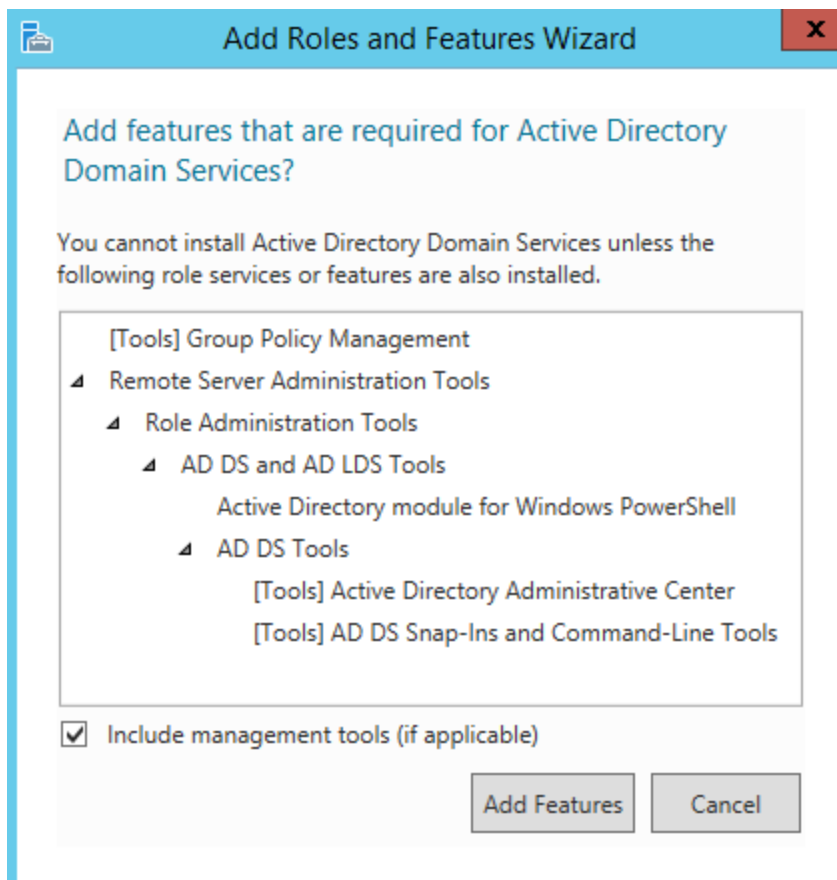
< Previous

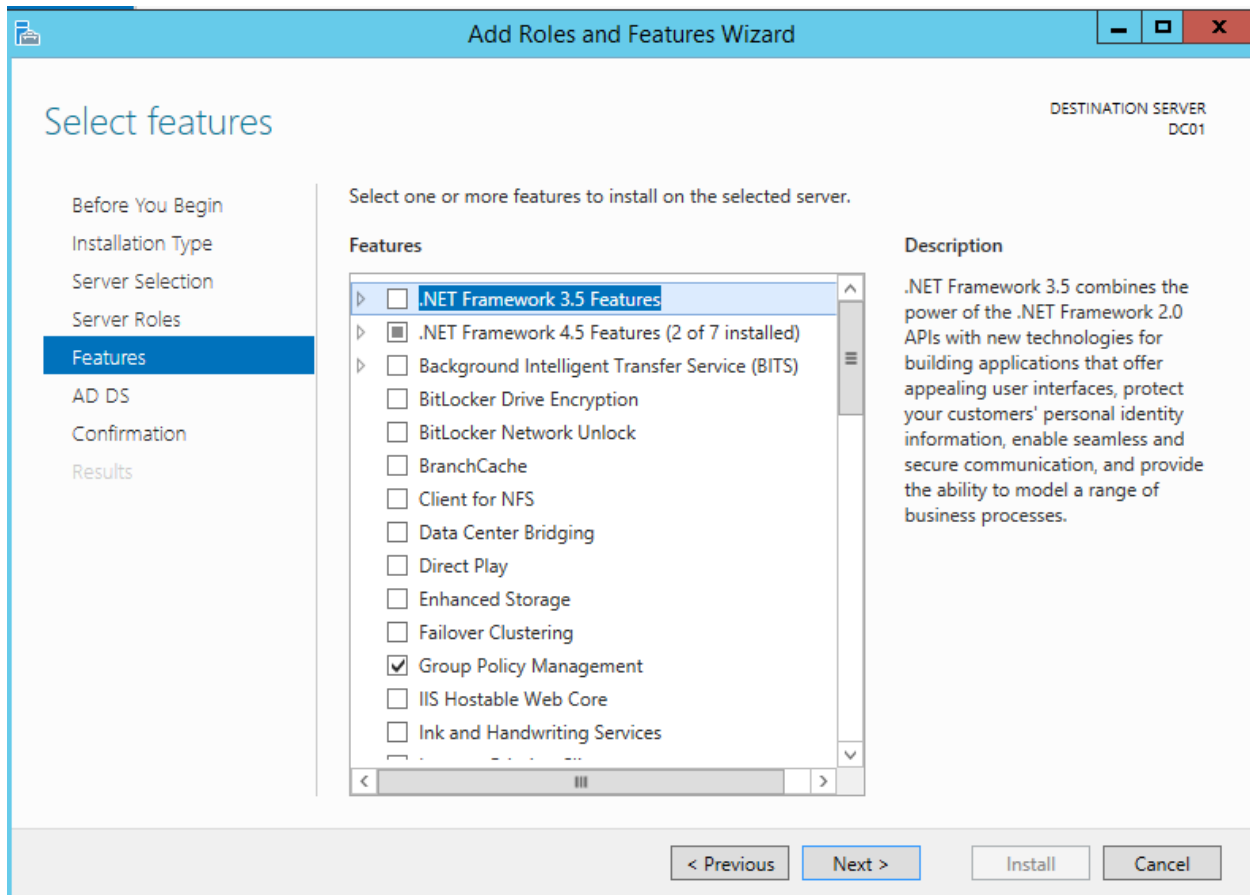
Next >

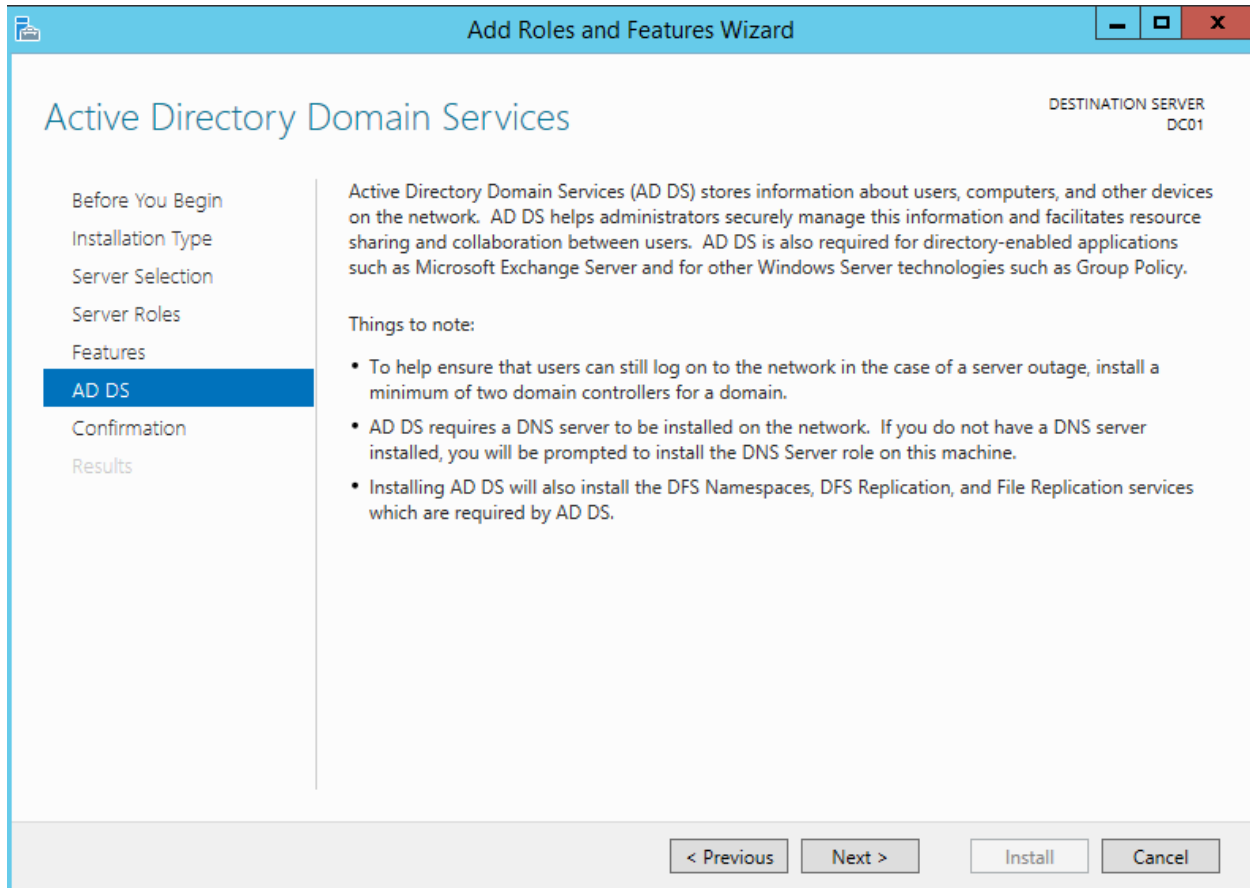
Install

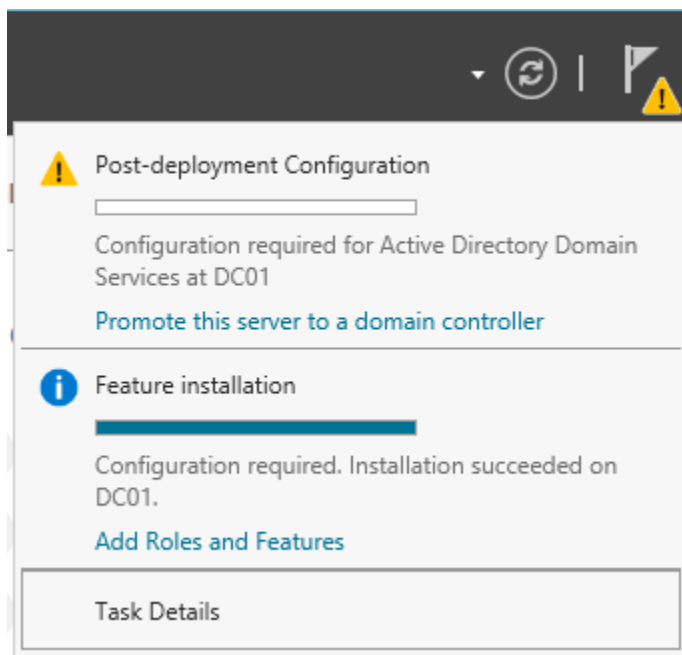
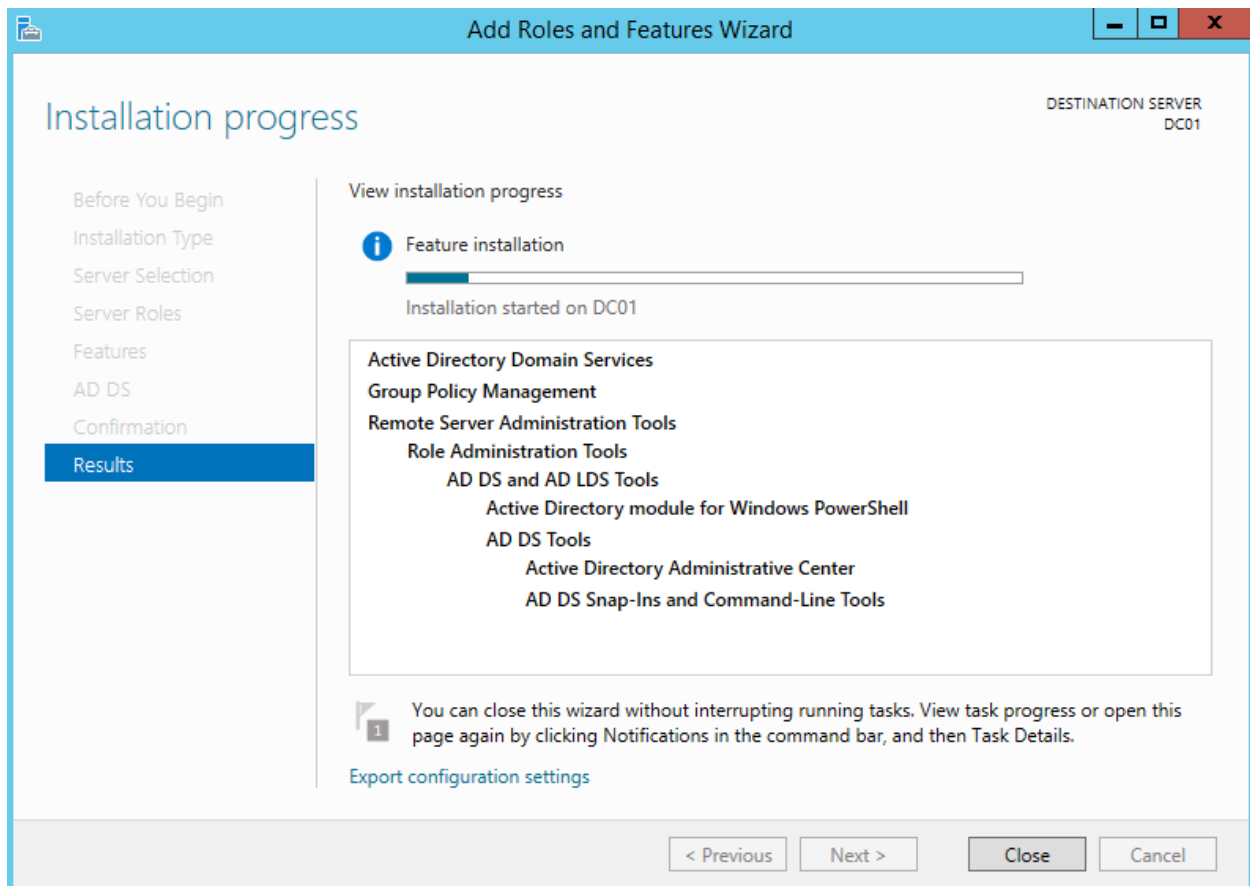
Cancel











Active Directory Domain Services Configuration Wizard

TARGET SERVER
DC01

Deployment Configuration

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

☒ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☐ Add a new forest

Specify the domain information for this operation

Domain:

Select...

Supply the credentials to perform this operation

<No credentials provided>

Change...

[More about deployment configurations](#)

< Previous

Next >

Install

Cancel

Active Directory Domain Services Configuration Wizard

TARGET SERVER
DC01

Deployment Configuration

Deployment Configuration

Domain Controller Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select the deployment operation

☐ Add a domain controller to an existing domain

☐ Add a new domain to an existing forest

☒ Add a new forest

Specify the domain information for this operation

Root domain name:

[More about deployment configurations](#)

< Previous

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Install

Cancel

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
DC01

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Select functional level of the new forest and root domain

Forest functional level:Windows Server 2012 R2

Domain functional level:Windows Server 2012 R2

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:.....

Confirm password:.....

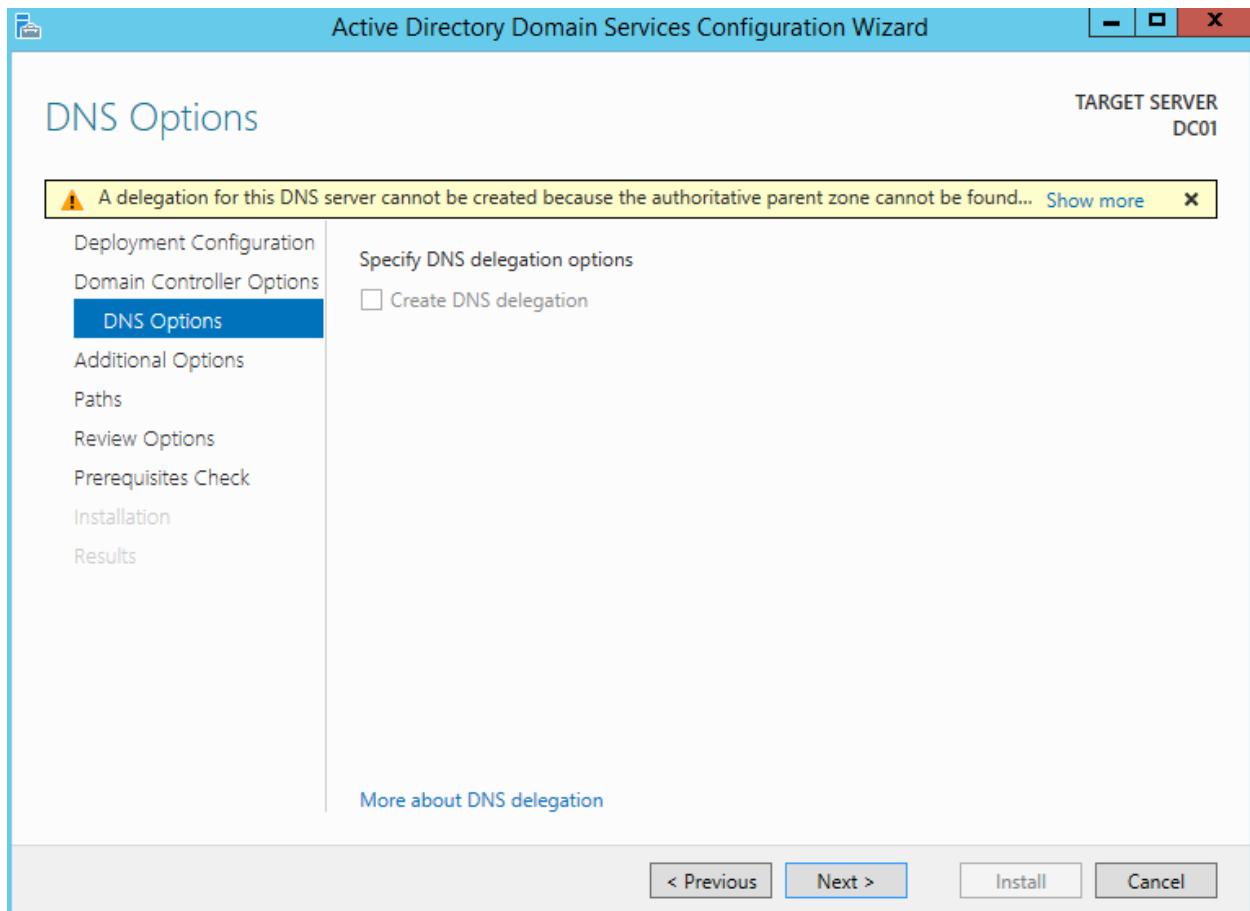
[More about domain controller options](#)

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Install

Cancel



Active Directory Domain Services Configuration Wizard

Additional Options

TARGET SERVER
DC01

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Verify the NetBIOS name assigned to the domain and change it if necessary

The NetBIOS domain name:

[More about additional options](#)

< Previous Next > Install Cancel

Active Directory Domain Services Configuration Wizard

Paths

TARGET SERVER
DC01

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Specify the location of the AD DS database, log files, and SYSVOL

Database folder:

C:\Windows\NTDS

...

Log files folder:

C:\Windows\NTDS

...

SYSVOL folder:

C:\Windows\SYSVOL

...

More about Active Directory paths

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Next >

Install

Cancel

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Active Directory Domain Services Configuration Wizard

Review Options

TARGET SERVER
DC01

Deployment Configuration

Domain Controller Options

DNS Options

Additional Options

Paths

Review Options

Prerequisites Check

Installation

Results

Review your selections:

Configure this server as the first Active Directory domain controller in a new forest.

The new domain name is "WidgetLLC.Internal". This is also the name of the new forest.

The NetBIOS name of the domain: WIDGETLLC

Forest Functional Level: Windows Server 2012 R2

Domain Functional Level: Windows Server 2012 R2

Additional Options:

Global catalog: Yes

DNS Server: Yes

Create DNS Delegation: No

These settings can be exported to a Windows PowerShell script to automate additional installations

View script

[More about installation options](#)

< Previous

Next >

Install

Cancel

